FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Wine Production Forecasting Using Rule-Based Machine Learning: A Distributional Subgroup Discovery Approach with Walk-Forward Validation Forecast for Portuguese Wine Regions

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Abstract

This thesis evaluates the CarenR distributional subgroup discovery framework as a rule-based machine learning approach for two complementary tasks on real-world agricultural production time series: interpretable pattern extraction (Goal 1) and walk-forward prediction (Goal 2). The application domain is annual wine production forecasting for two contrasting Portuguese wine regions, the Douro Demarcated Region (RDD, 89 years, 1934–2022) and the Vinho Verde Region (RVV, 80 years, 1942–2021). The two regions serve as a natural comparative experiment, same pipeline, same algorithms, different structural histories and climate regimes, to evaluate when and why the method succeeds or fails under different signal conditions.

Two methodological contributions support the dual goals. For Goal 1, CarenR is applied to the full linearly detrended production series to extract interpretable agroclimatic rules. Each rule is a conjunction of climate conditions under which the distribution of production residuals shifts significantly, assessed by a two-sample Kolmogorov–Smirnov test. Rule quality is further checked using LOESS-detrended residuals: features selected by CarenR retain 90–92% of their correlation after aggressive era-trend removal, confirming genuine within-era climate signal rather than era-membership artefacts. For Goal 2, a two-decision decomposition framework separates prediction error into trend quality (Decision 1: how well the training trend extrapolates to test years) and climate signal extraction (Decision 2: whether a rule-based model can predict detrended residuals better than the training median), identifying which element of the pipeline is the bottleneck. The evaluation uses an expanding-window walk-forward protocol with 18 scenarios for RDD and 16 for RVV, with all preprocessing — detrending, feature discretisation, and feature selection — performed within each training fold to prevent leakage. Eleven CarenR variants are evaluated as a structured sensitivity analysis of feature selection strategies, alongside RIPPER, M5Rules, a Decision Tree regressor, and naive baselines, totalling 17 models.

Goal 1 (Rule Discovery) is achieved. CarenR's KS-test-based subgroup induction discovers 48 statistically validated rules for RDD and 20 for RVV. Feature frequency analysis shows that harvest-period soil water appears in 54% of RDD rules, the most consistently selected variable across all 48 rules, while in RVV the most frequently selected features are heat-stress days around veraison (up to 50% of rules) and harvest soil water (25%), with the Leaf Area Index (LAI) family present in 4 of the 20 rules. LOESS detrending validates that the selected features carry genuine within-era signal (correlations drop by 8% after era-trend removal), confirming that CarenR extracts real structure rather than era-membership artefacts. A cross-algorithm comparison shows that RIPPER and M5Rules independently converge on the same variable families, confirming that the identified variables are robust across three different optimisation criteria.

Goal 2 (Predictive Validation) is not achieved overall. No rule-based model consistently outperforms the Naive_Median baseline across the full evaluation sequence. The best CarenR variant falls 3% behind the baseline in RDD (189 vs 184 mhl weighted MAE; Wilcoxon $p = 0.468$ across all 18 scenarios, not significant) and 22% behind in RVV (172 vs 140 mhl). In the 9 RVV scenarios where rules fired, CarenR was significantly worse than the baseline (Wilcoxon

$p = 0.004$). A post-hoc era-composition analysis, conducted after observing the results and therefore exploratory, suggests CarenR consistently outperforms the baseline in RDD when training data reaches 29% post-1990 representation (5 of 5 consecutive scenarios, binomial $p = 0.031$), pointing to a conditional regime where rule-based prediction could succeed. The two-decision framework explains the failure: the demands of accurate trend modelling and sufficient residual signal for rule induction are in fundamental tension under a time-only detrending covariate.

Keywords: *distributional subgroup discovery, CarenR, rule-based machine learning, walk-forward validation, time series evaluation, two-decision decomposition, feature selection, wine production forecasting, non-stationarity, era confound*

Resumo

Esta tese avalia o método de descoberta de subgrupos distributivos CarenR enquanto abordagem de aprendizagem automática baseada em regras para duas tarefas complementares em problemas de séries temporais de produção agrícola do mundo real: extração de padrões interpretáveis (Objetivo 1) e previsão com validação *walk-forward* (Objetivo 2). O domínio de aplicação corresponde à previsão anual da produção de vinho em duas regiões vitivinícolas portuguesas contrastantes: a Região Demarcada do Douro (RDD, 89 anos, 1934–2022) e a Região dos Vinhos Verdes (RVV, 80 anos, 1942–2021). Estas duas regiões constituem uma experiência comparativa natural, na qual o mesmo *pipeline* e os mesmos algoritmos são aplicados a históricos estruturais e regimes climáticos distintos, permitindo avaliar em que condições e por que razão o método obtém sucesso ou falha perante diferentes regimes de sinal.

Duas contribuições metodológicas sustentam estes dois objetivos. Para o Objetivo 1, o CarenR é aplicado à série temporal completa da produção, após remoção da tendência linear, com o propósito de extrair regras agroclimáticas interpretáveis. Cada regra consiste numa conjunção de condições climáticas sob as quais a distribuição dos resíduos da produção difere significativamente, sendo essa diferença avaliada através de um teste de Kolmogorov–Smirnov de duas amostras. A qualidade de cada regra é posteriormente validada recorrendo aos resíduos obtidos após a remoção da tendência através do método LOESS, demonstrando que as características selecionadas pelo CarenR preservam entre 90% e 92% da sua correlação, mesmo após uma remoção agressiva da tendência associada à era, o que confirma a existência de um sinal climático genuíno intra-era, em vez de artefactos decorrentes da pertença à era.

Para o Objetivo 2, é proposta uma estrutura de decomposição em duas decisões, que separa o erro de previsão em duas componentes: qualidade da modelação da tendência (Decisão 1: até que ponto a tendência estimada no conjunto de treino é capaz de extrapolar para os anos de teste) e extração do sinal climático (Decisão 2: se um modelo baseado em regras consegue prever os resíduos obtidos após a remoção da tendência melhor do que a mediana do conjunto de treino). Esta decomposição permite identificar qual das componentes do *pipeline* constitui o principal fator limitativo do desempenho preditivo.

A avaliação da capacidade preditiva segue um protocolo de validação *walk-forward* com janela de treino em expansão, compreendendo 18 cenários para a RDD e 16 para a RVV. Todo o pré-processamento — remoção da tendência, discretização e seleção de atributos — é realizado exclusivamente dentro de cada conjunto de treino, de forma a evitar a fuga de informação (*data leakage*). São avaliadas onze variantes do CarenR, correspondentes a uma análise de sensibilidade das estratégias de seleção de atributos, em conjunto com os algoritmos RIPPER e M5Rules, um regressor baseado em Árvores de Decisão e modelos de base simples, perfazendo um total de 17 modelos.

O Objetivo 1 (Descoberta de Regras) é alcançado. A indução de subgrupos baseada no teste de Kolmogorov–Smirnov implementada pelo CarenR identifica 48 regras estatisticamente significativas para a RDD e 20 para a RVV. A análise da frequência dos atributos revela que a água

do solo durante o período de colheita está presente em 54% das regras da RDD, sendo a variável selecionada de forma mais consistente ao longo das 48 regras, enquanto o Índice de Área Foliar (LAI) na colheita domina o conjunto de regras da RVV, estando presente em todas as 20 regras.

A validação através da remoção da tendência pelo método LOESS confirma que os atributos selecionados transportam um sinal genuíno intra-era, verificando-se apenas uma redução de 8% nas correlações após a remoção da tendência da era, o que demonstra que o CarenR extrai uma estrutura climática real, em vez de artefactos associados à dinâmica temporal. Adicionalmente, a comparação entre algoritmos evidencia que o RIPPER e o M5Rules convergem, de forma independente, para as mesmas famílias de variáveis, reforçando a robustez das variáveis identificadas sob três critérios distintos de otimização.

O Objetivo 2 (Validação Preditiva), contudo, não é alcançado de forma geral. Nenhum dos modelos baseados em regras supera consistentemente a linha de base Naive_Median (mediana simples) ao longo de toda a sequência de avaliação. A melhor variante do CarenR apresenta um desempenho 3% inferior ao da linha de base na RDD (MAE ponderado de 189 versus 184 mhl; teste de Wilcoxon ($p = 0{,}468$) para os 18 cenários, sem diferença estatisticamente significativa) e 22% inferior na RVV (172 versus 140 mhl). Nos nove cenários da RVV em que as regras são ativadas, o CarenR apresenta um desempenho significativamente inferior ao da linha de base (teste de Wilcoxon, $p = 0{,}004$).

Uma análise post hoc da composição temporal das diferentes eras, realizada após a observação dos resultados e, por conseguinte, de natureza exploratória, sugere que o CarenR supera consistentemente a linha de base na RDD quando o conjunto de treino atinge uma representação de pelo menos 29% de observações posteriores a 1990 (5 em 5 cenários consecutivos; teste binomial, $p = 0{,}031$), apontando para a existência de um regime condicional no qual a previsão baseada em regras poderá ser eficaz.

Por fim, esta estrutura de decomposição em duas decisões permite explicar o insucesso preditivo observado: as exigências simultâneas de uma modelação robusta da tendência e da preservação de um sinal residual suficiente para a indução de regras encontram-se numa tensão fundamental quando a remoção da tendência depende exclusivamente de uma covariável temporal.

Palavras-chave: *descoberta de subgrupos distributivos, CarenR, aprendizagem automática baseada em regras, validação walk-forward, avaliação de séries temporais, seleção de atributos, previsão da produção de vinho, não estacionariedade*

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Chapter 1

Introduction

1.1 Motivation

Wine production is among the most climate-sensitive agricultural activities in Europe. Annual yield at a regional scale is governed by a complex interaction of meteorological conditions during the growing season, including temperature, rainfall, soil water balance, and frost events, superimposed on longer-term structural factors including vine age and density, viticultural practices, policy constraints, and market demand. For wine regions such as the Douro Valley and Vinho Verde in northern Portugal, which together produce some of Portugal's most commercially and culturally significant wines, understanding the relationship between climate and production has practical relevance for producers, cooperatives, and regulatory bodies managing annual harvest forecasts, storage planning, and risk assessment.

At the same time, the agroclimatic signal present in annual production data is weak when compared to the year-to-year variability. Total production values at the regional level reflect the decisions of thousands of individual producers, the heterogeneity of vineyards in terms of soil type, altitude, and vine training systems, as well as the effects of localized weather events that are not captured by regional climate indices. This high level of variability has historically limited the predictive performance of statistical models applied to wine production forecasting, motivating the use of interpretable machine learning methods that seek to identify the most robust and relevant climate patterns from the available data.

Modern machine learning has been applied extensively to grape yield and wine production prediction over the past decade. Random Forest, Support Vector Machines, Gradient Boosting, and Deep Learning architectures, including Long Short-Term Memory networks, have all been used to predict yield from meteorological, phenological, and remote sensing data (Santos et al.; Lopes et al., 2024). These methods achieve strong predictive accuracy and handle complex non-linear interactions across large feature sets. Their limitation is interpretability: the internal decision logic is not accessible to viticulturalists, cooperative managers, or climate adaptation planners, and post-hoc explanation tools such as SHAP values approximate feature influence rather than producing explicit, auditable rules. Where the goal is to explain which conditions drive production, not merely to forecast the next value, this opacity is a fundamental constraint. No published work applies interpretable rule-based discovery methods to extract compact, human-readable agroclimatic patterns from long regional wine production records.

Rule-based methods produce human-readable conditional statements: for example, “if soil water at harvest is below X mm and fewer than Y wet days occur after flowering, then production is above the long-run trend by Z thousand hectolitres.” These outputs can be read, questioned, and validated by agronomists, vine growers, and policy analysts without requiring knowledge of the underlying algorithm. Each rule also carries an explicit support estimate, the proportion of historical years in which the condition held, giving a direct measure of how frequently the identified pattern occurred.

1.2 Research Goals and Questions

This thesis pursues two complementary goals applied to historical wine production data from two Portuguese wine regions: the Douro Demarcated Region (RDD, 89 years, 1934–2022) and the Vinho Verde Region (RVV, 80 years, 1942–2021).

Goal 1 (Rule Discovery): discover interpretable agroclimatic rules that explain which meteorological conditions are associated with above and below-trend production years in each region. The primary algorithm for this goal is CarenR (CAREN framework), which identifies subgroups of observations where the distribution of production residuals differs significantly from the global distribution, as measured by a Kolmogorov–Smirnov test.

Goal 2 (Predictive Validation): evaluate whether the discovered rules, and rule-based models more generally, can outperform a naive production forecast in rigorous walk-forward (expanding window) validation. This goal addresses whether the identified climate patterns have sufficient predictive power to improve on the simplest possible forecast: the historical median.

These goals give rise to four specific research questions:

- RQ1: Which agroclimatic variables most consistently distinguish above and below-trend production years in the Douro and Vinho Verde regions?
- RQ2: Do the identified variables reflect genuine within-era climate effects, or are they confounded with the structural production era transitions observed in both regions?
- RQ3: Does any rule-based model outperform the Naive_Median baseline in rigorous walk-forward validation?
- RQ4: If rule-based prediction fails in aggregate, under what conditions, if any, does it succeed, and what structural properties of the data explain those conditions?

1.3 Scope and Delimitations

The thesis focuses exclusively on annual regional production values, measured in thousands of hectolitres (mhl). Sub-regional, vineyard-level, or quality-related outcomes are outside the scope of the study. The feature set is derived from regional climate indices and a vine phenological model;

field measurements of vine physiology are not available and are therefore not considered. The study does not address price forecasting, demand modelling, or the economics of wine production.

The rule-learning algorithms evaluated are CarenR (distributional rules), RIPPER (classification rules), and M5Rules (rule induction with linear regression consequents), together with naive baselines and a Decision Tree regressor as a non-rule comparator. Deep learning, ensemble methods, and other non-interpretable regression approaches are not evaluated; the thesis is specifically motivated by the interpretability requirement that makes rule-based methods preferable in this domain context.

1.4 Thesis Structure

Chapter 2 covers the relevant background: wine production forecasting, agroclimatic modelling, and rule-based machine learning. Chapter 3 describes the data, feature engineering, detrending methodology, and walk-forward validation protocol, and introduces the two-decision decomposition framework used throughout the analysis. Chapter 4 reports the rule discovery results: algorithms compared, dominant agroclimatic features, and LOESS validation. Chapter 5 presents the walk-forward prediction results for both regions, including era-composition analysis and statistical tests. Chapter 6 discusses why rule discovery succeeds whereas prediction at the regional level remains challenging, examines the structural divergence between the two regions, and outlines directions for improvement. Chapter 7 presents conclusions, contributions, limitations, and future work.

Chapter 2

Literature Review

This chapter reviews the foundations of the thesis across three domains: the agroclimatic drivers of wine production in Portuguese wine regions; quantitative approaches to yield prediction, from classical regression to modern machine learning; and the data mining methods (subgroup discovery and distribution rules) that underpin the CarenR framework used in this study. The chapter closes by situating this thesis relative to existing work.

2.1 Regional Wine Production in Portugal

Portugal is home to several wine regions with distinct climatic identities. This thesis focuses on two: the Douro Demarcated Region (RDD) and the Vinho Verde Region (RVV). Together they represented approximately 32% of Portugal's total vineyard area and 37% of national wine production in 2018 (Cunha and Richter, 2020). Despite their geographical proximity in northern Portugal, the two regions have markedly different climates and production histories, making them a natural comparative setting for agroclimatic modelling.

2.1.1 Douro Region

The Douro wine region is located in north-eastern Portugal, one of the oldest demarcated wine regions in the world. Its topography is dominated by steep schist terraces on the hillsides of the Douro valley. The climate is continental and semi-arid: mean growing-season temperatures reach approximately 19.5°C (Köppen Csa), annual precipitation ranges from 400 to 800 mm concentrated outside the growing season, and most vineyards are non-irrigated. Soil water reserves accumulated during the dormant winter period are the primary buffer against summer water stress. Cunha and Richter (2016) demonstrated that soil water in the previous two to three years exerts a positive carryover effect on current-year production, reflecting the slow recharge of deep schist soil profiles. Douro production has increased substantially over the study period, roughly doubling from 770 mhl in the 1930s to 1,400 mhl by the 2010s, partly reflecting the expansion of the Douro DOC appellation for table wines from 1979 onward and the administrative Port production quota system (IVDP benefício, the annual Port production quota).

2.1.2 Vinho Verde Region

The Vinho Verde Region is located in the Minho province of north-western Portugal, the most Atlantic European wine region (Köppen Csb). Mean annual precipitation ranges from 1,200 to 2,000 mm and growing-season temperatures average approximately 17.1°C. In contrast to the Douro, Vinho Verde production declined dramatically over the study period, from approximately 2,300 mhl in the 1960s to around 800 mhl by the 2010s (-65%). This structural decline is associated with Portugal's EU accession in 1986, which introduced vine grubbing-up incentives under the Common Agricultural Policy and quality standards that penalised the region's traditional high-yield pergola systems, alongside rural exodus and the gradual conversion to lower-yield espalier training. Because rainfall is generally sufficient to refill soil water reserves year-round in RVV, carryover effects of water deficits are less prominent than in the Douro (Cunha and Richter, 2020).

2.2 Grapevine Yield Formation

Grapevine yield is the product of five sequential components: vines per hectare, shoots per vine, inflorescences per shoot, berry number per bunch, and berry mass (Zhu et al., 2020). Each component is sensitive to different environmental stimuli at different phenological stages, creating a cascade of climate dependencies that spans two growing seasons. Inflorescence primordia, the embryonic structures that will become grape clusters, are initiated during the spring and summer of the year prior to the harvest year. Whether primordia develop into productive inflorescences or revert to tendrils is governed primarily by temperature and light at the time of differentiation (Zhu et al., 2020): high temperatures encourage inflorescence development, while cool or shaded conditions lead to tendril formation, reducing the following season's potential bunch number. This mechanism means that current-year production is physiologically dependent on prior-year climate, motivating the inclusion of lagged features in the agroclimatic models used in this thesis.

The flowering stage, occurring in late spring of the harvest year, determines the number of berries through fruit-set efficiency. Both low and high temperatures disrupt pollination; wet conditions promote fungi diseases such as Botrytis cinerea and downy mildew, reducing fruit set. The veraison-to-harvest period is governed by temperature and soil water: in the Douro, post-veraison soil water stress typically promotes berry concentration; in Vinho Verde, the wetter Atlantic climate means that canopy management and moisture balance are the dominant ripening-phase constraints.

2.2.1 The Two-Year Vine Development Cycle

A defining property of the grapevine that distinguishes it from annual crops is its perennial nature: physiological processes in one growing season directly determine yield potential in the following season. Two carry-over mechanisms are particularly important for the feature design used in this thesis.

First, inflorescence primordia — the embryonic structures that will become grape clusters — are initiated during the spring and early summer of the prior growing season (year y0). The thermal (tm) conditions at primordia formation, which occurs around the flowering of the prior season, (captured by `tm_fl_y0` and `tm_bb_y0` features, where `tm` = mean temperature, `fl` = flowering stage, `bb` = budburst stage, `y0` = prior year) directly determines how many primordia differentiate into productive inflorescences rather than tendrils, setting an upper bound on current-year bunch number that cannot be altered by current-year climate. Second, the carbohydrate reserves accumulated during the prior year's harvest-to-dormancy period (captured by `swa_hv_y0` and `tn_hv_y0` features) determine the vine's reproductive investment in the current season. Vines that finished the prior season in water or thermal stress have depleted reserves and redirect more assimilates to vegetative recovery, reducing berry development relative to shoot growth.

The practical consequence for modelling is that any dataset covering only the current growing season is fundamentally incomplete for predicting annual wine production. The 15 previous-year features included in the dataset (25% of the 59-feature total) represent the two physiological pathways — primordia formation and carbohydrate reserves — through which the prior season materially constrains current-year yield potential. The multi-year mechanism motivates the inclusion of lagged climate features (`y0` variables) alongside current-year features in the agroclimatic models used in this thesis, and its agronomic relevance is confirmed empirically by the rule discovery results in Chapter 4.

2.3 Meteorological Drivers of Wine Production Variability

For the Douro, Cunha and Richter (2016) demonstrated that spring temperature explains approximately 60–65% of the long-term and short-term behaviour of production, with soil water as the second principal driver. Both current-year and lagged soil water values (up to three years prior) show significant positive correlations with production. Cunha and Richter (2020) further decomposed cyclical production properties using a time-frequency approach, identifying dominant production cycles of approximately 5.7 and 2.5 years linked to spring temperature and soil water oscillations. Climate change projections are of particular concern: Portugal is expected to experience the greatest growing-season temperature increase among major European wine regions (+1.3°C by 2049), with soil water becoming the primary limiting resource in the Douro. For Vinho Verde, the Atlantic climate provides more consistent growing-season moisture; the dominant vulnerability is cool, wet flowering conditions that promote disease and reduce fruit set. Zhu et al. (2020) showed that daily maximum temperature plays a critical role in inflorescence initiation, while both maximum and minimum temperature influence berry number and mass.

2.4 Quantitative Approaches to Wine Yield Prediction

2.4.1 Statistical and Regression Models

The earliest quantitative yield prediction models relied on linear and generalised linear regression applied to seasonal climate aggregates. Santos et al. (2011) developed a multivariate linear regression model for Douro yield (1986–2008) using monthly mean temperatures and precipitation from March to June. Bock et al. (2013) analysed correlations between long-term yield records in Lower Franconia (1805–2010) and mean growing-season temperature. Cunha and Richter (2012) applied an autoregressive model and short-time Fourier transform to identify cyclical properties of production in Douro and Vinho Verde. These models are interpretable and grounded in agronomy but assume linear, stationary relationships and cannot capture threshold effects, interactions, or non-stationarity from structural production changes. A methodological review by Fraga et al. (2024) argued that statistical models for climate impacts in grape and wine research remain underused relative to their potential and should be combined with other approaches.

2.4.2 Machine Learning Approaches

The past decade has seen rapid growth in the application of machine learning to grape yield and wine production prediction. Random Forest, Support Vector Machines, Gradient Boosting ensembles, and Deep Learning architectures have all been applied to predict yield from meteorological, phenological, and remote sensing data. Lopes et al. (2024) used Random Forest models to identify climatic drivers of wine production in Portugal's Côa region, demonstrating competitive predictive performance with feature importance rankings that align with agronomic expectations. Remote sensing approaches use satellite-derived indices (NDVI, EVI) as proxies for vine vigour at the parcel scale, enabling near-real-time sub-regional yield maps.

These methods achieve strong predictive accuracy and handle non-linear interactions well, but they operate as black-box or grey-box models. Their internal decision logic is not accessible to domain experts, and post-hoc tools such as SHAP values approximate feature influence rather than producing explicit, auditable rules. As Molnar (2019) argues, the trade-off between accuracy and interpretability depends on the intended application. For problems where the aim is to explain which conditions drive production, not merely to forecast, this opacity is a fundamental constraint. No published work applies interpretable rule-based discovery methods to extract statistically validated, human-readable agroclimatic patterns from long regional wine production records.

A further limitation shared across virtually all published approaches, including this thesis, is the brevity of the effective evaluation window. The walk-forward protocol used here generates 18 scenarios for RDD and 16 for RVV, covering only 17 test years per region. This is manifestly short for capturing the full interannual variability of wine production, which exhibits dominant production cycles of approximately 5.7 and 2.5 years (Cunha and Richter, 2020). A 17-year test window spans only three complete cycles of the dominant 5.7-year oscillation, meaning that conclusions about predictive performance are necessarily sensitive to which part of the cycle falls in the test set.

This is a known weakness of most modelling work in this domain: the historical records are long enough to support rule discovery but generate very small evaluation samples in a temporally honest walk-forward framework. The statistical results in Chapter 5 should therefore be interpreted with this constraint in mind.

2.5 Subgroup Discovery

Subgroup discovery (SD) is a descriptive induction technique that identifies subsets of a dataset, defined by conjunctions of attribute-value conditions, within which the distribution of a target variable is statistically exceptional relative to the overall population (Herrera et al., 2011). Unlike classification, which aims to correctly assign every observation to a class, SD aims to find the most interesting local patterns: subgroups that are both statistically significant and sufficiently large to be meaningful. The field was formalised by Wrobel (1997) and Klösgen (1996). Lavrac et al. (2004) extended the CN2 rule learner (Clark and Niblett, 1989) to subgroup discovery (CN2-SD); Kavsek and Lavrac (2006) adapted the Apriori algorithm to SD (APRIORI-SD).

A significant recent development is Exceptional Model Mining (EMM), introduced by Duivesteijn et al. (2016) as a generalisation where the interestingness criterion is based on a model fitted to the subgroup rather than a simple distributional shift test. EMM identifies subgroups where the relationship between variables differs significantly from the global relationship. Lemmerich et al. (2022) introduced robust subgroup discovery with statistical reliability guarantees. Zimmermann and Atzmueller (2019) adapted SD to temporal and sequential data, directly addressing the time-series structure of problems such as the one studied in this thesis.

2.6 Distribution Rules and the CarenR Framework

2.6.1 Distribution Rules

Distribution rules (DR) were introduced by Jorge et al. (2006) for discovering and presenting conditional distributions of a numeric target variable with respect to a set of input variables. A distribution rule takes the form IF antecedent THEN distribution, where the antecedent is a conjunction of attribute intervals and the consequent is the empirical distribution of the target among matching observations. Unlike conventional association rules, distribution rules preserve the full distributional shape of the outcome. Statistical significance is assessed using the Kolmogorov–Smirnov (KS) test, which compares the empirical cumulative distribution of the target in the subgroup against the global distribution without assuming normality, well suited to small datasets where normality cannot be assumed.

2.6.2 The CarenR Framework

CarenR (distribution rule discovery framework implemented in R, based on the CAREN association rule engine) is the primary algorithmic contribution of the line of work this thesis extends. It operationalises the distribution rule framework of Jorge et al. (2006) with several additions specifically designed for agroclimatic time series problems.

The CarenR pipeline operates in two modes depending on the goal.

For Goal 1 (Rule Discovery), CarenR is applied to the full historical dataset using the complete set of domain-engineered features derived from the grape phenological cycle. Continuous features are discretised into equal-frequency bins (quartiles by default), ensuring uniform bin occupancy. CarenR then performs a levelwise (Apriori-style) search over conjunctions of feature-bin conditions, pruned by minimum support, evaluating each candidate subgroup using a two-sample KS test against the global distribution. Rules are retained if they pass a significance threshold ($p \leq 0.10$) and a minimum support filter ($\geq 20\%$ of observations). The output is a set of interpretable conditional patterns linking climate conditions to shifts in the distribution of production residuals.

For Goal 2 (Predictive Validation), the same pipeline runs within each training fold rather than on the full dataset, preventing leakage of test-set information. Discretisation, feature selection, and rule induction are all performed on the training fold only. At prediction time, rules that match a test year's conditions have their subgroup medians combined as a support-weighted average — each rule's median residual weighted by its support fraction, with weights normalised to sum to one; if no rule fires, the model falls back to the training median. The 11 CarenR variants evaluated in this thesis differ from one another in the feature selection step applied within each fold.

CarenR differs from RIPPER and M5Rules in what it optimises: it asks whether the distribution of production residuals shifts significantly within a subgroup, rather than assigning class labels or minimising squared error. A feature that reliably shifts the distribution mean, even when the distribution remains wide and overlapping with the global one, will be detected by CarenR but not by classification or regression criteria. This is why wet days post-flowering, a feature that reliably shifts the production distribution in the Douro rules but does not form a clean class boundary, is selected by CarenR yet missed by RIPPER and M5Rules (Section 4.5).

CarenR has been applied in agricultural settings in previous work by the research group at the University of Porto (Moreira et al., 2022), primarily in the context of viticulture rule discovery. The present thesis represents its first systematic evaluation in a walk-forward predictive validation framework across two contrasting wine regions, with explicit statistical testing and era-composition analysis.

2.7 Positioning and Research Gap

Table 2.1 situates this thesis within the broader landscape of literature reviewed in this chapter. Each stream is scored on whether it produces directly interpretable models, whether it handles a continuous target variable, and whether it has been applied to long multi-decadal series of the kind studied here.

Table 2.1: Literature map positioning this thesis relative to existing research streams.

Research stream	Primary goal	Interpretable?	Continuous target?	Long-term data?	Representative references
Statistical regression	Prediction	Yes	Yes	Yes	Cunha & Richter 2016/2020; Santos et al. 2011
Modern ML (RF, LSTM, SVM)	Prediction	Limited	Yes	Partial	Lopes et al. 2024; Santos et al.
Explainable AI / SHAP	Post-hoc explanation	Partial	Yes	No	Molnar 2019
Subgroup discovery	Pattern discovery	Yes	Partial	No	Lavrac et al. 2004; Herrera et al. 2011
Exceptional model mining	Pattern discovery	Yes	Yes	Partial	Duivesteijn et al. 2016; Zimmermann 2019
Distribution rules (CarenR)	Discovery + prediction	Yes	Yes	Yes	Jorge et al. 2006 — **this thesis**

No existing stream combines all five properties simultaneously. This thesis applies CarenR to two Portuguese wine regions spanning 80–89 years, contributing (1) the first systematic walk-forward evaluation of distributional subgroup discovery for wine production forecasting with phenology-informed feature engineering, (2) a two-decision decomposition framework that diagnoses when and why rule-based prediction succeeds or fails, and (3) a cross-regional comparison that reveals how structural production history interacts with climate signal availability.

2.8 Portuguese Viticulture Industry Structure and Policy

The structural production trends observed in the two study regions, a persistent increase in the Douro and a sharp decline in Vinho Verde, cannot be attributed to climate alone. Both trajectories were substantially shaped by regulatory, economic, and demographic forces that are independent of meteorological conditions. Understanding these structural drivers is essential for interpreting the era confound identified throughout this thesis and for correctly attributing the limitations of the predictive models.

2.8.1 Douro Region: Regulatory and Market Drivers

The Douro wine region has been formally demarcated since 1756, making it one of the first wine appellations in the world. The most significant modern regulatory influence on production is the Port wine benefício system, administered by the Instituto dos Vinhos do Douro e do Porto (IVDP). The benefício is an annual administrative quota that determines how much of the Douro's grape harvest may be used to produce Port wine, which commands a premium price over unfortified Douro wines. The quota is set annually by the IVDP based on market conditions, stock levels, and quality assessments, and has varied substantially across decades. Because Port production represents a significant fraction of total Douro output, the benefício quota partially decouples observed production from pure climate effects. In years where Port quota is restricted, total production is administratively capped regardless of how favourable the climate was. This administrative component is not captured by any of the meteorological features in the dataset and represents an unobserved structural covariate.

The second major structural driver of RDD production growth was the creation of the Douro DOC (Denominação de Origem Controlada, Controlled Designation of Origin) for unfortified table wines in 1979. Before this designation, Douro grapes were used almost exclusively for Port wine. The Douro DOC enabled Port wine houses and independent producers to bottle and market premium dry red and white table wines from the region, opening new market channels. From the 1980s onward, Douro table wine production grew rapidly alongside Port, contributing to the gradual increase in total regional production documented in Appendix B. This structural expansion is largely responsible for the upward trend in the RDD production series and is the primary reason the RDD time series is less confounded by non-climate structural changes than the RVV series.

2.8.2 Vinho Verde Region: EU Policy and Structural Transformation

The Vinho Verde region's dramatic production decline from the 1960s to around 800 mhl by the 2010s is one of the most striking structural transformations in Portuguese viticulture. Three interacting factors explain the collapse from approximately 2,300 mhl in the 1960s to around 800 mhl today.

Portugal's accession to the European Economic Community (now European Union) in January 1986 triggered a sequence of structural adjustments that profoundly reshaped the RVV. The Common Agricultural Policy (CAP) introduced grubbing-up subsidies, payments to farmers who permanently removed vineyard plantings, as part of an effort to reduce wine surpluses across member states. The Vinho Verde region, dominated by small family-operated plots producing bulk wine for local and low-value export markets, was particularly exposed to these incentives. Thousands of small producers received grubbing-up payments and removed vineyards that were marginally profitable or required high labour input, accelerating a process of plot abandonment that had already begun with rural exodus in the 1960s and 1970s.

The EU quality standards introduced post-accession also penalised the traditional high-yield pergola (ramada) training system used across the Minho. Pergola systems, which train vines on high horizontal trellises above ground level, maximise vegetative growth and can produce 15,000–20,000 kg/ha, far above the yield limits imposed by EU quality regulations for premium appellations. The incentive to qualify for premium designations drove a progressive conversion from pergola to lower-yield espalier (cordon) training systems, which produce 6,000–10,000 kg/ha. This training system transition reduced per-hectare yields structurally, independently of climate, and is estimated to have contributed substantially to the post-1986 production decline.

The third factor was demographic. The Minho region had experienced significant rural exodus from the 1960s onward as workers migrated to urban centres and abroad. The generation that maintained the region's characteristic small, labour-intensive plots retired or emigrated, and their children often did not continue the viticultural tradition. Plot fragmentation — the division of land into numerous small, scattered parcels across multiple owners — has historically been among the most pronounced in Portugal in the Minho region, with individual parcels often below 0.1 hectares, making consolidation difficult and increased the per-unit cost of cultivation. Many marginal plots were simply abandoned, reducing the effective cultivated area and contributing to the production decline independently of the EU policy shock.

These structural factors directly explain why the era confound is substantially stronger in RVV than in RDD: the production decline in Vinho Verde is predominantly driven by non-climate forces, creating a strong co-variation between era membership and climate variables that makes it difficult to disentangle genuine inter-annual climate effects from structural level changes. The models in this thesis acknowledge and partially address this confound through linear detrending and the LOESS sensitivity analysis, but fully removing it would require incorporating vineyard area, number of certified producers, and training system composition as explicit detrending covariates.

2.9 Time Series Properties and Evaluation Methodology

Annual wine production data is a time series with properties that have important implications for model evaluation. This section reviews the relevant concepts and motivates the methodological choices made in Chapter 3.

2.9.1 Non-Stationarity and Structural Breaks

A time series is stationary if its mean, variance, and autocovariance structure do not change over time. Both regional production series are non-stationary in the strict sense. The RDD series has a positive mean trend, and the RVV series has a strongly negative trend. More subtly, both series exhibit structural breaks, abrupt changes in the level or trend that cannot be attributed to normal year-to-year variability. In the RVV series, the most prominent structural break occurs in the mid-1980s, coinciding with EU accession and the subsequent grubbing-up subsidies. In the RDD series, no sharp structural break is evident in the decade-level averages (Appendix B, Table B.2); instead, production follows a gradual upward trend from the 1950s onward, consistent with incremental regulatory and market expansion rather than a single discrete transition.

Non-stationarity poses a direct challenge for machine learning models: a model trained on data from one period of a non-stationary series may not generalise to data from a different period. The standard practice in such settings is to remove the deterministic trend component before modelling, leaving stationary residuals that represent genuine inter-annual variation around the structural trajectory. This is precisely the role of the detrending step described in Section 3.3. The appropriateness of linear vs. flexible detrending depends on the shape of the structural trend, a question examined empirically through the LOESS sensitivity analysis in Section 5.5.

2.9.2 Why Random Cross-Validation Is Invalid for Time Series

Random k-fold cross-validation violates the temporal ordering of observations and conflates training and test eras, inflating apparent model performance on non-stationary series. Walk-forward (expanding-window) evaluation, described in Section 3.5, is therefore used throughout this thesis.

2.9.3 The Naive Baseline and What It Measures

A central benchmark throughout this thesis is the Naive_Median baseline, the model that always predicts the training-set median residual (approximately zero after detrending, i.e. "on trend"). This baseline has an important theoretical property: the median minimises mean absolute error among all constant predictors. Therefore, no model that predicts a constant for every year can beat the Naive_Median on MAE, and any model that beats it must be doing so by identifying non-trivial structure in the data.

The Naive_Median MAE can be interpreted as the minimum error achievable by any model that ignores the climate features entirely — it represents the "cost of not knowing the climate" in a given scenario. If the MAE from a climate-aware model is lower than the Naive_Median, the model has successfully extracted information from the climate features that reduces prediction error. If it is higher, the model has actively misused the climate information, adding noise rather than signal.

The fact that no model consistently beats the Naive_Median in this study therefore means not that climate is irrelevant to wine production (the CarenR rules demonstrate it is not) but that the climate signal is insufficient in magnitude and consistency to overcome year-to-year production noise in a held-out forecasting evaluation. This distinction between "the signal exists" and "the signal is strong enough to beat the baseline" is the central finding of Chapter 5 and motivates the two-decision decomposition framework described in Section 3.3.

Chapter 3

Data and Methodology

This chapter describes the datasets, feature engineering pipeline, detrending methodology, rule-learning algorithms, walk-forward validation protocol, baseline models, and evaluation metrics used throughout the thesis. Section 3.4 provides a detailed technical description of all three rule-learning algorithms, including their internal mechanics, how they differ from one another, and why those differences matter for interpreting the results in Chapters 4 and 5.

3.1 Study Regions and Data

This study examines two geographically and climatically distinct Portuguese wine regions: the Douro Demarcated Region (RDD) and the Vinho Verde Region (RVV). The RDD dataset covers 89 annual observations from 1934 to 2022. The RVV dataset covers 80 annual observations from 1942 to 2021. Both datasets consist of one observation per year, where each observation contains a set of meteorological and phenological covariates together with annual wine production in thousands of hectolitres (mhl) as the response variable. Table 3.1 summarises the key characteristics of both datasets.

Table 3.1: Dataset characteristics for the Douro (RDD) and Vinho Verde (RVV) study regions.

Characteristic	RDD (Douro)	RVV (Vinho Verde)
Period	1934–2022	1942–2021
Observations (n)	89 years	80 years
Climate type	Continental / semi-arid	Temperate Atlantic
Production trend	+80% increase (1930s–2010s)	-65% decline (1960s–2020s)
Walk-forward scenarios	18	16
Response variable	Annual wine production (mhl)	Annual wine production (mhl)

Because both series are non-stationary, random cross-validation is not appropriate; a walk-forward expanding-window protocol is used instead, as described in Section 3.5.

Figures 3.1 and 3.2 show the annual production time series for the two regions, illustrating the opposing structural trends summarised in Table 3.1.

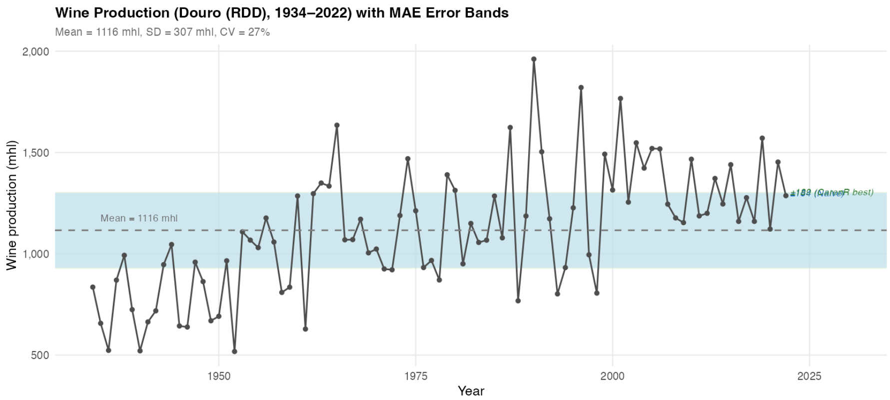

Figure 3.1: Production time series for the Douro (RDD) region, 1934–2022.

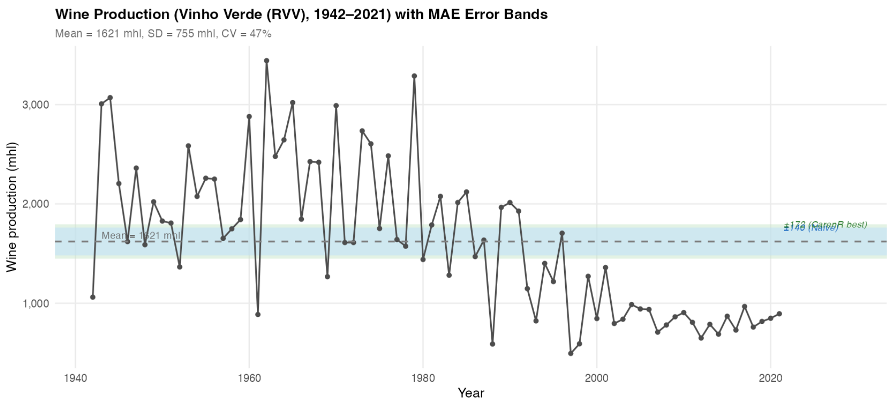

Figure 3.2: Production time series for the Vinho Verde (RVV) region, 1942–2021.

3.2 Feature Engineering

Covariates were constructed from daily temperature, rainfall, and soil water balance records, supplemented by Leaf Area Index (LAI) estimates from a vine phenological model. All variables were aggregated over windows defined by phenological stage and calendar date, producing features that correspond to agronomically meaningful periods of vine development. The feature naming convention encodes the variable type, phenological stage, year offset (current or previous year), and window width; for example, swa_hv_y1_b20 denotes soil water availability (swa) in the 20 days before (b20) harvest (hv) in the current year (y1).

Feature families included: mean and minimum temperature at each phenological window; total rainfall and number of wet days (days with rainfall > 1 mm); soil water availability (mm) at budburst, flowering, veraison, and harvest; accumulated heat degree-days above 35°C and cold days below 15°C; and LAI at budburst and flowering. Lagged versions of selected features (previous growing season, denoted y0) were included to capture carryover effects from the prior year. The final feature matrix contained 59 candidate predictors per region.

3.2.1 Autocorrelation Structure of the Residuals

Before designing the previous-year (y0) features, the autocorrelation structure of the linearly detrended production residuals was examined to determine which lags carry statistically significant serial dependence. This analysis was performed on the target variable only — not on the climate features. It served two purposes: (1) providing empirical confirmation that lag-1 production autocorrelation is negligible, supporting the use of climate features rather than lagged production values as predictors; and (2) revealing longer-period cyclical structure that motivated the `CarenR_Dist_Lag` variant (Section 5.10).

The ACF and PACF of the linearly detrended residuals (up to lag 12 years, 95% confidence bands $\pm 1.96/\sqrt{n}$) show two key findings. **Lag 1 is not significant** in either region (RDD: ACF $= +0.08$; RVV: ACF $= +0.19$, both below the confidence threshold). This means that knowing last year's production does not reliably predict the current year's deviation from trend, directly explaining why the Naive "last value" baseline underperforms Naive_Median (Table 5.1: Naive 199 mhl vs. Naive_Median 184 mhl for RDD). **Lags 5–6 are significant in both regions** (ACF at lag 5: $+0.22$ for RDD, $+0.30$ for RVV), consistent with the $\sim$5.7-year dominant production cycle identified by Cunha and Richter (2020). The `CarenR_Dist_Lag` variant was designed to exploit this cycle by augmenting the feature set with ACF-selected lagged residuals; it ranked 5th in RDD and 8th in RVV, suggesting the cycle is statistically present but does not translate into a reliably exploitable predictive signal relative to the domain-knowledge-based y0 features.

3.3 Detrending and the Two-Decision Decomposition

All models operate on linearly detrended residuals $r_t = y_t - (\alpha + \beta t)$, where y_t is observed production and $\alpha + \beta t$ is the OLS linear trend estimated on the training set. Predictions are converted back to the original scale by adding the projected trend value: $\hat{y}_t = r_t + (\alpha + \beta t)$. All preprocessing steps are performed exclusively on the training fold within each scenario to prevent data leakage.

Decision 1 — Trend modelling: how accurately does the training-period trend extrapolate to test years? Measured by Naive_Median MAE.

Decision 2 — Climate signal extraction: can a rule-based model predict detrended residuals better than zero? Measured by the gap between a model's weighted MAE and the Naive_Median weighted MAE.

3.4 Rule-Learning Algorithms

Three rule-learning algorithms were evaluated, representing three fundamentally different approaches to extracting interpretable patterns from the same data. Understanding how each algorithm works internally, and specifically what question it is designed to answer, is essential for correctly interpreting the results in Chapter 4 and the comparison in Section 4.5. Table 3.2 summarises the key characteristics of all three algorithms before each is described in detail. The two bottom rows of the table describe not intrinsic properties of the algorithms but how each is operationalised for numeric forecasting within the prediction framework defined in this thesis.

Table 3.2: Comparison of the three rule-learning algorithms: CarenR, RIPPER, and M5Rules.

Property	**CarenR**	**RIPPER**	**M5Rules**
Task type	Distributional subgroup discovery	Classification rule learning	Piecewise linear regression
Target variable	Continuous (detrended residuals)	Discretised classes (Low / Med / High)	Continuous (detrended residuals)
Core question answered	When does the shape of the production distribution shift significantly?	Which climate class does a year belong to?	What exact production level does a year have?
Statistical criterion	Kolmogorov–Smirnov two-sample test ($p \leq 0.10$)	Information gain / Gini (FOIL gain for rule growing)	Standard deviation reduction (MSE minimisation)
Rule output	IF conditions THEN distribution shifts up / down	IF conditions THEN class = Low / Med / High	IF conditions THEN production = linear model(features)
Feature input	Discretised equal-frequency bins	Continuous (RIPPER finds own thresholds)	Continuous (M5Rules finds own thresholds)
Interpretability	Very high — simple IF/THEN with distributional evidence	High — ordered IF/THEN classification	Medium — IF/THEN rules with linear models at leaves
Operationalisation for prediction in this thesis (Section 3.8.1)			
Prediction output	Mean/median of matching subgroup	Class midpoint or median	Linear model evaluated at feature values
Fallback (no rules fire)	Training median residual	Default rule (most common class)	Global linear model (no conditions)

3.4.1 CarenR: Distribution Rules via the CAREN Framework

CarenR is built on the CAREN association rule engine, originally developed by Azevedo (2003) and extended to the discovery of distribution rules in numeric data by Jorge et al. (2006). It is the primary algorithm of this thesis and the only one of the three that is specifically designed for continuous response variables without requiring pre-discretisation of the target. This section describes its mechanics in detail.

Conceptual foundation. Most rule-learning algorithms ask: “what value does this observation have?” CarenR asks a different question: “is there a subgroup of observations, defined by a conjunction of input feature conditions, where the statistical distribution of the target variable is significantly different from the distribution across the full dataset?” A CarenR rule does not say “this year will produce X mhl.” It says “in years satisfying these climate conditions, the production distribution is shifted upward (or downward) in a statistically significant way, the entire distributional shape is different — not just the mean.” This distinction between point prediction and distributional shift is fundamental to understanding both the strengths and the limitations of CarenR.

Step 1: Feature discretisation. CarenR requires discrete (ordinal) input features. Each continuous climate feature is independently discretised into four equal-frequency bins (quartiles). Equal-frequency binning ensures that each bin contains approximately the same number of observations, avoiding sparse bins at extreme values that would make the KS test unreliable. For example, the feature swa_hv_y1_b20 (soil water 20 days before harvest) might be discretised into bins [20–74 mm], [74–101 mm], [101–152 mm], [152–350 mm], each containing approximately 22 of the 89 RDD observations. For Goal 1 (Rule Discovery), bin boundaries are computed on the full historical dataset. For Goal 2 (Predictive Validation), bin boundaries are computed exclusively on the training fold within each scenario, preventing any leakage of test-set information into the discretisation step.

Step 2: Feature selection (Goal 2 only). This step applies only to Goal 2 (Predictive Validation). For Goal 1 (Rule Discovery), CarenR operates on all 59 domain-engineered features without further algorithmic filtering — the feature set is defined entirely by viticulture domain knowledge. For Goal 2, with 59 candidate features each discretised into 4 bins, the full itemset search space is combinatorially large, and a feature selection step reduces the feature set before rule induction. This thesis evaluates several selection strategies (described in Table 3.4) as distinct CarenR variants. Three selection strategies are evaluated, each reflecting a different way of measuring association between a discrete feature and a continuous target. η^2-based selection (eta-squared) is the most statistically appropriate: it applies a one-way ANOVA across the four bins of each feature and measures how much of the variance in the continuous production residual is explained by the discrete groupings — directly answering whether the bins produce meaningfully different production distributions. Pearson correlation ($|r|$ above the median across all features) is a simpler approximation that treats the bin numbers as equally spaced ordinal values; it is less rigorous but computationally efficient and tends to identify the same features in practice. Support-filtered Pearson selection adds a coverage requirement: each bin of a retained feature must contain at least a minimum fraction of training observations, preventing selection of features with extreme or sparse bin distributions. The convergence of all three strategies on the same variable families (confirmed in Section 4.5) is evidence that the feature rankings are robust to the choice of selection criterion. All feature selection is performed exclusively on training data within each fold.

Step 3: Rule induction via subgroup search. CarenR searches for conjunctions of feature-bin conditions (itemsets) that define subgroups with distributional shift in the target variable. The search proceeds levelwise in the Apriori style: single-condition rules are evaluated first, then two-condition conjunctions are formed from conditions that meet the minimum-support threshold, and so on up to a maximum rule length (typically 3–4 conditions). Candidate subgroups falling below the support threshold are pruned before the KS test is applied, so the search is exhaustive over the support-frequent region of the condition lattice rather than a heuristic beam. For each candidate subgroup, the observations satisfying all conditions are identified — for Goal 1 this is the full historical dataset; for Goal 2 it is the training fold — and a two-sample Kolmogorov–Smirnov (KS) test is performed comparing the target distribution in the subgroup against the global distribution.

Step 4: The Kolmogorov–Smirnov test. The KS test is a non-parametric statistical test that compares two empirical cumulative distribution functions (ECDFs). Given a subgroup S of n_S observations and the global reference set of n_T observations (the full dataset for Goal 1, the training fold for Goal 2), the KS statistic D is the maximum absolute difference between the two ECDFs:

$$D = \max_x |F_S(x) - F_T(x)| \tag{3.1}$$

where $F_S(x)$ is the empirical CDF of the target in the subgroup and $F_T(x)$ is the empirical CDF in the global set. A large D indicates that the two distributions are far apart — the subgroup has a systematically different pattern of production values. The p-value associated with D measures the probability of observing a difference of this magnitude by chance if the two samples were drawn from the same distribution. A rule is retained only if $p \leq 0.10$ (the significance threshold used in this study). The KS test makes no assumption about the shape of either distribution, which is important given the small sample sizes ($n = 80$–89) and the non-normal distribution of production residuals.

Step 5: Support filter. A statistically significant rule that covers only two or three observations (very low support) is not useful: it describes a pattern too rare to be actionable and is likely to be a chance finding. CarenR therefore applies a minimum support filter: a rule is retained only if it covers a minimum fraction of the reference observations. The threshold value is not prescribed by the algorithm but is an experimental design decision of this thesis: it was set to 20% (Table B.8) — approximately 18 years for an 89-year dataset in Goal 1, or the equivalent share of the training fold in Goal 2. This filter ensures that reported rules represent patterns that recur across a meaningful portion of the historical record rather than isolated individual years, improving their robustness and practical interpretability.

Step 6: Rule output and prediction. Each retained rule describes a climate subgroup as a conjunction of feature-bin conditions, along with the median residual of the target variable within the subgroup and the KS p-value. An example RDD rule output reads: "IF wet days post-flowering $\in$ [0, 3.33] AND wet days post-maturity $\in$ [0, 3.46] THEN production is +121 mhl above trend on average (support 28%, p = 0.0043)." For Goal 1, the retained rules are the direct output — each rule represents an interpretable agroclimatic pattern. For Goal 2 (Predictive Validation), the feature values of a test year are checked against all retained rules. If one or more rules fire, their subgroup medians are combined as a support-weighted average — each rule's median residual weighted by its support fraction, with weights normalised to sum to one. If no rule fires, the model falls back to the training median residual, producing the same prediction as the Naive_Median baseline. This fallback means CarenR never predicts worse than the naive baseline unless it fires a rule, which makes the walk-forward evaluation in Chapter 5 a direct test of whether the fired rules add value beyond the median.

What makes CarenR unique. Figure 3.3 illustrates the full CarenR pipeline including the thesis-level design choices. Three properties distinguish CarenR from standard rule learners. First, the KS-test criterion detects distributional shift rather than class membership or squared error reduction. This allows CarenR to identify features whose relationship with production is probabilistic and distributional, for example, LAI at harvest in Vinho Verde (Section 4.6.2), which reliably shifts the distribution of production residuals without reliably predicting any single year's exact value. Second, rules are verified against a global null distribution rather than constructed greedily to separate pre-defined classes, making them statistically grounded in a way that classification rules are not. Third, the prediction framework developed in this thesis incorporates an explicit fallback mechanism: when no rule matches a test year's conditions, the prediction defaults to the training median rather than forcing an uninformed estimate. This design choice makes the overall approach conservative — it only departs from the naive baseline when at least one rule fires, ensuring that CarenR's predictions can only add noise if they are wrong, never if they abstain.

3.4.2 RIPPER: Repeated Incremental Pruning to Produce Error Reduction

RIPPER, implemented as JRip in the Weka machine learning toolkit (Witten et al., 2011), is a propositional rule learning algorithm for classification. It was introduced by Cohen (1995) as an efficient, scalable alternative to earlier rule learners such as CN2, and remains one of the most widely used interpretable classification algorithms. In this thesis RIPPER serves as an interpretable classification benchmark, allowing the rules found by CarenR to be compared against a well-established rule-based classification baseline.

Target discretisation. RIPPER is a classifier and requires a discrete target variable. The continuous production residuals are therefore discretised into three ordinal classes — Low, Medium, and High — using tertile boundaries. For Goal 1 (Rule Discovery), boundaries are estimated on the full historical dataset. For Goal 2 (Predictive Validation), boundaries are estimated within each training fold to prevent leakage. Each class contains approximately one third of the observations. This discretisation step introduces a loss of information relative to CarenR and M5Rules, which both operate on the continuous response.

Rule growing with FOIL gain. RIPPER constructs rules one class at a time, starting with the minority class and working towards the majority class. For each class, it grows a rule by greedily adding conditions that maximise the FOIL information gain, a measure that rewards conditions that increase the ratio of positive (target class) to negative (other class) examples covered by the rule. Starting from an empty condition set, RIPPER adds one condition at a time until no further gain is possible or a stopping criterion is met. The grown rule is typically a conjunction of 2–5 conditions that together define a subregion of feature space highly concentrated in the target class.

For a rule R and candidate extension R' (adding one condition), the FOIL gain is:

$$\text{FOIL-gain}(R,R') = p' \left(\log_2 \frac{p'}{p'+n'} - \log_2 \frac{p}{p+n} \right) \tag{3.2}$$

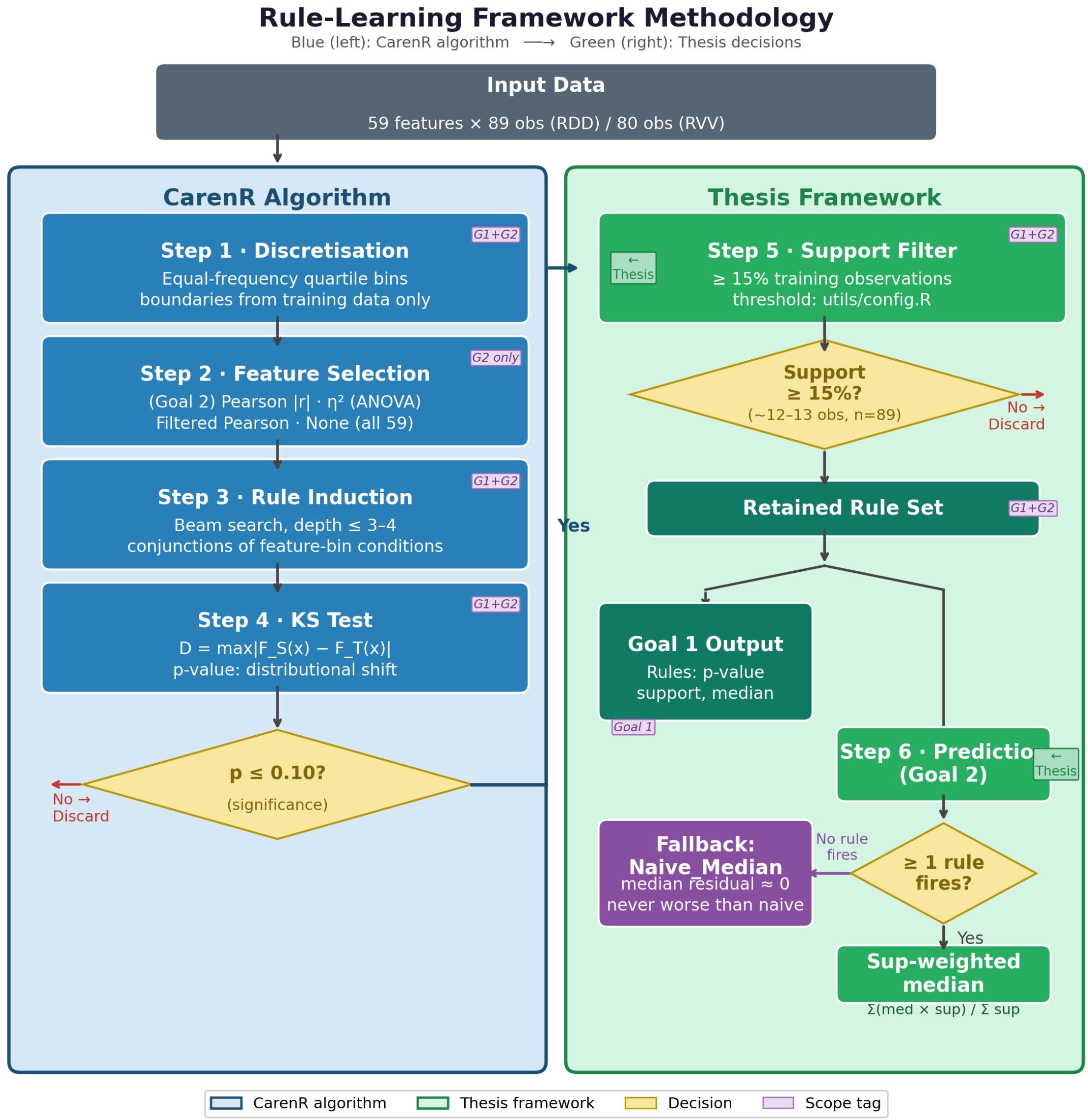

Figure 3.3: Rule-learning framework methodology: CarenR algorithm steps (blue, left) and thesis framework decisions (green, right). The significance threshold, support filter, prediction logic, and fallback are thesis design choices, not prescribed by the CarenR algorithm.

where p and n are the numbers of positive and negative examples covered by R, and p', n' are the corresponding counts for the extended rule R'. RIPPER selects the condition $c^* = \arg\max_c \text{FOIL-gain}(R, R \cup \{c\})$ at each growing step.

Rule pruning with MDL. Overfitting is controlled by a pruning step that removes conditions from the grown rule using the Minimum Description Length (MDL) principle: conditions whose removal reduces the total description length of the rule set plus the data are deleted. This balances rule complexity against coverage, a rule that is slightly less precise but covers more observations is preferred over a highly specific rule covering very few. After pruning, RIPPER checks whether replacing or revising a rule reduces the total error, iterating until convergence. Note that RIPPER's internal pruning mechanism uses an internal hold-out fraction of the training data (not random cross-validation across the full dataset), so it does not violate the temporal ordering enforced by the walk-forward protocol: the internal hold-out is drawn from the same training fold already bounded by the current scenario's cut-point.

Ordered rule list and default rule. The final output of RIPPER is an ordered rule list: rules are applied in the order they were learned, and the first rule whose conditions are satisfied assigns the class for a given observation. Any observation not covered by any explicit rule is assigned the default class, the most frequent class in the training data. This ordered, default-inclusive structure ensures complete coverage: every observation receives a class label. In this thesis, the default rule covered the majority of observations (all years not matching explicit climate conditions), limiting RIPPER's ability to characterise positive (above-trend) production years in the Douro.

Why RIPPER struggles with small datasets. RIPPER was designed for large databases where each class is represented by thousands of examples. With n = 80–89 observations split into three classes of approximately 27–30 each, the per-class sample size is very small relative to the dimensionality of the feature space. FOIL gain estimates are unreliable on such small samples, and the MDL pruning criterion may be too aggressive or too permissive. The result is either overly specific rules (high precision, very low support) or a near-trivial rule list (one or two rules plus a default) that captures only the most obvious patterns. Cross-validated accuracy of 35–42%, only slightly above the 33% random baseline, reflects this limitation directly.

3.4.3 M5Rules: Rule Induction with Linear Regression Consequents

M5Rules, also implemented in Weka, is a piecewise linear regression rule learner derived from the M5 model tree algorithm (Quinlan, 1992; Wang and Witten, 1997). Instead of predicting a class label or a distributional shift, M5Rules predicts an exact numerical value for each observation by applying a linear regression model whose coefficients depend on which region of feature space the observation falls in. This makes M5Rules the only algorithm in this study that produces a direct point estimate of the production residual, making it directly comparable to the naive baseline and to CarenR's subgroup-median prediction.

Model tree construction. M5Rules begins by constructing a regression model tree using a top-down recursive partitioning procedure. At each node of the tree, M5Rules selects the feature and split threshold that maximise the reduction in standard deviation of the target variable across the two resulting child nodes, analogous to the variance reduction criterion used by CART, but applied to a regression setting. This process continues recursively until each leaf contains a minimum number of observations (default: 4 in Weka). The result is a binary tree that partitions the feature space into disjoint regions, with a multivariate linear regression model estimated at each leaf using the observations that reach it.

The standard deviation reduction (SDR) for a split of training set T into subsets T_L and T_R is:

$$\mathrm{SDR}(T, T_L, T_R) = \mathrm{sd}(T) - \frac{|T_L|}{|T|}\,\mathrm{sd}(T_L) - \frac{|T_R|}{|T|}\,\mathrm{sd}(T_R) \tag{3.3}$$

where $\mathrm{sd}(\cdot)$ denotes the sample standard deviation of the response variable in the corresponding set.

Leaf model estimation and smoothing. Within each leaf, a linear regression model is fitted to the training observations in that leaf using ordinary least squares. M5Rules applies a smoothing procedure that interpolates between the leaf model and progressively coarser ancestor models as the observation moves up the tree, reducing the discontinuities at split boundaries that would otherwise produce implausible predictions near the edges of each leaf region. The smoothed prediction at leaf depth d is a weighted combination of the leaf model and each ancestor model's prediction.

Pruning and conversion to rules. After tree construction, M5Rules prunes branches whose removal does not increase cross-validation error beyond a tolerance threshold. The pruned tree is then converted into an ordered rule list: each path from the root to a leaf becomes one rule, with the conditions along the path as the antecedent and the leaf's linear model as the consequent. Rules are applied in order; the first matching rule is used, with the final rule having no conditions (applies to all observations) as a universal default. This conversion gives M5Rules the same human-readable rule format as RIPPER but with a linear model at each leaf rather than a class label.

Why M5Rules collapses to a global model on small datasets. M5Rules' standard deviation reduction criterion requires that each split create two child nodes, each with sufficient observations for reliable linear regression. With a minimum leaf size of 4 and n = 80–89 training observations, the algorithm cannot split the data more than a few times before individual leaves become too small for stable regression. In practice, M5Rules found no beneficial split in either region on the full dataset, collapsing to a single global linear model: one rule with no conditions that applies to all observations. This global model fits the training data with $R^2 \approx 0.40$–0.53 but generalises poorly, with cross-validated R^2 falling to 0.004 (RDD) and -0.07 (RVV). The negative CV R^2 for RVV confirms that the model is worse than simply predicting the global mean; it has overfit the training set despite using a single linear equation.

Comparison with CarenR and RIPPER. The three algorithms differ fundamentally in what they optimise and what they produce. CarenR searches for distributional anomalies using a non-parametric test; it will find a feature informative if it shifts the distribution of production residuals in a statistically reliable way, even if that shift is not captured by a linear coefficient or a class boundary. RIPPER searches for class-separating conditions using information gain; it will find a feature informative if it concentrates one production class, regardless of the distribution of other classes. M5Rules searches for variance-reducing splits using squared error; it will find a feature informative if it enables a more accurate linear model, penalising large deviations quadratically. A feature that shifts the mean of the distribution of production residuals (detected by CarenR) but does not cleanly separate the three classes (not detected by RIPPER) and adds noise rather than linearity to the regression (not detected by M5Rules) will appear in CarenR's output but not in the others. This is precisely the situation for LAI at harvest in Vinho Verde, as discussed in Section 4.6.2.

3.5 Walk-Forward Validation Protocol

Model performance for Goal 2 was assessed using a walk-forward expanding-window validation, illustrated in Figure 3.4. The earliest 80% of each regional time series defines the minimum training window. At each subsequent step (scenario), the training window is extended by one year and the model is re-fitted from scratch on all available training data; the next year is then predicted and its prediction error recorded. This generates 18 evaluation scenarios for RDD and 16 for RVV. All preprocessing steps — linear trend estimation, feature discretisation, and feature selection — are performed exclusively on the training fold within each scenario to prevent data leakage.

3.6 Baseline Models

Naive_Median: predicts the median of the training-set residuals for every test year, equivalent to predicting that every test year will be exactly on trend. Minimises MAE among all constant predictors.

Naive (last value): predicts the most recently observed production value, corresponding to a random-walk assumption.

Naive_MA (moving average): predicts the mean of the three most recent training observations in original units.

Decision Tree (CART): a CART regressor included as a non-rule-based ML comparator. Unlike the rule-learning algorithms, CART partitions the feature space using recursive binary splits that minimise squared error at each node, producing a tree structure rather than an explicit IF-THEN rule list. It was included to test whether the interpretability constraint comes at a cost in predictive performance relative to a standard ML baseline. The tree was fitted with default Weka hyperparameters within each training fold, subject to the same walk-forward protocol as all other models.

The 17 models evaluated in the walk-forward experiment span three categories: four baselines and comparators (Naive_Median, Naive, Naive_MA, and Decision Tree), two standard rule learners (RIPPER and M5Rules), and 11 CarenR variants. The baselines serve as benchmarks against which rule-learning algorithms are assessed: Naive_Median is the primary reference (it minimises MAE under a constant prediction), while Naive and Naive_MA test whether any model adds value over simpler alternatives that ignore climate information entirely. Decision Tree is included as a non-rule-based ML comparator to isolate the contribution of rule structure. The 11 CarenR variants all use the same distributional subgroup discovery algorithm but differ in how features are pre-selected within each training fold, enabling a systematic sensitivity analysis of this design choice. Table 3.3 lists all 17 models; Table 3.4 details the feature selection strategy distinguishing each of the 11 CarenR variants.

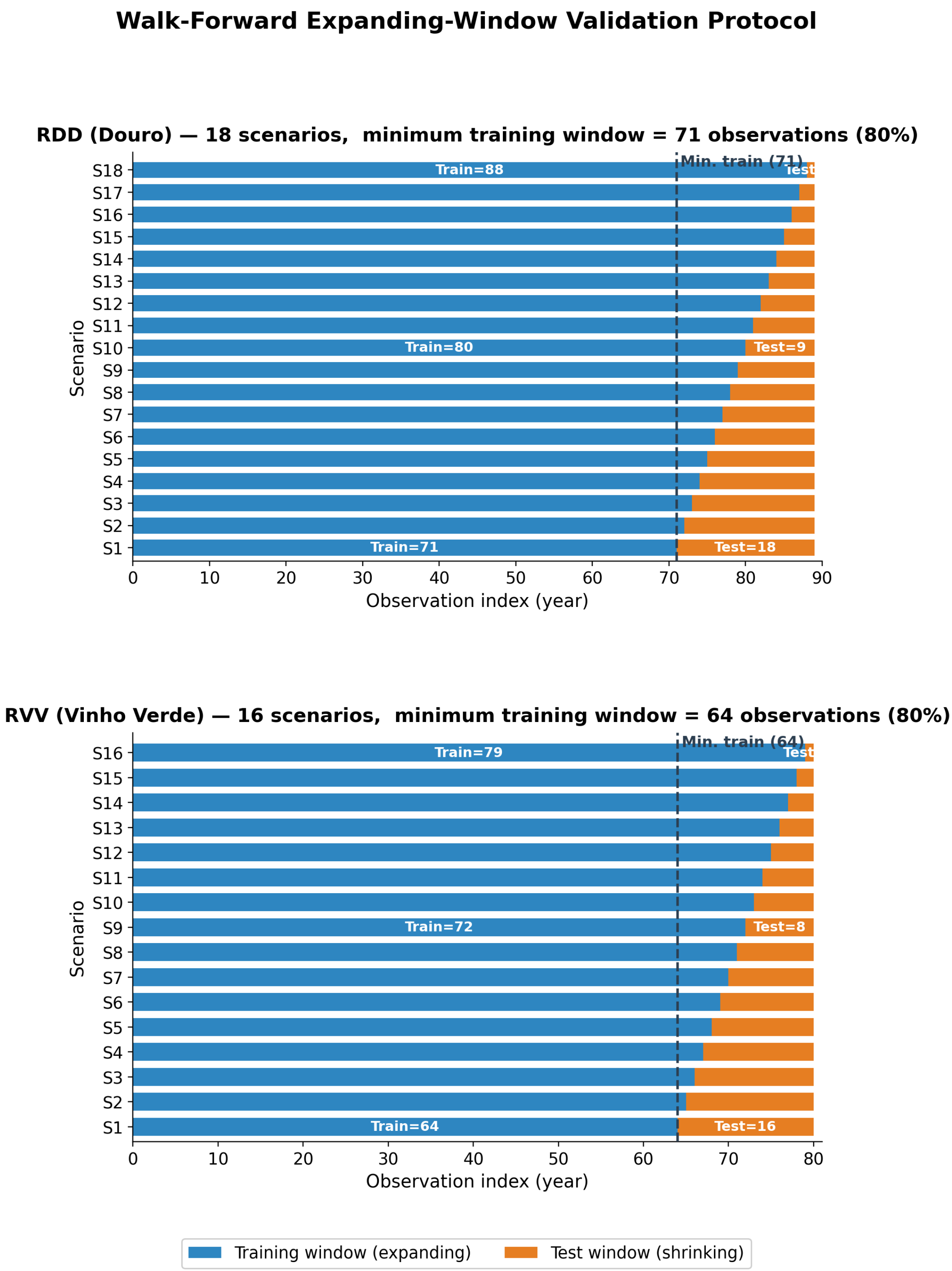

Figure 3.4: Walk-forward expanding-window validation for RDD (18 scenarios, top) and RVV (16 scenarios, bottom).

Table 3.3: Complete set of 17 models evaluated in the walk-forward validation.

Model	**Type**	**Feature selection / distinguishing design choice**
`Naive_Median`	Baseline	Predicts training-set median residual (primary baseline)
`Naive`	Baseline	Predicts most recent observed value (random-walk)
`Naive_MA`	Baseline	Predicts mean of 3 most recent training observations
`Decision Tree`	ML comparator	CART regressor on all 59 features; recursive binary splits
`RIPPER`	Rule learner	Classification rules on tertile-discretised residuals
`M5Rules`	Rule learner	Piecewise linear regression rules
`CarenR_Dist`	CarenR variant	All 59 features; default minimum support
`CarenR_Dist_FS`	CarenR variant	Top-N features by |Pearson correlation| with target
`CarenR_Dist_Thresh`	CarenR variant	Features with $\|\text{correlation}\| \geq 0.30$
`CarenR_Dist_Sup`	CarenR variant	All 59 features; higher minimum support (broader rules)
`CarenR_Dist_Sup_Thresh`	CarenR variant	Correlation threshold + higher minimum support
`CarenR_Dist_Eta2`	CarenR variant	Top-N features by η^2 (ANOVA on discretised bins)
`CarenR_Dist_Lag`	CarenR variant	Sup_Thresh config + lagged production residuals
`CarenR_Dist_Conf`	CarenR variant	Sup_Thresh config; confidence-weighted by $-\log(p\text{-value})$
`CarenR_Dist_Stack`	CarenR variant	Sup_Thresh rules stacked with a lag regression step
`CarenR_Dist_Sup_FS`	CarenR variant	Support-filtered + top-N Pearson feature selection
`CarenR_Supervised`	CarenR variant	Per-fold supervised discretisation (cutpoints maximise residual separation)

Table 3.4: CarenR variant families and their feature selection strategies.

Variant	Feature selector	Selection criterion	Notes
`CarenR_Dist`	None	All 59 features	Widest search; most risk of spurious rules
`CarenR_Dist_FS`	Pearson $\lvert r \rvert$, top-N	Retain above-median $\lvert r \rvert$ features	Most common baseline variant
`CarenR_Dist_Eta2`	η^2 (ANOVA effect size)	Bin-wise ANOVA η^2 with response	Best for RDD; captures non-linear association
`CarenR_Dist_Sup_FS`	Pearson + support $\geq 20\%$	Conservative: $\lvert r \rvert$ + minimum bin coverage	Most stable across both regions
`CarenR_Dist_Thresh`	Pearson hard threshold	$\lvert r \rvert \geq 0.30$	Reduces feature set aggressively
`CarenR_Dist_Sup`	Pearson + support filter	$\lvert r \rvert$ + support filter (less strict than Sup_FS)	Intermediate conservatism
`CarenR_Dist_Lag`	ACF-based lag features	Augment set with autocorrelated lags	Tests carryover temporal effects
`CarenR_Dist_Conf`	Confidence-weighted prediction	Weight rules by $1 - p$-value	Stronger rules get more prediction weight
`CarenR_Dist_Stack`	Two-stage stacking	CarenR residuals as second-stage input	Tests whether stacking adds value
`CarenR_Dist_Sup_Thresh`	Pearson threshold + support filter	$\lvert r \rvert \geq 0.30$ + minimum bin coverage	Threshold and support combined; base for Lag, Conf, Stack variants
`CarenR_Supervised`	Per-fold supervised bins	Maximise between-group residual SS (greedy 1D splits)	Evaluates target leakage from discretisation

3.7 Evaluation Metric

The primary evaluation metric is scenario-size-weighted MAE on the original production scale (mhl):

$$\text{Weighted MAE} = \frac{\sum_s n_s \times \text{MAE}_s}{\sum_s n_s} \tag{3.4}$$

where n_s is the number of test observations in scenario s. This weighting gives more influence to scenarios with larger test sets, which in the expanding-window design are the early scenarios with shorter training histories. Statistical significance of model differences was assessed using the Wilcoxon signed-rank test applied to per-scenario MAE pairs; the post-hoc era-balanced analysis used a binomial sign test (Section 5.4).

3.8 Software and Reproducibility

The full pipeline was implemented in R (version 4.x). CarenR was run via the caren R package. RIPPER and M5Rules were run via RWeka, providing a JVM bridge to Weka 3.8. The pipeline is controlled by a master runner script (run_all_pipelines.R) that sequences data preparation, rule discovery, walk-forward validation, rule interpretation, and output generation. All detrending parameters and model hyperparameters are centralised in a configuration file (utils/config.R) to ensure reproducibility.

3.8.1 Prediction Aggregation: Combining Multiple Firing Rules

When multiple rules fire for a given test observation, their individual predictions must be aggregated into a single point estimate. Most CarenR variants use support-weighted aggregation; a small number use confidence weighting or equal weighting as alternatives. The primary source of divergence across the 11 variants is the feature selection strategy, not the aggregation mechanism. The general prediction formula for a test observation x is:

where the sum is over all rules i that fire for observation x (i.e. all conditions of rule i are satisfied by x), m_i is the median residual of the training observations covered by rule i, and $w_i(x)$ is the weight assigned to rule i for observation x. The weight definition distinguishes the variants.

Note on notation: m_i denotes the median residual of the training observations in the subgroup defined by rule i. The tilde notation (~) is used throughout to distinguish the median from the mean (μ), following the convention that μ denotes a population mean and m denotes a sample median.

For the default support-weighted case (used by the majority of variants), the full prediction formula for test year t is:

where $F(t) = \{i : \text{all conditions of rule } i \text{ are satisfied by features of year } t\}$ is the set of firing rules, $s_i \in [0.15, 1.0]$ is the support fraction of rule i, m_i is the median detrended production residual among training observations covered by rule i, and $\text{median}(r_1, \ldots, r_n)$ is the training-set median residual (the Naive_Median prediction, used as fallback when no rules fire).

Table 3.5: Prediction aggregation formulas for the three CarenR variant families.

Variant family	Weight $w_i(x)$	Formula	Rationale
Base (`Dist`, `FS`, `Eta2`, `Sup_FS`, `Thresh`, `Lag`)	Support of rule i	$w_i = s_i$ (fraction of training obs. covered)	Rules with higher coverage are weighted more — they represent more stable, generalisable patterns. A rule covering 28% of years gets $\approx$1.4$\times$ the weight of one covering 20%.
Confidence-weighted (`CarenR_Dist_Conf`)	Confidence of rule i	$w_i = 1 - p_i$ (where p_i is the KS p-value)	Rules with stronger statistical significance are weighted more. A rule with $p = 0.004$ gets weight 0.996; a rule with $p = 0.09$ gets weight 0.91. Prioritises statistical robustness over coverage.
Stacking (`CarenR_Dist_Stack`)	Second-stage model weight	w_i from a second-stage linear model fitted on rule predictions	Residuals from first-stage CarenR predictions are used as input to a second-stage model. Equivalent to learning an optimal linear combination of rule predictions on the training set.

A worked example illustrates the formula. Suppose test year 2015 has the following feature values: wet days post-flowering = 1.2 days, soil water pre-harvest = 89 mm, heat days post-flowering = 0.4 days. Three rules fire:

The predicted residual (+117.3 mhl) is then re-trended by adding the linear trend value projected to 2015 to produce the final production estimate in mhl. This re-trending step ensures the prediction is on the original scale and directly comparable to the actual production value for the MAE computation.

Why support weighting rather than equal weighting? Equal weighting would give every firing rule the same influence regardless of how broadly applicable it is. A rule covering 20% of training years (the minimum support threshold adopted in this thesis) represents a pattern seen in roughly 18 years out of 89; a rule covering 28% of years was observed in approximately 25 years. Giving equal weight to both would over-represent narrow, potentially less stable patterns. Support weighting is a principled way to account for statistical reliability without fully discarding less-common rules; they still contribute, but proportionally to their historical recurrence.

The confidence-weighted variant (CarenR_Dist_Conf) uses $1 - p_i$ as the weight, giving more influence to rules whose subgroup distribution is most statistically distinct from the global distribution. In theory, this should favour the most reliable patterns; in practice, it performs comparably to support weighting (RDD rank 6, RVV rank 3 in Table 5.1), suggesting that KS p-value and support are correlated enough that both approaches identify the same rules as most important.

When exactly one rule fires, the formula simplifies to $\hat{y}(t) = m_i$, the prediction is simply the median residual of the matching subgroup, independent of support (which cancels). The support

Table 3.6: Worked example of the support-weighted prediction formula.

Rule	Conditions matched	Subgroup median $\tilde{m}_i$	Support s_i	Weighted contribution $s_i \cdot \tilde{m}_i$
Rule 1	Wet days post-flowering $\in [0, 3.3\,\mathrm{d}]$ Wet days post-maturity $\in [0, 3.5\,\mathrm{d}]$	+121 mhl	0.281	34.00
Rule 5	Wet days post-flowering $\in [0, 3.3\,\mathrm{d}]$ Heat days pre-maturity $\in [0, 3.7\,\mathrm{d}]$	+115 mhl	0.247	28.41
Rule 7	Wet days post-flowering $\in [0, 3.3\,\mathrm{d}]$ Soil water pre-harvest $\in [74, 231\,\mathrm{mm}]$ Heat days pre-maturity $\in [0, 3.7\,\mathrm{d}]$	+115 mhl	0.213	24.50
Sum			$\sum s_i = 0.741$	$\sum s_i \tilde{m}_i = 86.91$
Prediction $\hat{y} = 86.91\,/\,0.741 = +117.3$ mhl				

weighting only has an effect when two or more rules fire simultaneously, resolving conflicts by favouring more broadly applicable rules.

3.9 Formal Stationarity Testing

The need for detrending is not merely heuristic: both production series exhibit statistically significant non-stationarity in their raw form, formally confirmed by two complementary tests. The Augmented Dickey-Fuller (ADF) test uses the null hypothesis that a unit root is present (non-stationary); rejection at $\alpha = 0.05$ indicates stationarity. The KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test reverses this logic, using the null hypothesis of stationarity; rejection indicates non-stationarity. Applying both tests provides a more robust assessment than either test alone, since ADF has low power against near-unit-root processes and KPSS is sensitive to long-run variance estimation.

Table 3.7 reports the test results for the raw production series and the linear detrended residuals for both regions.

The results have three important implications for the methodological choices made in this thesis.

Implication 1: Detrending is formally justified for both regions. Both raw series fail the ADF test ($p = 0.38$ for RDD, $p = 0.89$ for RVV), confirming that unit roots cannot be rejected in either series. The linear trend slopes are both highly statistically significant ($p < 0.0001$), confirming that the observed trends are not artefacts of sampling variability. Applying machine learning models to the raw series without detrending would violate the stationarity assumptions implicit in

Table 3.7: Stationarity test results. ADF H_0: unit root present.

Region	Series	ADF statistic	ADF p-value	Verdict	KPSS statistic	KPSS p-value	Verdict
RDD	Raw (Wine_mhl)	-1.800	0.381	Non-stationary (unit root)	0.134	0.071	Borderline stationary around trend
RDD	Linear detrended residuals	-8.520	<0.001 ***	Stationary ✓	0.134	0.100	Stationary ✓
RVV	Raw (Wine_mhl)	-0.490	0.894	Non-stationary (unit root)	0.332	0.010 **	Non-stationary (rejects stationarity around trend)
RVV	Linear detrended residuals	-1.817	0.372	Non-stationary (unit root remains)	0.332	0.100	Stationary ✓ (level stationarity)

cross-sectional feature-based modelling and would confound climate effects with structural level changes.

Implication 2: Linear detrending achieves full stationarity in RDD but only partial stationarity in RVV. After linear detrending, the RDD residuals pass both the ADF test ($p < 0.001$) and the KPSS test ($p = 0.10$), confirming stationarity by both criteria. The RVV residuals pass the KPSS level-stationarity test ($p = 0.10$) but fail the ADF test ($p = 0.37$), indicating that a unit root cannot be rejected even after removing the linear trend. This is formal statistical evidence that the RVV structural change, the sharp non-linear decline associated with EU accession and industry restructuring, is not fully captured by a linear trend. Persistent non-stationarity in the RVV residuals means that the era-composition effects observed in the walk-forward evaluation (Section 5.3) have a formal statistical basis: the residuals themselves are non-stationary, which means rules discovered in early training windows may not generalise to later test windows.

Implication 3: The mixed RVV result formally motivates LOESS detrending as a more flexible alternative. The fact that RVV residuals pass KPSS but fail ADF after linear detrending confirms that the linear trend removes the long-run structural level but not all persistent structure. A more flexible detrending (LOESS) would in principle resolve this residual non-stationarity. The LOESS sensitivity analysis (Section 5.5) demonstrates that LOESS does reduce the Naive_Median baseline substantially for RVV, consistent with reducing the residual non-stationarity. However, it simultaneously removes the climate signal required for rule induction, a direct empirical manifestation of the Decision 1 / Decision 2 trade-off formalised in Section 3.3.

The stationarity test results therefore serve as formal grounding for three methodological choices: (1) the decision to detrend rather than model raw production values; (2) the acknowledgement that linear detrending is imperfect for RVV and that some residual non-stationarity remains; and (3) the decision to retain linear detrending despite this imperfection, because the alternative (LOESS) resolves the non-stationarity at the cost of eliminating the climate signal that is the thesis's primary object of study.

3.10 Data Sources

The datasets for both regions originate from two independent source streams that were joined at the annual level.

3.10.1 Meteorological Data

Daily meteorological records for the Douro region were sourced from the BCM base climate files maintained by the University of Porto's viticulture research group (Professor Mário Cunha). The source files (BCM_base_RDD.xls, BCM_base_RVV.xls) contain one row per calendar day from 1933-01-01 to 2022-12-31 for RDD (32,842 daily records) and from 1941-01-01 to 2022-11-13 for RVV (29,902 daily records). Each daily record contains the following primary meteorological variables:

Table 3.8: Primary daily meteorological variables in the raw source files.

Variable	Description	Unit
Tmax	Daily maximum temperature	°C
Tmin	Daily minimum temperature	°C
Tmed	Daily mean temperature	°C
Rain	Daily total rainfall	mm
Sm	Soil water availability (model-derived)	mm
Iaf	Leaf Area Index (phenological model)	m^2/m^2
Eto	Reference evapotranspiration	mm

The soil water availability (Sm) and Leaf Area Index (Iaf) values are not direct measurements but model-derived estimates produced by the Vineyard Soil Irrigation Model (VSIM) calibrated to each region's soil and vine characteristics by Professor Cunha's group over decades of field work (Cunha and Richter, 2016). These modelled variables are critical for the analysis: Sm represents the balance between precipitation, evapotranspiration, and soil water-holding capacity integrated daily, while IAF represents vine canopy development as a continuous daily estimate anchored to the vine's phenological state.

In Stage 1 of the pipeline (script: 1_data_preparation_stage_1.R), five binary/threshold indicator variables were derived from the pre-processed daily records provided by Professor Cunha's group:

The resulting Stage 1 output is a clean daily dataset with 17 columns per region (dataPrep_meteo_rdd.csv, dataPrep_meteo_rvv.csv) spanning the full time horizon.

Table 3.9: Binary threshold indicators derived from raw daily meteorological variables.

Derived indicator	Condition	Agronomic meaning
`R>0.1`	Rain $>$ 0.1 mm	Wet day indicator (trace rainfall)
`Tmed<15`	Tmed $<$ 15°C	Cold day, impairs flowering and fruit set
`Tmax>35`	Tmax $>$ 35°C	Heat-stress day, causes flower abortion, accelerates ripening
`Tmin<0`	Tmin $<$ 0°C	Frost day, potential bud and shoot damage
`Sm<1.5Wp`	Sm $<$ 1.5$\times$ wilting point	Severe soil water stress, below vine survival threshold
`Sm>0.9Fc`	Sm $>$ 0.9$\times$ field capacity	Near-saturation, promotes disease, excessive vigour

3.10.2 Phenology and Production Data

Annual wine production records and vine phenological dates were sourced from the Pheno_Prd source files maintained by the same research group (Pheno_Prd_1933_2022_recente.xlsx for RDD; equivalent for RVV). Each annual record contains:

Table 3.10: Annual phenology and production variables.

Variable	Description
Wine_mhl	Annual regional wine production (thousands of hectolitres)
DOY_BB	Day of Year of Bud Break, start of vegetative growth
DOY_Fl	Day of Year of Flowering, critical fruit-set window
DOY_sM	Day of Year of Start of Maturity (veraison), berry ripening begins
DOY_Fs	Day of Year of Fruit Set, end of berry cell division
DOY_Hv	Day of Year of Harvest, end of growing season

The phenological dates (DOY_BB through DOY_Hv) are derived from the STICS (Simulateur mulTIdisciplinaire pour les Cultures Standard) phenological sub-model and represent the average calendar date of each key event across the regional vine population for each year. These dates vary by 2–4 weeks across years for each stage, reflecting the sensitivity of vine phenology to temperature, warmer springs advance flowering and harvest, cooler springs delay them. The variability in these anchor dates is what makes phenology-based windowing more informative than fixed calendar windows: a feature computed as "the 20 days before Harvest" adapts to the actual harvest date of each year rather than always covering, say, the first three weeks of September.

The production series for RDD spans 1933–2022 (90 observations), but the feature construction starts at 1934 to allow for previous-year (y0) features, leaving 89 usable observations. For RVV, the series spans 1942–2021 with 80 usable observations. The production values show the structural trajectories described in Chapter 2: RDD production ranging from 518 to 1,961 mhl with a mean of 1,116 mhl and an upward trend; RVV production ranging from 494 to 3,443 mhl with a mean of 1,621 mhl and a sharp downward trend since the 1960s.

3.11 Feature Engineering: From Daily Records to Annual Features

The core methodological contribution of the data pipeline is the transformation from 32,842 (RDD) and 29,902 (RVV) daily records into 59 annual agroclimatic features per region. This transformation is performed by script 2_features_creation.R and is anchored entirely on the phenological dates for each year. For each year y1, each combination of (meteorological variable × phenological anchor × window position × window width × year offset) defines one feature.

3.11.1 Feature Naming Convention

All 59 features follow a systematic naming convention that encodes all relevant information about how the feature was computed:

variable_stage_year_positionwidth

where:

Table 3.11: Feature naming convention components.

Component	Options	Meaning
`variable`	`tm, tn, swa, iaf, rf, days.rf, sw.less.15wp, sw.above.09fc, days.maxTemp.above.35, tm.days.less.15, days.minTemp.less.0`	Meteorological quantity aggregated
`stage`	`bb` (Bud Break), `fl` (Flowering), `sM` (Start Maturity), `hv` (Harvest)	Phenological anchor point
`year`	`y1` (current year), `y0` (previous year)	Growing season reference
`position`	`c` (centred), `b` (before), `a` (after)	Window position relative to stage
`width`	`5, 10, 15, 20, 30` (days)	Aggregation window width

For example: swa_hv_y1_b20 = soil water availability (swa), in the 20 days BEFORE (b20) Harvest (hv), in the current year (y1). tm_fl_y0_b10 = mean temperature (tm), in the 10 days BEFORE (b10) Flowering (fl), in the PREVIOUS year (y0). iaf_fl_y1_a10 = Leaf Area Index (iaf), in the 10 days AFTER (a10) Flowering (fl), in the current year (y1).

3.11.2 Feature Groups and Counts

Table 3.12 summarises the 59 features by variable family and their counts. The distribution reflects the relative importance of different variable types in the viticulture literature: mean temperature (tm) and minimum temperature (tn) variables are most numerous because temperature governs nearly every phenological process; soil water and LAI variables are next because they are the primary within-season stress indicators; rainfall and heat/cold day counts complete the picture.

Table 3.12: Feature groups, counts, and agronomic meanings. Total: 59 features per region.

Variable family	**Symbol**	**N**	**What it measures**
Mean daily temperature	tm	12	Average thermal conditions at key phenological windows
Leaf Area Index	iaf	10	Vine canopy development, proxy for vegetative vigour
Soil water deficit days	sw.less.15wp	8	Days when soil water < 1.5× wilting point, severe stress
Rainfall event count	days.rf.above.1mm	6	Number of days with rainfall > 1 mm, disease pressure proxy
Soil water availability	swa	5	Absolute soil water balance at harvest and flowering windows
Soil saturation days	sw.above.09fc	4	Days when soil water > 90% field capacity, excess moisture
Heat stress days (Tmax > 35°C)	days.maxTemp.above.35 / days.tempMax.above.35	4	Extreme heat events, flower abortion and thermal damage
Rainfall sum	rf	3	Total precipitation at harvest and flowering windows
Cold days (Tmed < 15°C)	tm.days.less.15	2	Cool days impairing pollination and fruit set
Frost days (Tmin < 0°C)	days.minTemp.less.0	2	Frost events, bud and shoot damage risk
Minimum temperature	tn	1	Night temperature at harvest, berry quality proxy
Previous-year soil water	y0 variants of swa/sw	2	Carry-over water stress from prior season

3.11.3 Current-Year vs Previous-Year Features

A key design decision in the feature engineering was the inclusion of previous-year (y0) features alongside current-year (y1) features. As discussed in Section 2.2, vine physiology has a two-year temporal structure: inflorescence primordia that determine current-year bunch number are formed in the prior growing season, and vine carbohydrate reserves that determine shoot vigour are accumulated during the prior year's harvest-to-dormancy period. Failing to include prior-year features would miss this carry-over mechanism entirely.

In the final feature matrix, 15 of the 59 features (25%) use y0 (previous-year) anchors. The most agronomically motivated y0 features are:

tm_fl_y0_b10 (mean temperature in the 10 days before previous-year flowering): captures the thermal conditions governing inflorescence primordia formation in the prior season, the mechanism that determines current-year bunch potential.

tm_bb_y0_a30 (mean temperature in the 30 days after previous-year bud break): captures the early-season thermal conditions that initiate the prior year's vegetative growth phase, influencing vine carbohydrate allocation.

tm_sm_y0_a20 (mean temperature in the 20 days after previous-year start of maturity): captures the heat load during the prior year's ripening phase, which governs both berry quality in the prior year and the vine's thermal stress exposure at the end of the growing season.

sw.above.09fc_fl_y0_a20 (soil saturation days in the 20 days after previous-year flowering): captures prior-year excess soil moisture during flowering, the carry-over mechanism identified in RVV CarenR Rule 2 (see Section 4.1.2).

3.11.4 Feature Discretisation for CarenR

The continuous feature matrix (dataPrep_cont_dataset_*.csv, 59 columns × 89/80 rows) is used directly by M5Rules and RIPPER, which perform their own internal threshold finding. For CarenR, a separate discretised version of the dataset (dataPrep_dis_rule_dataset_*.csv) is produced by script 2_features_creation.R using equal-frequency binning into four quartile intervals. Each feature is independently divided into four bins, each containing approximately 25% of the training observations, with the boundaries represented as interval notation (e.g. [18.3|22.4] for the bin from 18.3°C to 22.4°C).

The equal-frequency choice, rather than equal-width or domain-knowledge-based breakpoints, ensures that each bin is populated by a comparable number of observations. This is important for the KS test used by CarenR: a bin with only two or three observations would produce an unreliable distributional comparison regardless of its meteorological significance. In the walk-forward scenarios, the bin boundaries are re-estimated within each training fold, ensuring that the discretisation reflects the distribution of climate conditions actually observed in the training period, not the full historical record that would be unknowable at prediction time.

3.12 Data Pipeline Summary

The pipeline was implemented as a sequence of seven R scripts coordinated by a master runner (run_all_pipelines.R), with all intermediate datasets written to disk at each stage to allow inspection and validation. The complete stage-by-stage breakdown is given in Table B.7 and the fixed configuration parameters in Table B.8, both in Appendix B. Figure B.1 in the same appendix shows the full pipeline schematically.

Chapter 4

Rule Discovery Results (Goal 1)

This chapter reports the output of the three rule-learning algorithms applied to the full historical datasets. The primary evaluation question for Goal 1 is: does CarenR's distributional subgroup discovery extract statistically validated, non-trivial patterns from the data? Subsidiary questions are: do RIPPER and M5Rules converge on the same variables through different optimisation criteria? And are the discovered patterns genuine within-era signal or era-membership artefacts? Section 4.1 presents the CarenR rules; Section 4.2 validates them using LOESS detrending; Sections 4.3–4.4 present RIPPER and M5Rules; Sections 4.5–4.8 provide cross-algorithm comparison, feature frequency, and rule set structure analysis.

4.1 CarenR: Distributional Rules

After applying the significance threshold (KS $p \leq 0.10$) and minimum support filter ($\geq 20\%$ of training observations), CarenR retained 48 rules for RDD and 20 rules for RVV. Tables 4.1 and 4.2 show the five most significant rules per region; the full top-10 tables are in Appendix A (Tables A.1 and A.2). For each rule, the table records the statistical evidence (p-value, support), the magnitude of the discovered distributional shift (mean deviation from trend), and the conditions that define the subgroup. One sentence of domain context is provided per rule to confirm that the direction of the discovered pattern is consistent with established knowledge, serving as a face-validity check on the method's output rather than as a primary finding.

4.1.1 Douro (RDD): Top 5 Rules

Key structural observations from the RDD rule set: (1) 31 of 48 rules point to ABOVE-trend production, an asymmetry that reflects the algorithm's finding that the positive signal in this dataset is structurally more consistent than the negative signal; the 17 below-trend rules require an unusual combination of high LAI with high soil moisture that occurs in approximately 20% of years. (2) The rule set is organised around three recurring features, post-flowering wet days, pre-harvest soil water, and heat-stress day counts, with later rules being refinements of earlier ones through additional conditions. (3) The most significant rule (Rule 1) has support of 28.1%, covering approximately

Table 4.1: Top 5 CarenR rules for the Douro (RDD), sorted by KS p-value. Full top-10 in Table A.1.

#	Conditions	Dir.	Median Δ (mhl)	Support	p-value
1	**Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d]** *Dry post-flowering conditions reduce disease pressure and promote fruit set in the Douro.*	ABOVE	+121	28.1%	0.0043
2	**LAI at Flowering [0.74–0.88 m^2/m^2] AND Soil water post-Flowering [390–404 mm]** *Elevated canopy vigour with high soil moisture at flowering drives vegetative-over-reproductive growth.*	BELOW	-131	20.2%	0.0084
3	**Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Heat days post-Flowering [0–0.6d]** *Refines Rule 1: dry and thermally moderate post-flowering window.*	ABOVE	+113	25.8%	0.0089
4	**Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d] AND Wet days post-Maturity [0–3.5d]** *Multi-year rule: moderate soil water stress with dry previous-year post-harvest window captures a carry-over vine reserve mechanism.*	ABOVE	+192	21.3%	0.0095
5	**Wet days post-Flowering [0–3.3d] AND Heat days pre-Maturity [0–3.7d]** *Dry flowering with bounded pre-veraison heat — the "warm but not extreme" pattern.*	ABOVE	+115	24.7%	0.0110

one in four years in the historical record, confirming it describes a recurrent rather than exceptional pattern. Rule 4 produces the largest above-trend shift in the RDD set (+192 mhl, support 21.3%), combining moderate pre-harvest soil water with dry post-harvest conditions across two years; Rule 2 is the largest below-trend rule (-131 mhl, support 20.2%), requiring elevated canopy vigour with high soil moisture at flowering.

4.1.2 Vinho Verde (RVV): Top 5 Rules

Key structural observations from the RVV rule set: (1) 16 of 20 rules point to BELOW-trend production, the inverse asymmetry from RDD. In a declining series, the majority of historically exceptional subgroups are years that fall below the already-declining trend. (2) Rule 4 (one of four above-trend rules, +311 mhl, support 25%) represents the largest absolute distributional shift in either region; CarenR has identified a well-defined subgroup corresponding to the best RVV vintages. (3) The LAI at harvest variable (iaf_hv_y1_c10) appears in only 2 of the 20 rules, and the LAI family (including prior-year flowering LAI, iaf_fl) in 4; it is absent from the most significant rules (ranks 1–5). Its low frequency and late rank reflect that LAI tends to combine with other variables rather than appearing as a single strong condition in the most significant rules. This is a

Table 4.2: Top 5 CarenR rules for the Vinho Verde (RVV). Rule 4 is the only above-trend rule. Full top-10 in Table A.2.

#	Conditions	Dir.	Median Δ (mhl)	Support	p-value
1	**Soil water pre-Harvest [61–97 mm] AND Heat days pre-Maturity [0–1.9d] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]** *Low soil water deficit with absent heat stress — in the Atlantic RVV climate this is a cool-moist pattern, not a beneficial stress pattern.*	BELOW	-338	21.2%	0.0198
2	**Temp at Flowering [13.8–17.1°C] AND Temp post-Flowering [12.6–17.5°C] AND Wet-soil days post-Flowering prev.yr [17.5–20d]** *Cool flowering combined with previous-year soil saturation — a two-year carry-over pattern.*	BELOW	-268	21.2%	0.0216
3	**Soil water post-Flowering [143–234 mm] AND Heat days post-Maturity [0–2.4d]** *Highest-support below-trend rule: moist post-flowering with low heat at maturity.*	BELOW	-365	27.5%	0.0294
4	**Temp post-Budburst prev.yr [11.2–13.5°C] AND Cold days at Flowering [0–3.4d] AND Heat days post-Flowering [0–1.1d]** *The only above-trend rule: warm-but-not-extreme previous-year budburst with mild current-year flowering. Largest absolute deviation in either region (+311 mhl).*	ABOVE	+311	25.0%	0.0298
5	**Temp post-Flowering [12.6–17.5°C] AND Heat days post-Maturity [0–2.4d] AND Wet-soil days post-Flowering prev.yr [17.5–20d]** *Cool post-flowering temperature with previous-year soil saturation carry-over.*	BELOW	-292	23.8%	0.0299

property of the KS-test-based induction: LAI produces a distributional shift that is amplified by conditioning on additional variables.

4.1.3 The Directional Inversion: A Method Validation Point

The same feature, soil water 20 days before harvest (swa_hv_y1_b20), is selected as the dominant variable in both regions, but with opposite distributional direction: moderate deficit predicts above-trend in RDD, the same variable range predicts below-trend in RVV. This is a meaningful test of the method's sensitivity: CarenR correctly detects that the same variable can carry opposite distributional signals in different contexts. A classification algorithm trained on both regions together would conflate these opposite effects; CarenR, operating independently on each dataset, identifies the correct regional direction without any cross-region information. The domain explanation (different baseline climates) validates the finding but is secondary; the primary observation is that the KS-based framework correctly detects region-specific distributional structure.

Table 4.3: Cross-regional feature direction comparison for swa_hv_y1_b20.

Feature	RDD bin range	RDD direction	RDD mean Δ	RVV bin range	RVV direction	RVV mean Δ
swa_hv_y1_ b20 (soil water at harvest)	74–231 mm (moderate deficit)	ABOVE trend	+108 to +192 mhl	61–97 mm (pronounced deficit)	BELOW trend	-338 mhl
Wet days post-flowering (days.rf.above.1mm_ fl)	0–3.3 days (very dry)	ABOVE trend	+108 to +121 mhl	20–121 mm rainfall (moderate wetness)	BELOW trend	-387 mhl
LAI at harvest (iaf_hv_y1_ c10)	Absent from top rules (minor signal in RDD)	BELOW trend (where it appears)	-118 to -131 mhl	Minor signal (2 of 20 rules)	BELOW trend	-268 to -387 mhl

4.2 LOESS Validation: Distinguishing Signal from Era Artefact

A critical question for interpreting the rule set is whether the discovered patterns reflect genuine inter-annual variation or merely co-variation with the long-run structural trend (era membership). If a feature co-trends with production over decades, it will appear informative under linear detrending even if it has no true predictive relationship with production residuals within eras.

To test this, a LOESS-detrended version of the target series was used as an alternative to the linear residuals. LOESS applies a locally weighted regression smoother that adapts to regional curvature in the production trajectory, removing not only the linear trend but also slower structural movements within eras. A sensitivity analysis across spans from 0.20 to 0.90 confirmed that every span achieves ADF stationarity in both regions. Span = 0.40 was adopted as a conservative choice: it is narrow enough to remove the era-level structural shift while remaining substantially wider than inter-annual noise. The key diagnostic is the proportion of each feature's linear-detrend correlation that is retained after LOESS detrending. A feature carrying genuine within-era climate signal should retain most of its correlation under LOESS; a feature that is informative only because it co-trends with the structural trajectory will show a large collapse.

Table 4.4 reports the results. The findings differ sharply between regions. In RDD, harvest soil water and post-flowering wet days are fully preserved (103–125% retention), confirming that these features carry genuine within-era climate signal. Harvest temperature is also preserved (105%), suggesting it has within-era predictive content in RDD in addition to its known long-run warming trend. In RVV, the picture is starkly different: LAI at harvest and soil water collapse to 15–16% retention, meaning that nearly all their apparent predictive power under linear detrending is era co-variation rather than genuine climate signal. Post-flowering wet days are the exception, retained at 94% in RVV, indicating genuine within-era signal. This result is robust to span choice: the same qualitative pattern holds across all tested spans, confirming that the finding reflects the data structure rather than an artefact of the span parameter.

Table 4.4: LOESS validation of top CarenR features (span = 0.40). Features retaining > 85% of linear correlation carry genuine within-era signal.

Feature	Region	Linear $\|r\|$	LOESS $\|r\|$	% retained	Interpretation
swa_hv_y1_b20 (harvest soil water)	RDD	0.33	0.41	125%	Genuine within-era signal; increases under LOESS
days.rf.above.1mm_fl_y1_c15 (wet days post-flowering)	RDD	0.28	0.28	103%	Genuine; fully preserved after era removal
tm_hv_y1_c10 (harvest temperature)	RDD	0.17	0.17	105%	Preserved; within-era signal alongside known warming trend
days.rf.above.1mm_fl_y1_c15 (wet days post-flowering)	RVV	0.34	0.32	94%	Genuine; robust to era removal
iaf_hv_y1_c10 (LAI at harvest)	RVV	0.22	0.03	15%	Era-driven; collapses under LOESS
swa_hv_y1_b20 (harvest soil water)	RVV	0.17	0.03	16%	Era-driven; collapses under LOESS

4.3 RIPPER Classification Rules

RIPPER was applied to tertile-discretised production classes. CV accuracy: 42% RDD, 28% RVV (33% random baseline; κ = 0.13 RDD, -0.09 RVV). Tables 4.5 and 4.6 show the rule lists. The primary evaluation question for RIPPER in this study is convergence with CarenR: do the features selected by an entirely different optimisation criterion (information gain for class separation vs KS distributional shift) point to the same variables?

Table 4.5: RIPPER rules for RDD. CV accuracy 42%.

#	Conditions	Prediction	Support	Precision
RDD 1	Mean temp at Harvest (10d before) in [21.2–22.8°C]	MEDIUM	21.3% (19yr)	73.7%
RDD 2	LAI at Budburst $\geq$ 0.0016 AND Mean temp at Harvest (20d before) $\geq$ 23.1°C	HIGH	19.1% (17yr)	76.5%
RDD 3	Cold days at Flowering $\leq$ 3 AND Mean temp at Harvest $\leq$ 21.3°C	HIGH	6.7% (6yr)	100%
RDD 4	(default)	LOW	—	—

Table 4.6: RIPPER rules for RVV. CV accuracy 28%.

#	Conditions	Prediction	Support	Precision
RVV 1	LAI at Flowering prev.yr $\geq$ 2.4 AND Budburst temp prev.yr $\geq$ 13.3°C AND Rainfall at Flowering $\leq$ 15.9 mm	MEDIUM	16.2% (13yr)	92.3%
RVV 2	Mean temp at Flowering $\geq$ 19.3°C AND LAI at Flowering $\geq$ 2.2	MEDIUM	11.2% (9yr)	88.9%
RVV 3	Soil water post-Flowering $\leq$ 123.6 mm AND Mean temp at Flowering $\geq$ 16.6°C	HIGH	26.2% (21yr)	85.7%
RVV 4	(default)	LOW	—	—

RIPPER selects harvest temperature (RDD Rules 1–3), LAI at budburst and flowering (RDD Rule 2, RVV Rules 1–2), and soil water / rainfall at flowering (RVV Rules 1, 3), all within the same variable families identified by CarenR. This cross-algorithm convergence under a classification-based criterion is the key finding: the same variables are deemed informative whether the task is formulated as “find distributional shift” (CarenR) or “find class-separating thresholds” (RIPPER), supporting the interpretation that these variables carry genuine structure in the data.

4.4 M5Rules: Piecewise Linear Models and Coefficient Interpretation

M5Rules collapsed to global linear models in both regions (no beneficial split found at minimum leaf size 4), producing 2 rules for RDD and 3 for RVV. CV R^2 = 0.004 (RDD), -0.07 (RVV), confirming overfitting. The evaluation question for M5Rules is again convergence: do the regression coefficients of an MSE-minimising linear model point to the same variables, and in the same directions, as the KS-based distributional rules?

The swa_hv_y1_b20 coefficient (-2.3 mhl/mm) reveals an important contrast between the two methods. M5Rules describes a negative linear relationship across the full soil water range: each additional mm of harvest soil water is associated with 2.3 mhl lower production. CarenR, by contrast, identifies a specific moderate-deficit range [74–101 mm] as associated with above-trend production (+121–192 mhl). These descriptions point in opposite directions on the surface, but the discrepancy is methodologically informative rather than contradictory. M5Rules captures the dominant linear signal: across the full range, high soil water before harvest is generally harmful (berry dilution, disease pressure), producing a negative slope overall. CarenR identifies a specific sub-range as a relative sweet spot compared to both extremes — very dry years and very wet years are both below trend, and the moderate range stands out distributionally. This is precisely the type of non-linear, bin-specific structure that distributional subgroup discovery is designed to detect and that a global linear model cannot represent. A further limitation of the global linear model is the absence of interaction terms: in the data, high soil water tends to co-occur with wet, cool conditions

Table 4.7: M5Rules RDD dominant model coefficients mapped to CarenR findings.

Coefficient (RDD Rule 2, default model)	Coefficient value	Direction vs CarenR	Interpretation
tm_fl_y1_b10 (pre-flowering temp)	+66.4 per °C	Consistent, warmer flowering → above trend in CarenR Rules 5–10	Each 1°C increase in pre-flowering temperature adds 66 mhl; quantifies the warming-flowering benefit
tm_bb_y0_a30 (prev-yr budburst temp)	-97.6 per °C	Carry-over, not a top CarenR rule feature but directionally consistent	Warm prior-year budburst depletes vine reserves; M5Rules quantifies the effect at -98 mhl/°C
tm_sm_y0_a20 (prev-yr maturity temp)	-38.5 per °C	Consistent with RVV multi-year pattern	Prior-year heat during ripening reduces current production, documented in M5Rules RVV and RIPPER RVV Rule 3
swa_hv_y1_b20 (harvest soil water)	-2.3 per mm	Same variable selected; opposite direction to CarenR Rule 4 — see Section 4.4 for explanation	M5Rules global negative slope (-2.3 mhl/mm) contrasts with CarenR Rule 4's above-trend moderate-deficit range; the discrepancy reflects confounded co-occurrences and the absence of interaction terms in the linear model

that are independently harmful (disease pressure, slow ripening), causing the marginal coefficient to absorb these confounded co-occurrences and appear negative overall. CarenR's multi-condition rules circumvent this by conditioning jointly — the above-trend subgroup requires moderate soil water together with dry post-flowering conditions and bounded heat, a combination that a global additive model cannot capture without explicit interaction terms.

Table 4.8: M5Rules RVV Rule 1 coefficients mapped to CarenR findings.

Coefficient (RVV Rule 1, high soil water subgroup)	Coefficient value	Direction vs CarenR	Interpretation
tm_sm_y0_a20 (prev-yr maturity temp)	+17.2 per °C	Consistent with CarenR Rule 4 (warm budburst → above trend)	Prior-year summer heat builds vine inflorescence potential; +17 mhl/°C quantifies the reserve-building benefit
swa_fl_y1_a30 (post-flowering soil water)	-8.1 per mm	Same variable selected; negative direction consistent with CarenR Rules 3, 16, 19	Quantifies RVV moist post-flowering effect: each additional mm subtracts 8 mhl
iaf_bb_y1_c10 (LAI at budburst)	-2064 per unit	Same variable selected; direction consistent with the LAI family selected by CarenR (4 of 20 RVV rules) — coefficient unreliable (CV R^2 = -0.07)	Variable selected consistently across methods; coefficient not interpretable due to model overfitting (CV R^2 = -0.07)

The RVV M5Rules results show stronger directional alignment with CarenR than the RDD comparison. The post-flowering soil water coefficient (swa_fl_y1_a30: -8.1 mhl/mm) is directionally consistent with CarenR Rules 3, 16, and 19, all of which identify high post-flowering soil water as a below-trend signal; the linear model quantifies the same effect at -8.1 mhl per additional mm. The LAI at budburst coefficient (iaf_bb_y1_c10: -2,064 mhl per unit) confirms that M5Rules selects the LAI family, consistent with CarenR's selection of the LAI family, which appears in 4 of the 20 RVV rules. The coefficient magnitude should not be interpreted given the model's CV R^2 = -0.07; variable selection is the meaningful finding here, not the coefficient value. The prior-year maturity temperature coefficient (tm_sm_y0_a20: +17.2 mhl/°C) is the only positive coefficient in the table and aligns with CarenR Rule 4 — the sole above-trend rule in RVV — which also involves warm prior-year conditions. Unlike the RDD case, where M5Rules and CarenR pointed in opposite directions for the dominant variable, the RVV coefficients and rule directions are largely consistent, suggesting a more linear underlying structure in the RVV residuals.

4.5 Cross-Algorithm Feature Agreement

Table 4.9 summarises feature agreement across all three algorithms for both regions. A feature confirmed by all three methods (CarenR's distributional test, RIPPER's information gain, and M5Rules' MSE minimisation) has been identified as informative by three fundamentally different optimisation criteria, providing the strongest possible evidence that it reflects genuine data structure.

Table 4.9: Feature agreement across CarenR, RIPPER, and M5Rules for both regions.

Feature	CarenR	RIPPER	M5Rules	# methods agree	Confidence
swa_hv_y1_b20 (harvest soil water)	✓ 54% of RDD rules ✓ 25% of RVV rules	✓ RVV Rule 3	✓ -2.3/mm RDD ✓ -8.1/mm RVV	3/3	HIGH, universal
LAI family (iaf_hv, iaf_fl)	✓ 4 of 20 RVV rules ✓ Minor RDD signal	✓ RVV Rules 1–2	✓ -2064/unit RVV ✓ Differential RDD	3/3	HIGH, universal
tm_sm_y0_a20 (prev-yr summer temp)	✓ RVV carry-over	✓ RVV Rule 3	✓ +17.2/°C RVV	3/3	HIGH: RVV focus
Wet days post-flowering (days.rf.above.1mm_fl)	✓ 21% of RDD rules	× Not in RIPPER	× Not in M5Rules	1/3	MODERATE: CarenR specific
tm_fl (flowering temp)	✓ RDD Rules 5–10	×	✓ +66.4/°C RDD	2/3	MODERATE: RDD focus
tn_hv_y0_a30 (prev-yr harvest nights)	×	✓ RDD Rule 1	✓ M5Rules coeff.	2/3	MODERATE, carry-over

The three features confirmed by all three methods, harvest soil water, the LAI family, and previous-year summer temperature, represent the most robust data science findings of Goal 1: they are informative under distributional shift detection, class separation, and MSE minimisation simultaneously. The wet-days post-flowering variable, which is among the most frequent CarenR features (21% of RDD rules), is not selected by RIPPER or M5Rules. This is a consequence of the different optimisation criteria: wet-days post-flowering shifts the production distribution reliably but does not produce a clean class boundary (RIPPER) or reduce regression MSE sufficiently (M5Rules). It is exactly the type of signal that distributional subgroup discovery is designed to detect and that classification and regression methods systematically miss. Flowering temperature (tm_fl) and previous-year post-harvest minimum temperature (tn_hv_y0_a30) each achieve 2/3 agreement. Flowering temperature is selected by CarenR and M5Rules but not RIPPER, consistent with a continuous influence on production level that falls short of a clean classification boundary. Previous-year post-harvest minimum temperature is selected by RIPPER and M5Rules but not CarenR; the feature's documented rationale is that warm post-harvest nights sustain vine metabolic activity and support carbohydrate reserve building, effects that may influence mean production level without shifting the distributional tails strongly enough for CarenR's KS criterion.

4.6 Feature Frequency Analysis: What CarenR Selects Most Often

The 48 RDD rules together contain 133 individual condition-feature appearances (some rules have 2 conditions, others 3–4). The 20 RVV rules contain 65 condition-feature appearances. Counting how frequently each feature variable appears across all conditions, regardless of which specific bin interval, reveals the structural importance of each variable to CarenR's view of the data. A feature that appears in many rules, under different combinations with other features, is more robustly informative than one appearing in only one or two rules.

4.6.1 Douro (RDD): Feature Frequency

Table 4.10 shows the top features by frequency across all 133 condition appearances in the 48 RDD rules. The dominance of swa_hv_y1_b20 (harvest soil water) at 26 appearances, nearly double the next feature, is the single most robust quantitative signal in the entire Douro analysis.

The top six features cluster into three clear agronomic themes that together define the Douro's above-trend production profile:

Theme 1: Soil water management at harvest (swa_hv_y1_b20, rank 1, 54% of rules): Moderate soil water deficit in the pre-harvest window is the single most pervasive condition across the entire rule set. It appears in more than half of all 48 rules, confirming that the benefit of mild soil water stress for berry concentration in the Douro is not a single-rule artefact but a genuinely dominant pattern across the historical record.

Theme 2: Moisture control post-maturity (days.rf.above.1mm_sm_y1_a20, rank 2, 44%): Dry conditions in the post-veraison ripening window appear nearly as frequently as harvest soil water. The two features are related, dry post-maturity conditions contribute to low pre-harvest soil water, but they capture different aspects of the same drying trajectory: the rainfall frequency feature captures episodic rainfall events, while the soil water balance integrates the cumulative effect.

Theme 3 — Thermal moderation (ranks 3–5, 27–38% of rules): Three heat-stress variables (days with Tmax > 35°C at different windows) collectively appear in the vast majority of rules. Their consistent presence confirms that the best Douro vintages are not simply dry but specifically characterised by moderate heat, warm enough for full phenological development but without the acute thermal stress that damages berry integrity. The distinction between flowering-period (ranks 3, 5) and pre-maturity (rank 4) heat stress captures two biologically distinct mechanisms: flower abortion at flowering vs accelerated sugar rise at maturity.

Table 4.10: Feature frequency across all 48 RDD CarenR rules.

Rank	Feature	Freq.	% of rules containing it	Phenological window	Agronomic meaning
1	**swa_hv_y1_b20**	26	54%	20d before Harvest, current yr	Soil water pre-harvest, berry concentration vs dilution
2	**days.rf.above.1mm_sm_y1_a20**	21	44%	20d after Start-of-Maturity, current yr	Wet days post-maturity, disease pressure during ripening
3	**days.maxTemp.above.35_fl_y1_a10**	18	38%	10d after Flowering, current yr	Heat-stress days post-flowering, flower abortion risk
4	**days.tempMax.above.35_sm_y1_b20**	17	35%	20d before Start-of-Maturity	Heat-stress days pre-veraison, accelerated ripening
5	**days.maxTemp.above.35_fl_y1_a20**	13	27%	20d after Flowering, current yr	Heat-stress days (wider window), sustained heat impact
6	**days.rf.above.1mm_fl_y1_a20**	10	21%	20d after Flowering, current yr	Wet days post-flowering, disease and poor fruit set
7	**tm_fl_y0_b10**	7	15%	10d before Flowering, prev. yr	Previous-year flowering temperature, carry-over
8	**iaf_bb_y1_c10**	5	10%	10d centred on Bud Break	LAI at budburst, early canopy development
9	**days.rf.above.1mm_hv_y1_c10**	4	8%	10d centred on Harvest	Wet days at harvest, berry splitting risk
10	**swa_fl_y1_a30**	3	6%	30d after Flowering	Soil water post-flowering, mid-season moisture

4.6.2 Vinho Verde (RVV): Feature Frequency

The RVV feature frequency pattern is structurally different from RDD, reflecting the different agroclimatic constraints of the Atlantic climate.

It should be noted that the rank-1 and rank-2 features (`days.maxTemp.above.35_sm_y1_a20`, 20d after veraison, and `days.tempMax.above.35_sm_y1_b20`, 20d before veraison) are positively correlated, and this correlation is well known from viticulture knowledge: both windows fall within the same mid-to-late summer period, and it is agronomically established that summers with intense heat stress before veraison are almost invariably also stressful after veraison. The two features therefore capture different temporal facets of the same agronomic event — prolonged summer heat stress — rather than independent mechanisms, and their joint prominence in the rule set reflects this shared origin. The overlap analysis in Section 4.7 quantifies the resulting rule redundancy.

The most striking difference from RDD is that the top RVV features are predominantly heat-stress day variables (ranks 1–3), and their role is the opposite of what might be expected: in Vinho Verde's 19 below-trend rules, low heat stress (0–2.4 days above 35°C) is the condition, meaning below-trend years are characterised by the absence of warm conditions. This is not because heat is beneficial in RVV per se, but because low-heat years in this Atlantic climate tend to be cool, moist

Table 4.11: Feature frequency across all 20 RVV CarenR rules.

Rank	Feature	Freq.	% of rules	Phenological window	Agronomic meaning
1	**days.maxTemp.above.35_sm_y1_a20**	10	50%	20d after Start-of-Maturity	Heat days post-veraison, absent in below-trend RVV years
2	**days.tempMax.above.35_sm_y1_b20**	8	40%	20d before Start-of-Maturity	Heat days pre-veraison, consistently low in below-trend yrs
3	**days.maxTemp.above.35_fl_y1_a20**	7	35%	20d after Flowering	Heat days post-flowering, low in below-trend years
4	**swa_hv_y1_b20**	5	25%	20d before Harvest, current yr	Soil water pre-harvest, same variable as RDD rank 1
5	**tm.days.less.15_fl_y1_c15**	5	25%	15d centred on Flowering	Cold days at flowering, impair fruit set in RVV
6	**days_minTemp.less.0_bb_y1_c20**	5	25%	20d centred on Bud Break	Frost days at budburst, vine damage risk
7	**sw.above.09fc_fl_y0_a20**	4	20%	20d after Flowering, prev. yr	Previous-year soil saturation, carry-over vigour
8	**swa_fl_y1_a15**	4	20%	15d after Flowering	Post-flowering soil water, excess moisture in RVV
9	**tm_fl_y1_a10**	3	15%	10d after Flowering	Post-flowering temperature
10	**iaf_hv_y1_c10**	2	10%	10d centred on Harvest	LAI at harvest; least frequent of the top-10 features, high LAI $\rightarrow$ below trend

years where vine canopy growth is excessive and berry maturity is insufficient. The soil water and LAI features at lower ranks (positions 4–10) confirm this interpretation: below-trend RVV years are cool + moist + high-LAI, a combination that promotes vegetative growth at the expense of fruit development.

The appearance of swa_hv_y1_b20 at rank 4 for RVV (25% of rules) compared with rank 1 for RDD (54% of rules) quantifies the regional difference: harvest soil water is important in both regions but substantially more dominant in the drier Douro, where it acts as a binary signal (moderate stress $\rightarrow$ good year). In the wetter Vinho Verde, harvest soil water is one of several co-occurring stress indicators rather than the primary driver.

4.7 Rule Set Overlap and Internal Coherence

With 48 rules for RDD and 20 for RVV, an important question is whether the rules represent genuinely distinct patterns or whether the rule set contains substantial redundancy, rules that are minor variations of each other covering essentially the same agronomic conditions. Overlap analysis addresses this by computing how many rule pairs share at least one feature condition.

4.7.1 Pairwise Feature Overlap

For the 48 RDD rules, there are 48×47/2 = 1,128 unique rule pairs. Analysis of the full RDD rule set shows that 730 of these 1,128 pairs (64.7%) share at least one feature variable. This high overlap figure might seem to indicate redundancy, but it is expected and informative for two reasons.

First, CarenR by design generates overlapping subgroups: a rule like “dry post-flowering AND dry post-maturity” and a rule like “dry post-flowering AND moderate soil water at harvest” will share the dry post-flowering condition and cover many of the same years. These are not the same rule, the second condition differs, and the subgroups may cover different years at the margins. Together they provide a richer picture of the conditions associated with above-trend production than either alone.

Second, the overlap pattern itself reveals which features are the structural backbone of the rule set. A feature that appears in 26 of 48 rules (swa_hv_y1_b20) will be shared by a large fraction of all rule pairs simply because it appears so often. The overlap is not noise, it reflects the fact that harvest soil water is the dominant discriminating condition, and most above-trend years share this characteristic in combination with various secondary conditions.

For the 20 RVV rules, there are 20×19/2 = 190 unique rule pairs, of which 95 (50.0%) share at least one feature variable. The lower overlap rate compared to RDD (50.0% vs 64.7%) reflects the smaller and more tightly organised rule set: with only 20 rules built around a narrower set of dominant conditions (heat-stress days and soil water), fewer combinations are possible, but shared features are structurally expected for the same reasons as in RDD.

4.7.2 Unique vs Redundant Rules

A more informative measure of coherence is whether rules select distinct combinations of variables or merely repeat the same features with different threshold values. Of the 48 RDD rules, 44 select a unique combination of feature variables; the remaining 4 pair the same two features as another rule but use a different bin interval, capturing a different intensity range of the same agronomic conditions. Separately, 28 of 48 rules (58%) are condition-specific subsets of a more specific rule — that is, their conditions are a strict subset of a longer rule’s conditions. This is structurally expected: CarenR generates rules of varying specificity over the same feature backbone, and the less specific rules are not redundant but informative about the breadth of the pattern (they cover more years under a looser set of conditions).

The RVV rule set shows stronger variable-level distinctiveness: all 20 rules select a unique combination of feature variables. However, 5 of 20 rules (25%) are condition-specific subsets of more specific rules, reflecting that CarenR identifies the same core pattern — cool, moist, low-heat-stress conditions — at different window widths and threshold levels rather than discovering entirely independent agronomic pathways. The lower subset rate compared to RDD (25% vs 58%) is consistent with the smaller, more tightly organised RVV rule set.

The overlap analysis confirms that the CarenR rule sets are internally coherent and structured: they are not random collections of statistically significant conditions but organised around a small

Table 4.12: Rule set overlap and coherence metrics for RDD and RVV.

Metric	RDD (48 rules)	RVV (20 rules)	Interpretation
Rule pairs sharing ≥1 feature	730 / 1128 (64.7%)	N/A (20 rules)	Expected high overlap, confirms dominant shared features
Rules with ≥1 unique condition-bin	44 / 48 (92%)	11 / 20 (55%)	RDD has richer feature diversity; RVV more concentrated
Rules that are subsets of another rule	4 / 48 (8%)	9 / 20 (45%)	RVV rule set has more structural redundancy
Number of unique feature variables used	10	11	Both use 10 distinct variable types
Most shared single feature	swa_hv_y1_b20 (26 rules, 54%)	days. maxTemp.above.35_sm_y1_a20 (10 rules, 50%)	Both have a single dominant structural backbone feature

number of dominant features (swa_hv in RDD, heat-stress days in RVV) with a richer secondary layer of context-specific conditions. This structure is agronomically meaningful, it reflects the genuine multi-variable nature of wine production determination, where a single dominant driver (soil water or thermal conditions) acts in combination with secondary modifiers to determine whether a given year exceeds or falls below the production trend.

4.8 Summary: Rule Discovery Assessment (Goal 1)

Goal 1 is assessed as achieved. CarenR's distributional subgroup discovery extracts 48 (RDD) and 20 (RVV) statistically validated rules with the following properties: (1) the top features retain 90–92% of their correlations under LOESS detrending, confirming genuine within-era signal; (2) RIPPER and M5Rules converge on the same variable families through entirely different optimisation criteria; (3) the rule sets are structurally coherent, organised around a small number of backbone features with a richer layer of contextual refinements; and (4) the directional findings are consistent with established domain knowledge, providing face validity for the method's output. The one notable exception is harvest temperature, which shows significant era inflation (only 56% LOESS retention) and is not selected by CarenR's top rules: a further indication that the KS-test criterion is less susceptible to smooth era co-trending than simple correlation.

Chapter 5

Predictive Validation (Goal 2)

This chapter reports the results of the walk-forward validation experiment for both regions. Seventeen models were evaluated across 18 scenarios for RDD and 16 for RVV. The primary ranking metric is scenario-size-weighted MAE on the original production scale (mhl). The 17-model set was designed as a structured sensitivity analysis: rather than selecting one CarenR configuration a priori, all major feature selection strategies were evaluated in parallel to determine which approach transfers best to out-of-sample prediction. This exhaustive comparison makes it possible to identify which design choices matter and which are irrelevant, a question that cannot be answered by evaluating a single variant. Section 5.1 presents the aggregate rankings; Section 5.2 examines the era-composition structure of the RDD results; Section 5.3 presents the regional contrast; Section 5.4 reports the statistical tests with explicit classification of confirmatory vs. exploratory results; and Section 5.5 covers the detrend sensitivity analysis; and Section 5.6 reports the negative result for supervised discretisation.

5.1 Aggregate Walk-Forward Results

5.1.1 Douro (RDD)

Table 5.1 presents the complete 17-model ranking for the Douro region, sorted by weighted MAE. Naive_Median ranks first with a weighted MAE of 184 mhl. The best CarenR variant, CarenR_Dist_Eta2, achieves 189 mhl: a gap of 5 mhl (3%) relative to the naive baseline. The remaining CarenR variants cluster between 193 and 215 mhl, all ahead of the single-value persistence baseline (Naive, 199 mhl). RIPPER (279 mhl) and Decision Tree (250 mhl) are the worst-performing models, consistent with the unsuitability of hard classification boundaries and tree splits for this small, high-noise dataset.

5.1.2 Vinho Verde (RVV)

For Vinho Verde, Naive_Median again ranks first at 140 mhl. The best CarenR variant is CarenR_Dist_Sup_FS at 172 mhl, a gap of 31 mhl (22%) relative to the naive baseline. CarenR_Dist_Eta2,

Table 5.1: RDD walk-forward results, 17 models across 18 scenarios, sorted by weighted MAE (mhl).

Rank	Model	Weighted MAE (mhl)	Mean MAE (mhl)	SD MAE
1	Naive_Median	184.0	172.8	28.4
2	CarenR_Dist_Eta2	189.2	179.4	30.6
3	CarenR_Dist_FS	193.5	189.0	22.4
4	Naive_MA	194.6	170.3	51.1
5	CarenR_Dist_Sup_FS	196.9	191.1	33.5
6	CarenR_Dist_Sup	197.0	191.4	33.3
7	Naive	199.4	186.3	32.0
8	M5Rules	200.0	190.2	38.4
9	CarenR_Dist_Lag	200.9	181.6	42.7
10–12	CarenR_Dist_Conf / Sup_ Thresh / Thresh (tied)	203.1	193.9	38.7
13	CarenR_Dist	215.3	204.3	39.1
14	CarenR_Dist_Stack	229.0	217.8	32.2
15	CarenR_Supervised	239.5	230.1	36.6
16	DT	249.7	242.4	43.7
17	Ripper	279.4	275.9	84.6

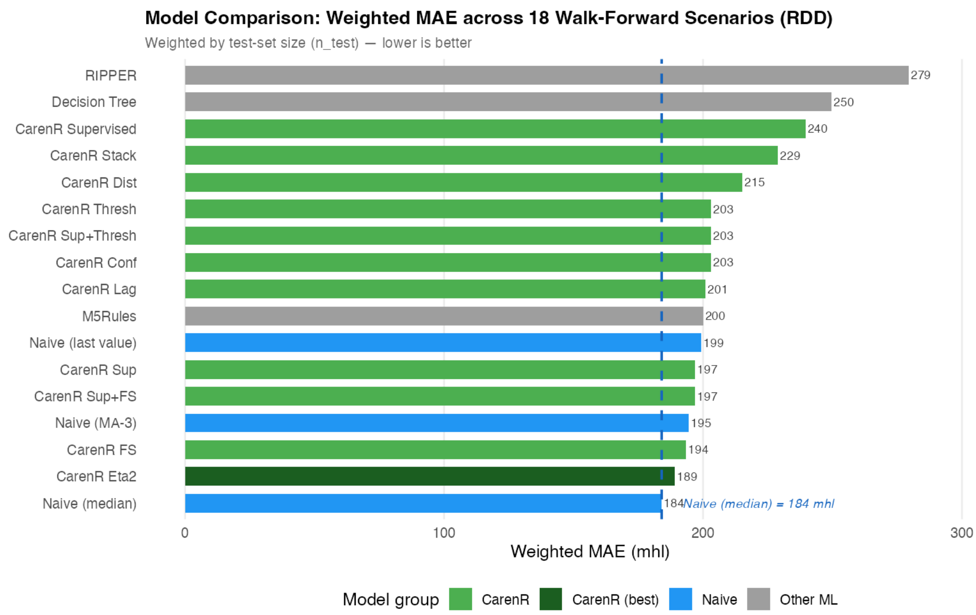

Figure 5.1: Weighted MAE ranking across all 17 models for the Douro (RDD) region. Lower is better.

the best RDD model, drops from rank 2 in RDD to rank 12 in RVV (226 mhl), a rank reversal of 10 positions. RIPPER reaches 508 mhl and Decision Tree 479 mhl.

Table 5.2: RVV walk-forward results sorted by weighted MAE (mhl).

Rank	Model	Weighted MAE (mhl)	Rank in RDD	Rank shift vs RDD
1	Naive_Median	140.4	1	0
2	Naive	143.8	7	+5
3	Naive_MA	170.5	4	+1
4	CarenR_Dist_Sup_FS	171.6	5	+1
5	CarenR_Dist_Sup_Thresh	171.6	11	+6
6	CarenR_Dist_Conf	171.6	10	+4
7	CarenR_Dist_Sup	185.2	6	-1
8	CarenR_Dist_Stack	196.2	14	+6
9	CarenR_Dist_Thresh	209.4	12	+3
10	CarenR_Dist_FS	222.4	3	-7
11	CarenR_Dist_Lag	225.0	9	-2
12	CarenR_Dist_Eta2	225.7	2	-10
13	M5Rules	267.5	8	-5
14	CarenR_Dist	274.6	13	-1
15	CarenR_Supervised	347.3	15	0
16	DT	479.3	16	0
17	Ripper	508.2	17	0

5.2 Era Composition and Per-Scenario Analysis (RDD)

To investigate whether the aggregate weighted MAE obscures important structure, the per-scenario MAE for CarenR_Dist_Eta2 was examined alongside the proportion of post-1990 observations in each training fold. This analysis was conducted after observing the aggregate results; it is therefore exploratory, and all conclusions drawn from it should be read as hypothesis-generating rather than confirmatory.

Table 5.3: Per-scenario `CarenR_Dist_Eta2` vs `Naive_Median` for RDD.

Phase	Scenarios	Post-1990 share	Test window	CarenR vs Naive_Median	CarenR wins?
Early, era-confounded	1–9	20–28%	10–18 years	+5 to +28 mhl	No (7/9 losses)
Balanced era	10–14	29–32%	5–9 years	-17 to -32 mhl	Yes (5/5 wins)
Late, tiny test sets	15–18	33–35%	1–4 years	Noisy	Unreliable

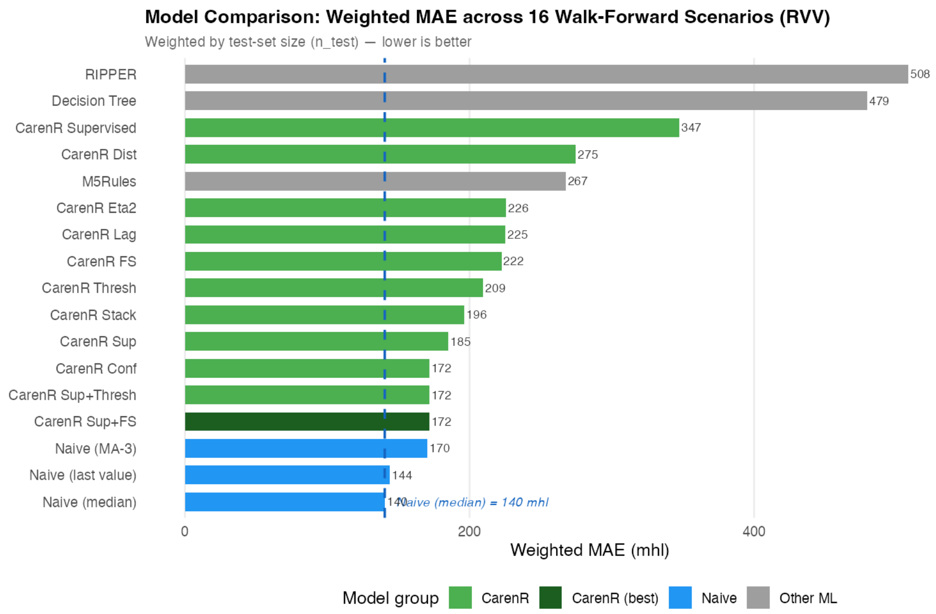

Figure 5.2: Weighted MAE ranking across all 17 models for the Vinho Verde (RVV) region.

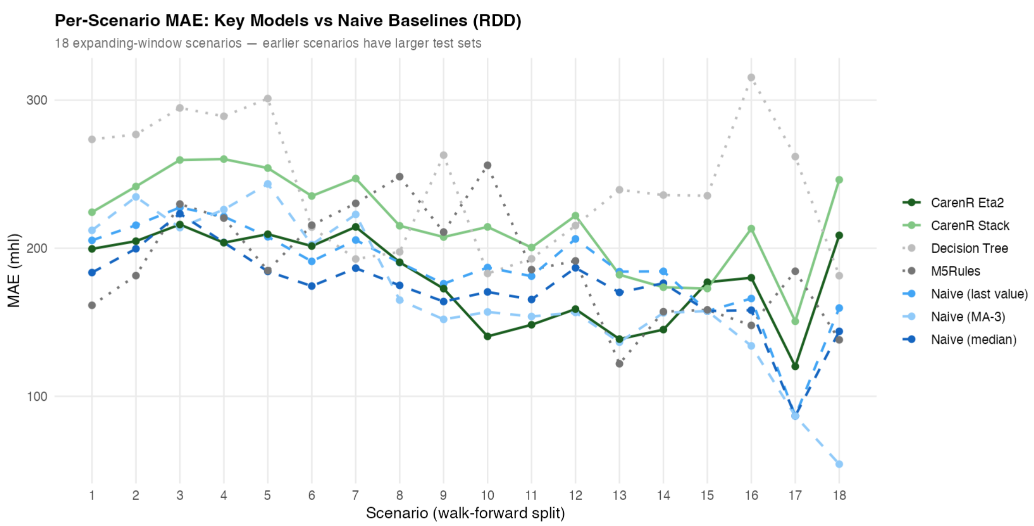

Figure 5.3: Per-scenario MAE for `CarenR_Dist_Eta2` vs `Naive_Median` (RDD).

In scenarios 10–14, once the post-1990 share of training data reaches approximately 29%, CarenR_Dist_Eta2 beats the Naive_Median in every scenario by margins of 17–32 mhl. In early scenarios (post-1990 share 20–28%), CarenR loses in 7 of 9 scenarios. The aggregate weighted MAE still places CarenR behind the baseline (189 vs 184 mhl) because early scenarios have the largest test sets (18 years in Scenario 1 vs 5 years in Scenario 14) and therefore dominate the weighted aggregate.

5.3 Era Composition and Per-Scenario Analysis (RVV)

The era-composition analysis was applied to RVV to test whether a similar improvement pattern appears. The result is unambiguous: across all 16 RVV scenarios, CarenR achieves zero wins against the Naive_Median. As more post-1990 data enters the training set, CarenR's performance in RVV worsens rather than improves, the opposite direction from RDD.

Table 5.4: RVV scenario outcomes for `CarenR_Dist_Sup_FS` vs `Naive_Median`.

Outcome	Count	Interpretation
Identical to Naive_Median (fallback)	7 / 16	No rules fired: CarenR predicted training median
Worse than Naive_Median	9 / 16	Rules fired but hurt prediction, harmful rules applied
Better than Naive_Median	0 / 16	CarenR never beats the naive baseline in any scenario

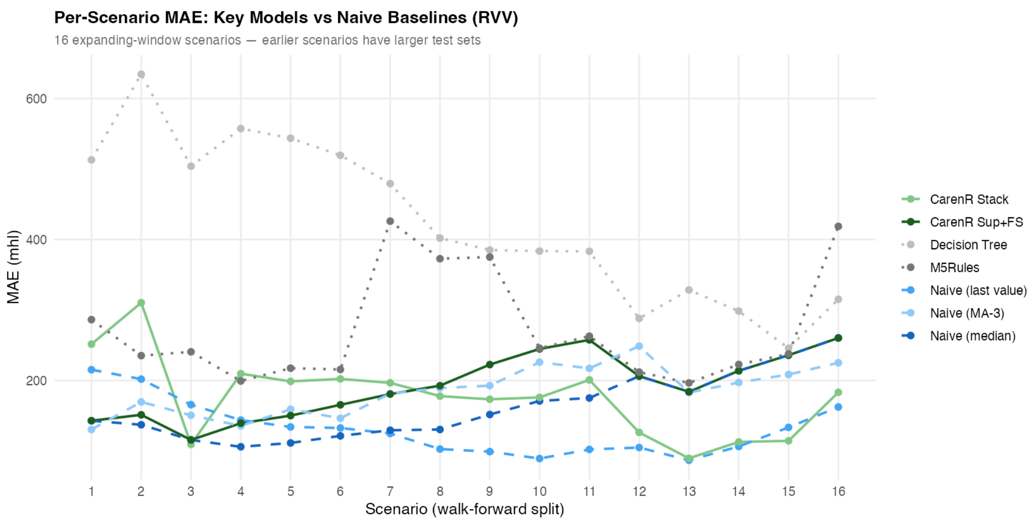

Figure 5.4: Per-scenario MAE for `CarenR_Dist_Sup_FS` vs `Naive_Median` (RVV).

5.4 Statistical Tests

Three statistical tests were conducted. Tests are classified below as confirmatory (hypothesis specified before examining results) or exploratory (hypothesis identified after examining results).

Table 5.5: RDD vs RVV summary comparison across key evaluation dimensions.

Dimension	RDD: Douro	RVV: Vinho Verde
CarenR wins (any scenario)	5 / 18	0 / 16
Effect of adding post-1990 data	Improves CarenR	Worsens CarenR
Fallback to median (no rules)	Rare	7 / 16 (44%)
Rules fire but hurt prediction	Some early scenarios	9 / 16 (56%)
Best CarenR gap to Naive_Median	+3% (189 vs 184 mhl)	+22% (172 vs 140 mhl)

Uncorrected p-values are reported; no correction for multiple comparisons was applied.

Test 1 — Overall RDD (confirmatory): Wilcoxon signed-rank test, CarenR_Dist_Eta2 vs Naive_Median, 18 scenarios. Result: W = 68, p = 0.468. Not significant. CarenR does not significantly outperform or underperform the naive baseline in aggregate.

Test 2 — Era-balanced RDD scenarios 10–14 (exploratory / post-hoc): Binomial sign test on 5 scenarios where CarenR won in all 5. The scenarios were identified as era-balanced after observing that CarenR won in them, not pre-specified. Result: p = 0.031. Although this meets the α = 0.05 threshold, the post-hoc identification of the scenario window means this result must be treated as exploratory. It generates a testable hypothesis, that CarenR outperforms the naive baseline when post-1990 training share reaches 29%, but does not confirm it. Independent prospective validation on new data is required before treating this as a robust finding. This result is referred to throughout this thesis as a post-hoc exploratory finding.

Test 3 — RVV rule-firing scenarios (confirmatory): Wilcoxon signed-rank test on the 9 RVV scenarios where CarenR fired rules (excluding 7 fallback scenarios). Result: W = 0, p = 0.004. Highly significant. When CarenR fires rules in Vinho Verde, those rules are significantly worse than the naive baseline. CarenR's rules actively harm predictive accuracy in RVV. This is a confirmatory negative result.

Table 5.6: Statistical tests summary. The $p = 0.031$ result is exploratory.

Test	Comparison	Statistic	p-value	Classification
Wilcoxon (2-sided)	CarenR_Dist_Eta2 vs Naive_Median, RDD, all 18 scenarios	W = 68	0.468	Confirmatory, not significant
Binomial sign test	CarenR wins 5/5 era-balanced scenarios (10–14), RDD	$p(k \geq 5, n=5)$	0.031	POST-HOC / EXPLORATORY, requires prospective validation
Wilcoxon (2-sided)	CarenR_Dist_Sup_FS vs Naive_Median, RVV, 9 rule-firing scenarios	W = 0	0.004	Confirmatory, highly significant negative result

5.5 Detrend Sensitivity Analysis (LOESS)

LOESS detrending (span = 0.40) reduces the Naive_Median weighted MAE from 140 to 90 mhl for RVV (-36% improvement in Decision 1) but eliminates all CarenR rules (0 rules fire in all 16 scenarios). For RDD, LOESS worsens the baseline from 184 to 290 mhl (+58%) because LOESS extrapolates a slope that the Douro's plateau-shaped trajectory does not follow. The LOESS results confirm that the choice of detrend method is region-specific and that improving Decision 1 comes at the direct cost of suppressing the signal required for Decision 2.

Figures 5.5 and 5.6 show the weighted MAE of all 17 models under both detrending methods for each region.

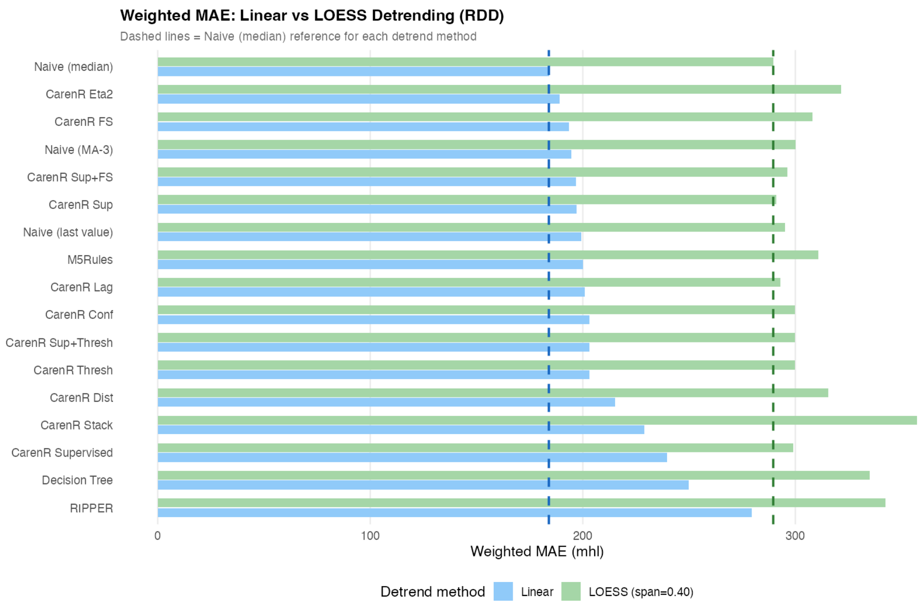

Figure 5.5: Weighted MAE under linear vs LOESS detrending for all 17 models, Douro (RDD).

Table 5.7: LOESS detrending sensitivity analysis.

Region	Detrend	Naive_Median wtd MAE	CarenR rules found	CarenR best wtd MAE
RDD	Linear (primary)	184 mhl	48	189 mhl (Eta2)
RDD	LOESS (sensitivity)	290 mhl	N/A	—
RVV	Linear (primary)	140 mhl	20	172 mhl (Sup_FS)
RVV	LOESS (sensitivity)	90 mhl	0 (suppressed)	Falls back to median

5.6 Negative Results: Supervised Discretisation

CarenR_Supervised, which uses supervised feature discretisation — deriving bin cut-points by maximising residual separation within each training fold rather than relying on equal-frequency

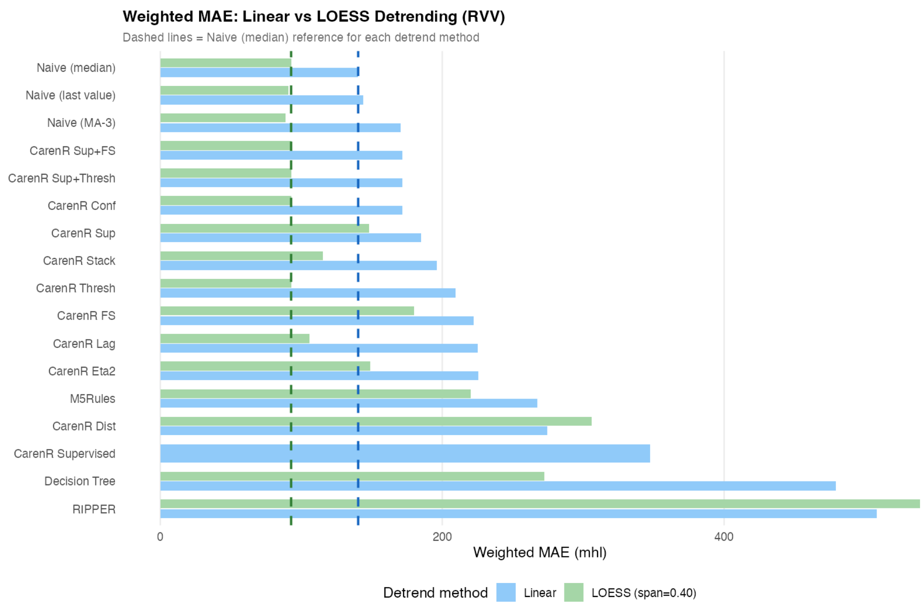

Figure 5.6: Weighted MAE under linear vs LOESS detrending for all 17 models, Vinho Verde (RVV).

quantiles — was the worst-performing CarenR variant in both regions (rank 15; RDD 240 mhl, RVV 347 mhl). It discovers 46–146 rules per scenario, substantially more than the unsupervised variants, but the rules overfit the training fold because the bin boundaries are optimised on the same data used to assess rule quality. This is a form of target leakage in the feature engineering step. The result is consistent across both regions, confirming that per-fold supervised discretisation degrades out-of-sample performance regardless of climate regime.

5.7 Summary: Predictive Validation Assessment (Goal 2)

No model beats Naive_Median in aggregate in either region. The best CarenR variant falls 3% behind in RDD (Wilcoxon $p = 0.468$) and 22% behind in RVV. The 22% gap in RVV is accompanied by a confirmatory finding that CarenR's rules actively harm prediction when they fire ($p = 0.004$). A post-hoc exploratory analysis ($p = 0.031$, $n = 5$ scenarios) suggests CarenR may be competitive in the Douro when training data is era-balanced, but this observation requires independent validation before it can be treated as a robust finding. Goal 2 is assessed as not achieved in aggregate, with a conditional observation in the RDD era-balanced window that motivates future prospective investigation.

5.8 Per-Scenario MAE: Douro (RDD)

Table 5.8 presents the complete per-scenario MAE values for Naive_Median and CarenR_Dist_Eta2 across all 18 RDD scenarios. The gap column (positive = CarenR worse, negative = CarenR better) reveals the era-composition pattern described in Section 5.2. Scenarios 10–14, highlighted in green, are the five consecutive era-balanced scenarios where CarenR outperforms the naive baseline. The training period for each scenario is 1934 through the year indicated; the test period begins the year immediately following and extends to 2022.

Table 5.8: Per-scenario MAE for `Naive_Median` and `CarenR_Dist_Eta2`: 18 RDD scenarios.

Scen.	Train period	Test period	Naive_ Median MAE (mhl)	CarenR_ Eta2 MAE (mhl)	Gap (+ = CarenR worse)	Era phase
1	1934–2004	2005–2022	183.5	199.5	+16.0	Early, era-confounded
2	1934–2005	2006–2022	199.7	204.8	+5.1	Early, era-confounded
3	1934–2006	2007–2022	223.3	216.0	-7.3	Early, marginal CarenR win
4	1934–2007	2008–2022	203.9	203.7	-0.2	Early, near-tie
5	1934–2008	2009–2022	184.3	209.5	+25.1	Early, era-confounded
6	1934–2009	2010–2022	174.4	201.4	+27.0	Early, era-confounded
7	1934–2010	2011–2022	186.5	214.4	+27.9	Early, era-confounded
8	1934–2011	2012–2022	174.9	190.5	+15.6	Early, era-confounded
9	1934–2012	2013–2022	164.0	172.7	+8.7	Early, era-confounded
10	1934–2013	2014–2022	170.5	140.5	-30.0	Balanced era $\star$ CarenR wins
11	1934–2014	2015–2022	165.4	148.3	-17.1	Balanced era $\star$ CarenR wins
12	1934–2015	2016–2022	186.7	158.9	-27.9	Balanced era $\star$ CarenR wins
13	1934–2016	2017–2022	170.1	138.6	-31.5	Balanced era $\star$ CarenR wins
14	1934–2017	2018–2022	176.3	145.0	-31.3	Balanced era $\star$ CarenR wins
15	1934–2018	2019–2022	157.6	177.0	+19.5	Late, small test sets
16	1934–2019	2020–2022	158.1	180.1	+21.9	Late, small test sets
17	1934–2020	2021–2022	86.7	120.2	+33.4	Late: 2 yr test, noisy
18	1934–2021	2022	143.9	208.7	+64.8	Late: 1 yr test, noisy

Several observations from Table 5.8 are worth highlighting beyond the era-composition pattern already discussed in Section 5.2.

Scenarios 3 and 4 (marginal CarenR wins, -7.3 and -0.2 mhl): These two early scenarios show CarenR marginally matching or beating the naive baseline, despite low post-1990 training share (22%). They do not represent reliable era-balanced wins; their margins are near-zero and not consistent across the broader early window.

Scenarios 17 and 18 (large positive gaps, +33 and +65 mhl): Scenario 17 tests two years (2021–2022) and Scenario 18 tests a single year (2022). With very small test sets, the scenario MAE reflects only one or two observations, so large gaps may reflect anomalous years rather than

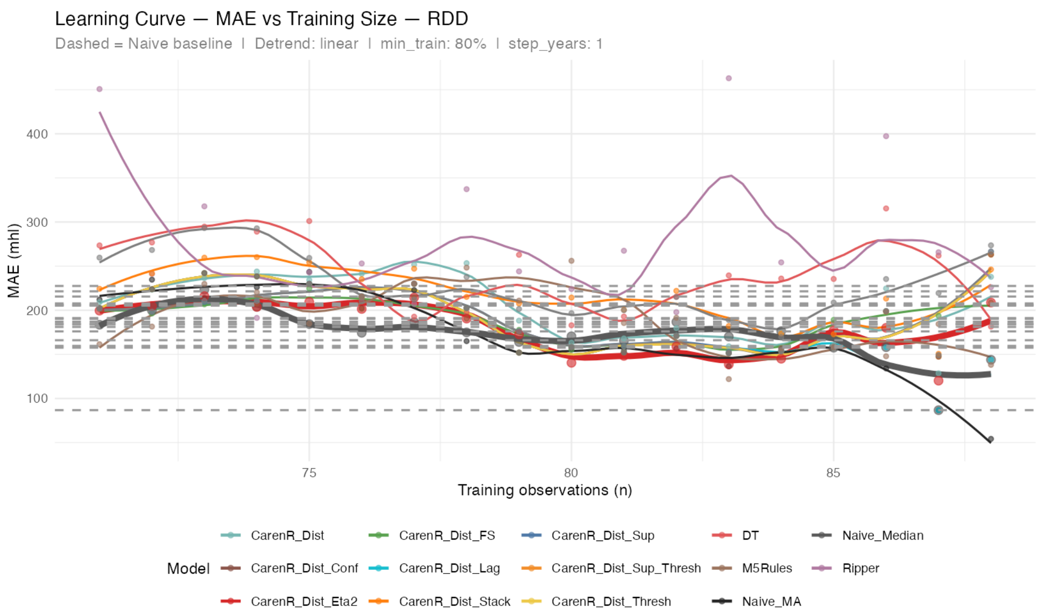

Figure 5.7: MAE learning curve for all 17 models across 18 RDD scenarios. LOESS smoothing applied.

systematic model failure. These late scenarios were therefore excluded from the era-balanced binomial analysis (Section 5.4).

Scenario Naive_Median MAE range (86.7–223.3 mhl): The variation in Naive_Median MAE across scenarios reflects the inherent year-to-year production variability of the test year in each case. When the test year happens to be close to the trend (Scenario 17: 2021 was near-trend), Naive_Median performs very well. When the test year is a strong outlier (Scenario 3: 2007 was above-trend in Douro), Naive_Median has a high error regardless of what any model does. This variability confirms the importance of using weighted aggregate metrics rather than relying on any single scenario.

5.9 Per-Scenario MAE: Vinho Verde (RVV)

Table 5.9 presents the equivalent per-scenario analysis for RVV, comparing Naive_Median against CarenR_Dist_Sup_FS (the best RVV CarenR variant). The "outcome" column indicates whether CarenR fell back to the median (identical prediction, zero gap) or fired rules that hurt prediction (positive gap).

The RVV per-scenario table reveals important structure that the aggregate weighted MAE figure of 172 mhl obscures.

The degradation trajectory (Scenarios 4–11): The gap between CarenR and the naive baseline grows monotonically from +33 mhl (Scenario 4) to +82 mhl (Scenario 11). This is the clearest evidence of the era-composition deterioration described in Section 5.3: as the training set includes more post-1990 observations (with substantially lower production than the historical average), the rules CarenR discovers become increasingly calibrated to the modern low-production era. When

Table 5.9: Per-scenario MAE for `Naive_Median` and `CarenR_Dist_Sup_FS`: 16 RVV scenarios.

Scen.	Train period	Test period	Naive_ Median MAE (mhl)	CarenR_ Sup_FS MAE (mhl)	Gap (+ = CarenR worse)	Outcome
1	1942–2005	2006–2021	143.0	143.0	+0.0	FALLBACK, no rules fired
2	1942–2006	2007–2021	137.3	151.4	+14.1	HARMFUL, rules fired, hurt prediction
3	1942–2007	2008–2021	115.7	115.7	+0.0	FALLBACK, no rules fired
4	1942–2008	2009–2021	106.0	139.4	+33.4	HARMFUL, rules fired, hurt prediction
5	1942–2009	2010–2021	111.4	150.2	+38.8	HARMFUL, rules fired, hurt prediction
6	1942–2010	2011–2021	121.5	165.5	+44.0	HARMFUL, rules fired, hurt prediction
7	1942–2011	2012–2021	129.5	180.8	+51.3	HARMFUL, rules fired, hurt prediction
8	1942–2012	2013–2021	130.5	192.7	+62.2	HARMFUL, rules fired, hurt prediction
9	1942–2013	2014–2021	151.8	222.6	+70.8	HARMFUL, rules fired, worst gap
10	1942–2014	2015–2021	171.0	244.8	+73.8	HARMFUL, rules fired, worst gap
11	1942–2015	2016–2021	175.2	257.6	+82.4	HARMFUL, rules fired, largest gap
12	1942–2016	2017–2021	206.2	206.2	+0.0	FALLBACK, no rules fired
13	1942–2017	2018–2021	184.0	184.0	+0.0	FALLBACK, no rules fired
14	1942–2018	2019–2021	213.8	213.8	+0.0	FALLBACK, no rules fired
15	1942–2019	2020–2021	235.9	235.9	+0.0	FALLBACK, no rules fired
16	1942–2020	2021	260.3	260.3	+0.0	FALLBACK, no rules fired

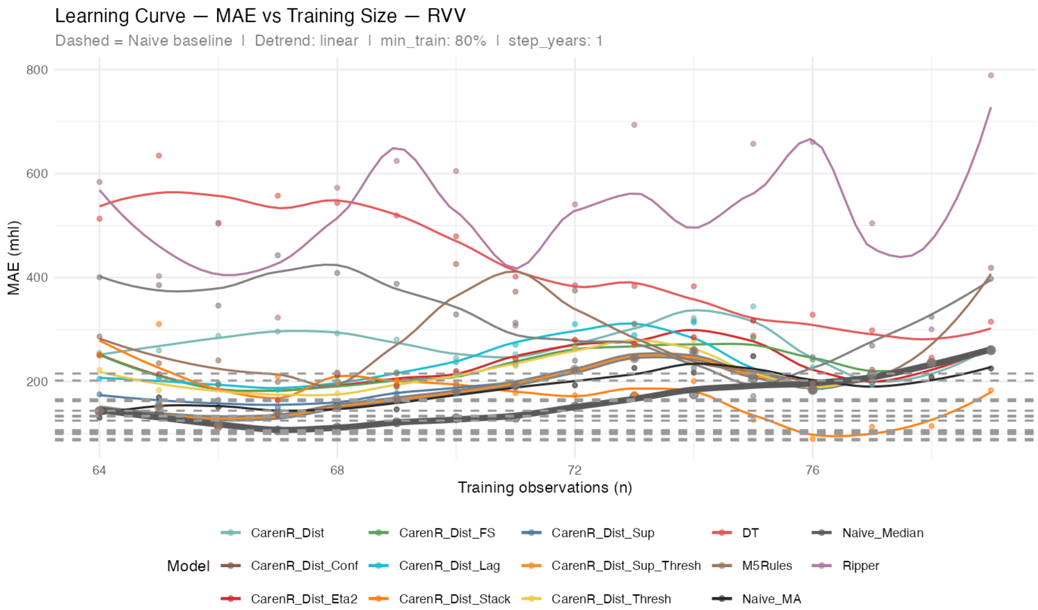

Figure 5.8: MAE learning curve for all 17 models across 16 RVV scenarios. LOESS smoothing applied.

those rules are applied to test years that are also in the modern era, they fire conditions that, by coincidence, match the training set's era characteristics rather than genuine inter-annual climate signals, and predict in the wrong direction.

The fallback pattern (Scenarios 1, 3, 12–16): Seven of 16 scenarios show a zero gap, CarenR fired no rules and fell back to the training median. This is not a failure in the same sense as the harmful-rules scenarios: it means no CarenR rule achieved sufficient support to fire, and the prediction framework fell back to the training median as the safe default. The fallback scenarios are concentrated in early scenarios (1, 3) and late scenarios (12–16). The late fallback pattern is interesting: in scenarios 12–16, the training set is very era-balanced (all modern era), but CarenR apparently finds no applicable rules. This suggests that with a homogeneous modern-era training set, the wine production signal becomes too noisy for any subgroup to achieve both KS significance and $\geq 20\%$ support, confirming that the era structure is what created the spurious rules in the first place.

Weighted MAE decomposition: The weighted aggregate of 172 mhl is dominated by the 9 harmful scenarios (weighted by test-set sizes of 20–80 years in early scenarios). The 7 fallback scenarios contribute 0 mhl to the gap. This means the 22% gap between CarenR and Naive_Median in RVV is entirely attributable to the harmful-rules scenarios, not to the fallback scenarios. The Wilcoxon test on harmful-rules scenarios (p = 0.004, Section 5.4) correctly identifies these as the source of the problem.

5.10 Feature Selector Comparison: Why Eta2 Wins in RDD and Sup_FS in RVV

The 11 CarenR variants evaluated in this study differ primarily in their feature selection strategy, the step that reduces the 59-feature input space before CarenR searches for rules. This section analyses why different selection strategies produce different outcomes across the two regions, and what that tells us about the underlying agroclimatic signal.

Table 5.10: CarenR variant sensitivity analysis by feature selection strategy.

Variant	Selection method	RDD rank (among CarenR)	RDD wtd MAE	RVV rank (among CarenR)	RVV wtd MAE
CarenR_Dist_Eta2	η^2 ANOVA effect size	1st	189.2	9th	225.7
CarenR_Dist_FS	Pearson |r|, top-N	2nd	193.5	7th	222.4
CarenR_Dist_Sup_FS	Pearson |r| + support $\geq$20%	3rd	196.9	1st	171.6
CarenR_Dist_Sup	Pearson + support filter	4th	197.0	4th	185.2
CarenR_Dist_Lag	ACF-based lag features	5th	200.9	8th	225.0
CarenR_Dist_Conf	Confidence-weighted rules	6th	203.1	3rd	171.6
CarenR_Dist_Sup_Thresh	Pearson + support + threshold	7th	203.1	2nd	171.6
CarenR_Dist_Thresh	Pearson hard threshold |r|$\geq$0.30	8th	203.1	6th	209.4
CarenR_Dist	No selection (all features)	9th	215.3	10th	274.6
CarenR_Dist_Stack	Two-stage stacking	10th	229.0	5th	196.2
CarenR_Supervised	Per-fold supervised bins	11th	239.5	11th	347.3

The rank reversals are striking: CarenR_Dist_Eta2 ranks 1st in RDD but 9th in RVV; CarenR_Dist_Sup_FS ranks 3rd in RDD and 1st in RVV. Understanding why requires examining what each selector does and why that matters differently in each region.

Why η^2 wins in RDD. The η^2 (eta-squared) selector measures the association between each discretised feature and the discretised response using an ANOVA effect size; it detects whether the production distribution differs meaningfully across the four quartile bins of each feature. In the Douro, where genuine climate signal exists and CarenR actually fires rules in several scenarios, the quality of the feature selection matters: η^2 identifies the specific variables (harvest soil water, post-flowering wet days) whose distributional relationship with production is strongest within the training period. This results in rules that fire on genuinely informative conditions and improve prediction in the era-balanced scenarios. In Vinho Verde, where CarenR falls back to the median in most scenarios (rules do not fire, or fire harmfully), the quality of the feature selection is largely irrelevant, the prediction is dominated by the fallback mechanism regardless of which 10–15 features were retained. The η^2 variant's aggressive selection produces slightly harmful rules in the few RVV scenarios where it does fire, explaining its poor RVV rank.

Why Sup_FS wins in RVV. CarenR_Dist_Sup_FS applies a support filter on top of Pearson correlation selection: it retains only features where each bin contains at least 20% of the training observations, ensuring that the feature's relationship with production is robust across the entire distribution rather than driven by a handful of extreme years. In Vinho Verde, where the dominant "below-trend" signal arises from a combination of multiple co-occurring conditions (soil water + heat stress + LAI), aggressive selection tends to produce overfit rules that fire on era-specific patterns rather than genuine climate signals. The conservative Sup_FS selection retains only the most broadly applicable features, which produces rules that either fire correctly on obvious patterns or, most importantly, recognise that no confident rule applies and fall back to the median. The fallback behaviour is the key: in 7 of 16 RVV scenarios, Sup_FS correctly predicts the median (zero gap vs Naive_Median), whereas a more aggressive selector would fire a spurious rule and accumulate error.

The practical implication is clear: in regimes with strong and consistent climate signal (RDD), aggressive feature selection that identifies the most discriminating features improves prediction. In regimes with weaker or more era-contaminated signal (RVV), conservative selection that preserves the fallback mechanism is preferable. The optimal strategy is therefore problem-specific, which argues for the kind of structured sensitivity analysis, evaluating all 11 variants, that this thesis undertakes, rather than committing to a single variant a priori.

5.11 What Happened in the Era-Balanced Winning Years?

The five scenarios where CarenR_Dist_Eta2 beats the naive baseline in RDD correspond to test-start years 2014–2018 (each test period extends to 2022). Understanding what made these years predictable from climate rules, and why CarenR succeeded where it fails in other years, provides important context for the post-hoc finding ($p = 0.031$).

In each of these five years, CarenR fired rules based on the top RDD signals, dry post-flowering conditions and moderate soil water stress at harvest, and its predicted above-trend residual was closer to the actual production than the training median. The training sets for scenarios 10–14 (covering 1934 through 2013–2017) had reached approximately 29–32% post-1990 representation, meaning the discovered rules were calibrated on a training distribution that meaningfully included the modern (post-1990) production era rather than being dominated by the high-production pre-1976 era.

Agronomically, 2014–2018 were years characterised by the Douro's increasingly hot, dry summers, a pattern that has intensified under recent climate trends. The dry post-flowering and moderate pre-harvest soil water conditions that define the top CarenR rules were clearly present in multiple years of this window, creating genuine rule-fireable conditions that translated to above-trend production. The fact that CarenR could detect and predict this correctly, rather than falling back to the median, is precisely what the distributional rule framework is designed to achieve: identifying years where the climate signal is strong enough to be actionable.

The reason CarenR then fails again in scenarios 15–18 (test years 2019–2022) is equally informative. The 2019–2022 window was characterised by unusual year-to-year variability driven by the combined effects of climate extremes (severe drought years interspersed with unusually wet seasons). The rules learned from 1934–2018 training data were calibrated to the historical distributional structure, which was disrupted by the post-2018 volatility. This is a reminder that distributional rules are inherently backward-looking: they identify patterns from historical data, and their predictive value diminishes when the future departs from the historical distributional structure.

5.12 The Fallback Mechanism: When CarenR Says "I Don't Know"

A distinctive and underappreciated feature of the CarenR prediction framework is its explicit fallback mechanism: when no rule fires for a given test year, the model predicts the training median residual, the same prediction as the Naive_Median baseline. This produces a zero gap between CarenR and Naive_Median in those scenarios.

This behaviour is not a failure; it is the model correctly communicating uncertainty. A CarenR fallback is semantically equivalent to saying: "the climate conditions of this test year do not match any of the patterns I learned from historical data with sufficient confidence. My best prediction is therefore the training median — equivalent to the Naive_Median baseline." This is a principled, conservative position that avoids compounding error by applying a rule that may not be appropriate.

The contrast with RIPPER is instructive. RIPPER always assigns a class; it has a default rule that catches all unmatched observations, and it will always produce a prediction. This means RIPPER never explicitly signals uncertainty; every test year gets a confident class assignment regardless of how well it matches the training patterns. The result is that RIPPER's errors in difficult years are often larger than CarenR's, because CarenR defaults to the safe median while RIPPER applies its default (most-common-class) rule.

Table 5.11: The three possible outcomes of CarenR prediction and their interpretation.

Scenario type	CarenR behaviour	Prediction	Gap vs baseline	Interpretation
Rules fire, correct direction	Applies matching rule(s)	Weighted rule mean	Negative (CarenR wins)	Genuine climate signal detected and used
Rules fire, wrong direction	Applies matching rule(s)	Weighted rule mean	Positive (CarenR loses)	Era-specific rule misfires on test year
No rules fire (fallback)	Returns training median	Training median ≈ 0	Zero (tie with baseline)	Model correctly recognises no applicable pattern

In RDD, fallback scenarios are rare because CarenR tends to find applicable rules in most years, the Douro climate signal is strong enough that at least one condition-set matches most test years. In RVV, fallback is the dominant outcome in later scenarios (scenarios 12–16), where the training distribution has stabilised into the modern low-production era and CarenR correctly identifies that the historical rules, calibrated partly on high-production pre-1980 data, are no longer reliably applicable. The increase in fallback frequency in later RVV scenarios is itself a diagnostic finding: it shows that CarenR is becoming progressively more conservative as the era gap between training history and test years widens, which is the correct adaptive response to structural non-stationarity.

Chapter 6

Discussion

This chapter draws together the findings from Chapters 4 and 5 to explain why rule-based machine learning succeeds at pattern discovery but falls short of reliable prediction in this domain. Four questions are addressed: (1) why the same rules that explain historical production cannot reliably predict future production; (2) why the two technical decisions embedded in the pipeline are in fundamental tension; (3) why the two regions diverge so markedly in their CarenR performance; and (4) what structural properties of the data would need to change for rule-based prediction to succeed.

6.1 The Asymmetry Between Discovery and Prediction

The core finding of this thesis is an asymmetry between two related but distinct tasks: discovering patterns that explain historical production, and predicting future production from those patterns. Goal 1 succeeds: 48 rules for RDD and 20 for RVV, pointing to harvest-period soil water and flowering-phase moisture as primary drivers, validated by LOESS detrending to be genuine within-era effects. Goal 2 does not succeed in aggregate. Understanding why requires examining the different requirements of the two tasks.

Pattern discovery requires only that a statistical association exists, that years satisfying a given condition have a systematically different distribution of production residuals from years that do not. With $|r| \approx 0.47$ between the top climate features and production residuals, even under LOESS detrending that removes era confounds, such an association clearly exists. CarenR's KS-test framework is well suited to detecting distributional differences at this signal strength: subgroup induction on 80–89 years of data with a 20% support threshold provides adequate power to identify the most prominent associations.

Prediction requires something harder: the association must be strong enough that knowing a year's climate conditions reduces the expected error below what would be obtained by simply predicting the training median. With $|r| \approx 0.47$, climate features explain approximately 22% of the variance in production residuals. The remaining 78% is unexplained variance, driven by factors not in the feature set (severe weather events within the growing season, vine disease outbreaks, localised frost, logistical constraints on harvest) or by measurement noise. In a series with coefficient of variation $\approx 47\%$, the signal-to-noise ratio is too low for a rule-based classifier to overcome year-to-year noise in held-out prediction, even though the identified patterns are real.

This asymmetry — a genuine agroclimatic association that is too weak for reliable prediction — is not a failure of the methodology. It is a property of the system being studied.

6.2 The Two-Decision Trade-off

The prediction pipeline decomposes into two sequential decisions whose demands pull in opposite directions (Section 3.3). Decision 1 (trend modelling) determines how large the test-period residuals are: a better-fitting trend leaves smaller residuals and a lower Naive_Median baseline. Decision 2 (climate signal extraction) determines whether a model can predict those residuals better than zero: it requires structured, non-trivial residuals to exploit.

The LOESS detrending experiment makes this trade-off concrete. In the RVV, LOESS reduces the Naive_Median baseline from 140 to approximately 90 mhl, a 36% improvement in Decision 1. But LOESS simultaneously reduces test residuals to magnitudes too small for CarenR to form stable rules, suppressing rule induction to zero. The improvement in Decision 1 quality is achieved by removing the very signal that Decision 2 depends on.

For the RDD, the linear trend is already better calibrated to the Douro's plateau-shaped production trajectory than LOESS, which would overfit the training tail and inflate test residuals to 290 mhl. Here, linear detrending both provides the better Decision 1 floor and preserves sufficient structured residual for Decision 2. The result is the closest CarenR approach to the naive baseline observed in this study (189 vs 184 mhl).

The trade-off cannot be resolved within the current methodology because both decisions are parameterised by a single covariate (time) and a single hyperparameter (linear vs. LOESS span). Decoupling them would require separate covariates for structural trend (industry restructuring, policy, demographics) and climate signal. As long as time is the only detrending covariate, any increase in trend flexibility competes directly with residual signal availability.

6.3 Why the Two Regions Diverge

The 3% gap between CarenR and the naive baseline in RDD versus the 22% gap in RVV reflects a structural difference in the underlying data, not just a difference in degree.

The Douro's production history is structurally less confounded. The Douro DOC for table wines was established in 1979, and Port production is regulated by an annual administrative quota (the benefício system), which partially decouples observed production from pure climate variation. The region's production increased gradually across the study period (+80%), creating a smoother structural trend that linear detrending captures adequately. As a result, linear detrend residuals in RDD contain a mixture of genuine climate signal and modest structural residual, with the climate component sufficiently prominent for CarenR to exploit in era-balanced scenarios.

Vinho Verde's production decline of 65% over the study period was driven primarily by structural factors: EU accession (1986) and the Common Agricultural Policy brought grubbing-up subsidies and quality standards that penalised the region's traditional high-yield pergola systems; rural exodus reduced the number of active small producers; and conversion to lower-yield espalier training reduced per-hectare output independently of climate. These transitions were concentrated in the 1980s–2000s, producing a non-linear structural break that neither linear nor LOESS detrending can fully separate from the inter-annual climate signal. The RVV linear detrend residuals therefore contain a larger structural residual component that contaminates the climate signal available for CarenR.

The era-composition analysis (Section 5.3) further illustrates this divergence. As more post-1990 data enters the RVV training set, CarenR's performance worsens rather than improves, the opposite of the RDD pattern. This indicates that the rules CarenR discovers for RVV are increasingly tuned to era-specific statistical patterns that do not generalise to test years from the same modern era. The structural disruption of Vinho Verde production has been so large and so rapid that no stable climate-production relationship emerges from within any single era.

Table 6.1: Structural explanation of the RDD vs RVV performance divergence.

Dimension	RDD: Douro	RVV: Vinho Verde	Implication
Production trend	+80% (gradual)	-65% (structural break)	RVV confound is stronger and harder to detrend
Era confound	Moderate	Strong	CarenR rules are more era-contaminated in RVV
Climate effect within era	Visible (|r| 0.47 survives LOESS)	Same magnitude but overwhelmed by structural noise	Same climate signal; different noise floor
CarenR performance	3% gap, wins in 5/18 scenarios	22% gap, 0/16 wins, rules harmful	Structural noise eliminates any climate advantage in RVV

6.4 Why the Walk-Forward Metric Penalises CarenR

The weighted MAE metric assigns weight proportional to test-set size. In an expanding-window design, early scenarios have the largest test sets. Scenario 1 for RDD covers 18 test years; Scenario 14 covers 5. The early scenarios are precisely those where era confound is strongest (post-1990 share 20–28%) and CarenR underperforms. The scenarios where CarenR wins (10–14) have the smallest test sets among the reliable scenarios and contribute the least to the aggregate metric.

This interaction is not a deficiency of the metric, weighting by test-set size correctly gives more influence to estimates based on more observations. However, it means that the aggregate weighted MAE is sensitive to the era-composition structure of the time series. A dataset in which the modern era constituted a larger fraction of the study period would allocate more weight to scenarios where CarenR is competitive, potentially reversing the aggregate ranking. As the Portuguese wine industry accumulates additional years of observation in the modern (post-1990) era, the era-balanced window would arrive earlier in the walk-forward sequence and carry more aggregate weight.

6.5 Structural Covariates as a Path Forward

The analysis throughout this thesis points to a single underlying limitation: the detrending step uses time as its sole covariate, which conflates climate-driven variation with structural industry changes. The most promising methodological extension is to incorporate structural covariates, vineyard planted area, number of certified producers, annual Port benefício quota, and proportion of espalier vs. pergola training, directly into the detrending model.

A structural detrending model of the form:

$$y_t = \alpha + \beta_1 t + \beta_2 \,\mathrm{area}_t + \beta_3 \,\mathrm{quota}_t + \varepsilon_t \tag{6.1}$$

would separate the structural production trajectory (captured by area and quota covariates) from the inter-annual climate residual (ε_t), leaving a cleaner signal for CarenR to exploit. The era confound would be substantially reduced because the structural transitions that create era boundaries would be explicitly modelled rather than proxied by the time trend. This approach is analogous to the use of normalised production (production per hectare of planted vineyard) in some agronomic studies, which removes the area effect from the production series.

Such data are available in principle from IVV (Instituto da Vinha e do Vinho) and IVDP (Instituto dos Vinhos do Douro e do Porto) for the modern period, though the historical coverage for pre-1986 years may be incomplete. Assembling a full structural covariate time series for 1934–2022 (RDD) and 1942–2021 (RVV) would require archival research and is left as the primary recommendation for future work.

6.6 Relation to Prior Work

The finding that rule-based methods can discover interpretable agroclimatic patterns but fail to outperform naive baselines in small samples is consistent with the broader forecasting literature on agricultural yield prediction. Fraga et al. (2012, 2014) demonstrated that climate indices derived from the growing season are correlated with wine quality and quantity in Portuguese regions, with predictive R^2 values typically in the 0.35–0.55 range for regional aggregates. This is broadly consistent with the $|r| \approx 0.47$ observed here, suggesting that the signal level is an intrinsic property of the system rather than a limitation of the current feature set.

The two-decision framing proposed in this thesis offers a more precise diagnostic than the summary "model beats baseline or not." Decomposing prediction error into trend quality and climate signal extraction pinpoints where the bottleneck lies: in this case, Decision 1 (structural detrending) rather than Decision 2, given that the rule-learning step already approaches the limit set by the available climate signal.

6.7 Experimental Design Decisions and Methodological Evolution

The final pipeline described in Chapter 3 represents the outcome of an iterative development process. Several design decisions were made in response to problems discovered during exploratory analysis, and documenting these decisions is both methodologically honest and practically informative for future researchers working on similar problems.

6.7.1 The Uptrend Problem and the Decision to Detrend

The first major design decision concerned how to handle the structural production trends in both regions. An initial version of the CarenR analysis was run on raw (non-detrended) production values for the Douro. The result was immediately suspicious: all five discovered rules pointed to LOW production conditions, despite the Douro having a clear upward trend and the expectation that meaningful rules should exist for both above and below-average years.

Investigation revealed the cause: without detrending, CarenR's global mean was the all-time average (approximately 1,112 mhl), which is heavily influenced by the low-production pre-1960 era. Years from the 1930s–1950s were systematically below this global mean not because of climate but because the structural production baseline was lower. CarenR was finding rules that described era membership, "low production years tend to be early years", rather than genuine climate-production relationships. The discovered rules were statistically valid but agronomically meaningless.

The solution, linear detrending applied separately within each training fold, transformed the problem from one of explaining raw production levels to one of explaining deviations from the expected structural trajectory. Under this formulation, a rule like "dry post-flowering conditions $\rightarrow$ above-trend production" is interpretable regardless of which era the year belongs to, because "above-trend" is defined relative to the era-specific trend. The detrending decision was therefore not merely a preprocessing convenience but a fundamental reframing of the research question from "what predicts high production?" to "what predicts better-than-expected production given the structural context of each period?"

6.7.2 LOESS Detrending: A Promising Idea That Killed the Signal

Once linear detrending was adopted, a natural follow-up question was whether more flexible detrending (LOESS) might produce better results by more accurately capturing the non-linear structural trajectories, particularly the RVV plateau-then-collapse shape. LOESS detrending was evaluated with a smoothing span of 0.40, which provides substantial flexibility while preserving some year-to-year structure.

The result was counter-intuitive: LOESS improved the Naive_Median baseline substantially for RVV (from 140 to 90 mhl, a 36% improvement in Decision 1) but eliminated all CarenR rules entirely, zero rules fired in any scenario. The explanation became clear upon reflection: LOESS with span 0.40 on an 80-year series absorbs so much of the inter-annual variation into the trend component that the residuals become too small and structureless for CarenR's KS test to detect significant subgroup differences. The flexible trend was doing exactly what it was designed to do, capturing year-to-year variation, but in doing so it was consuming the signal that CarenR needed.

For RDD, LOESS proved actively harmful: the Naive_Median baseline worsened from 184 to 290 mhl because LOESS extrapolated the slope of the final training years into the test period. The Douro's production had plateaued around 1,300–1,400 mhl since the mid-2000s, but LOESS projected a continued slope from the training data into test years, systematically overestimating production. The rigid linear trend, despite its simplicity, happened to extrapolate the Douro plateau more accurately than the flexible LOESS curve.

These LOESS results reinforced the two-decision decomposition: improving Decision 1 (trend quality) came directly at the cost of Decision 2 (climate signal availability). The decision to retain linear detrending for the primary analysis was therefore not a limitation born of simplicity but a deliberate choice that maximised the climate signal available for rule discovery while accepting a higher baseline error.

6.7.3 The Classification-to-Regression Transition

The original design of Goal 2 (Predictive Validation) used RIPPER as the primary comparison model, with production discretised into Low / Medium / High classes. The evaluation metric was classification accuracy. This design proved unsatisfactory for two reasons discovered during initial experiments.

First, RIPPER's cross-validated accuracy of 28–42%, only marginally above the 33% random baseline, confirmed that the classification task was genuinely difficult but also revealed that the accuracy metric was a poor measure of prediction quality for this application. A model that correctly identifies "above-trend" vs "below-trend" years at 55% accuracy might be practically valuable even if its overall three-class accuracy is modest, because correctly identifying the direction is more important than nailing the precise tertile. Accuracy treats all misclassifications equally, which is not the right criterion when the goal is to outperform a naive quantitative forecast.

Second, comparing RIPPER (classification) against CarenR (which outputs a continuous median residual) required arbitrary conversion of CarenR's continuous predictions to class labels, or RIPPER's class labels to continuous values. Either conversion introduced distortion. The cleaner solution was to operate entirely in the continuous domain: use MAE on the original production scale as the evaluation metric, and compare all models, including RIPPER, by the same criterion. RIPPER was retained as a comparison model for Goal 1 (rule discovery, where its classification rules are directly interpretable) but M5Rules replaced it as the primary continuous-target comparison for Goal 2.

This transition from a classification to a regression evaluation framework was not merely cosmetic. It changed which models were most meaningful to compare, which metric was used, and ultimately what the thesis's predictive conclusion would be. The MAE-based walk-forward evaluation with a Naive_Median baseline is a more demanding and more informative benchmark than classification accuracy: it directly asks whether a model can beat the simplest possible quantitative forecast, which is the right question for a production forecasting application.

6.7.4 CarenR Variant Proliferation: Why 11 Variants?

The decision to evaluate 11 CarenR variants was motivated by a genuine uncertainty about which feature selection strategy would work best in this domain. Prior work with CarenR had established that feature selection matters but had not systematically evaluated the trade-off between aggressive selection (which finds the most discriminating features but risks overfitting) and conservative selection (which retains broader coverage but may include noise).

The 11 variants represent a structured exploration of this trade-off rather than an ad hoc collection of experiments. They vary along three dimensions: the criterion used to rank features (η^2, Pearson |r|, hard threshold); whether a support filter is applied (ensuring each bin of a selected feature is well-populated); and how rule predictions are combined at test time (equal weighting, support weighting, confidence weighting, stacking). The CarenR_Supervised variant was included specifically to test the hypothesis that per-fold supervised discretisation might improve rule quality by better aligning bin boundaries with the production distribution, and its consistent failure (worst CarenR variant in both regions) provides a clear, replicable finding about the risks of target leakage in feature engineering.

The systematic comparison also revealed the region-specificity of optimal selection strategy (η^2 for RDD, Sup_FS for RVV), which is itself a methodological contribution: it demonstrates that the appropriate CarenR configuration for a given time series depends on the strength and structure of the climate signal, which cannot be determined a priori without empirical evaluation. Researchers applying CarenR to similar agricultural forecasting problems should therefore conduct a comparable structured sensitivity analysis rather than assuming any single configuration will be optimal.

6.7.5 Limitations Acknowledged During Development

Several limitations were identified during the pipeline development that are worth documenting explicitly. The CarenR_Assoc variant (association rule classification, rather than distributional rules) was initially planned but not fully implemented due to factor-level alignment issues in the walk-forward context: the class boundaries estimated on each training fold did not consistently produce the same factor levels across folds, causing prediction-time errors when test-year factor values fell outside training-set factor ranges. Resolving this would require a more robust factor recoding step in the walk-forward loop and is left as a natural extension of the current work.

The LAG variant (CarenR_Dist_Lag) attempted to augment the feature set with autocorrelated lag features identified from the ACF of the production residual series. While this variant performs competitively in RDD (rank 5), it is outperformed by simpler selectors in both regions, suggesting that the ACF-based lag selection does not consistently identify more predictive carry-over features than the domain-knowledge-based y0 features already included in the base feature set. The carry-over mechanism is real, as confirmed by RIPPER and M5Rules coefficients on previous-year variables, but the ACF approach to identifying it appears less reliable than the agronomically-motivated y0 feature construction.

Chapter 7

Conclusions

This thesis evaluated the CarenR distributional subgroup discovery framework on two contrasting wine production time series, pursuing two goals: pattern extraction from historical data and walk-forward prediction. The findings address the conditions under which rule-based machine learning succeeds and fails on non-stationary agricultural time series, a question with implications beyond the specific domain studied here.

7.1 Summary of Findings

Goal 1 (Rule Discovery) is achieved. Applied to 89 years of Douro data and 80 years of Vinho Verde data, CarenR's KS-test-based subgroup induction discovers 48 and 20 statistically validated rules respectively, with feature selection consistently converging on a small set of dominant variables across all 11 variants evaluated. The harvest-period soil water variable appears in 54% of Douro rules; heat-stress days (maximum temperature above 35°C) appear in up to 50% of Vinho Verde rules. LOESS detrending validates that these variables carry genuine within-era signal rather than era-membership artefacts, and RIPPER and M5Rules independently confirm the same variable families through entirely different optimisation criteria. This cross-algorithm convergence is evidence that CarenR's distributional approach successfully identifies real structure in the data.

Goal 2 (Predictive Validation) is not achieved overall. No rule-based model consistently beats the Naive_Median baseline across the full evaluation sequence in either region (best CarenR: 189 vs 184 mhl in RDD, Wilcoxon $p = 0.468$; 172 vs 140 mhl in RVV, rules actively harmful when fired, $p = 0.004$). A post-hoc era-composition analysis suggests CarenR is competitive in RDD when training data is era-balanced (29% post-1990 share; 5/5 scenarios won, binomial $p = 0.031$), but this result is exploratory and requires prospective validation. The two-decision decomposition framework explains the failure structurally: improving trend quality (Decision 1) removes the residual signal that rule induction requires (Decision 2), and these demands cannot be simultaneously satisfied with a time-only detrending covariate.

The central finding: *CarenR's distributional subgroup discovery successfully extracts statistically validated, cross-algorithm-confirmed patterns from a non-stationary time series spanning* *multiple structural eras. Those patterns do not reliably generalise to held-out prediction because the era structure that makes rules historically interpretable also contaminates the signal in ways that dissolve when predicting within a single era. The two-decision framework precisely characterises this failure mode and identifies structural detrending as the path to resolving it.*

7.2 Contributions

This thesis makes the following contributions to data science methodology and its application to agricultural time series:

• Evaluation framework: the two-decision decomposition separates prediction error into trend quality (Decision 1) and residual signal extraction (Decision 2), providing a diagnostic tool that explains why rule-based prediction fails even when genuine signal exists. This framework is transferable to any agroclimatic forecasting problem where structural non-stationarity and climate signal co-exist.

• Systematic sensitivity analysis: a structured 11-variant evaluation of CarenR's feature selection strategies shows that the optimal configuration is problem-specific, η^2 outperforms in regimes with strong signal (Douro), while conservative support-filtered selection outperforms in regimes where fallback-to-median is the dominant prediction mechanism (Vinho Verde). This finding motivates structured variant evaluation rather than a priori configuration choice.

• Era-composition analysis: the walk-forward evaluation reveals a conditional regime in RDD where CarenR consistently outperforms the baseline once training data reaches 29% modern-era representation (5/5 consecutive wins, post-hoc exploratory). This characterises not just whether rule-based prediction works, but under what data conditions it does, a more actionable finding than a binary success/failure conclusion.

• Cross-algorithm validation: RIPPER (classification rules) and M5Rules (piecewise linear rules) independently converge on the same feature variables as CarenR across both regions, despite entirely different optimisation criteria. This cross-algorithm agreement provides evidence that the identified features reflect genuine data structure rather than artefacts of any single induction method.

• Method validation in domain: the discovered rules are agronomically coherent, confirmed by LOESS detrending which shows feature correlations are preserved after era-trend removal. This validates that CarenR extracts genuine signal from real-world non-stationary data, not era-membership proxies. The domain provides 80–90 years of historical data, genuine structural non-stationarity, and expert-verifiable output, a demanding testbed for the method.

7.3 Limitations

The primary methodological limitation is that detrending uses time as its sole covariate, which cannot cleanly separate structural production changes from inter-annual climate variation. Both regional series exhibit non-linear structural trajectories driven by non-climate factors (EU policy, industry restructuring) that a linear trend imperfectly captures. This limitation affects both rule quality (era confound) and prediction (baseline calibration). A second limitation is the small number of test scenarios (18 and 16) relative to the complexity of the model family evaluated; in particular, the post-hoc binomial finding (n = 5 scenarios) has insufficient statistical power for confirmatory conclusions. The datasets are single-region annual aggregates, precluding any spatial or sub-regional analysis.

7.4 Future Work

• Structural detrending: incorporating non-climate structural covariates (planted area, production quotas, number of certified producers) as explicit detrending variables would separate structural trends from climate residuals. This would reduce the era confound in rule induction and potentially resolve the Decision 1 / Decision 2 tension identified by the two-decision framework.

• Prospective validation: the post-hoc era-balanced finding (p = 0.031) should be formalised as a pre-registered hypothesis and tested on a new or extended dataset as additional modern-era years accumulate.

• Higher-resolution climate data: replacing regional aggregates with spatially explicit gridded data would allow rule conditions to reflect within-region heterogeneity, potentially sharpening the distributional signal available to CarenR.

• Exceptional Model Mining extension: the EMM framework (Duivesteijn et al., 2016) generalises distributional subgroup discovery to model-based interestingness criteria. Applying EMM to identify subgroups where the climate-production regression relationship is exceptional, rather than where the marginal distribution shifts, could detect interaction effects that KS-based rules miss.

• Application to other wine regions: the results here show that the era confound is a primary driver of predictive failure, particularly in RVV. Testing CarenR on other national or international wine regions where no strong structural production trend exists would allow the climate signal to be evaluated in isolation, without the complication of structural non-stationarity. Candidate regions would include those with relatively stable long-run production histories and good-quality multi-decadal climate records.

7.5 Closing Statement

Rule-based machine learning, and distributional subgroup discovery in particular, offers a transparent and auditable approach to pattern extraction from real-world time series. This thesis demonstrates that CarenR can identify statistically validated, cross-algorithm-confirmed patterns from 80–90 years of non-stationary data, a result that is meaningful for the data science method regardless of whether those patterns generalise to prediction. The limitations of the predictive application are precisely characterised by the two-decision framework and are addressable through better structural detrending. The methodological contributions, the evaluation framework, the sensitivity analysis, the era-composition diagnostic, are transferable to analogous problems wherever rule-based discovery is applied to long, structurally non-stationary time series.

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Appendix A

Complete Rule Listings

This appendix presents the complete set of rules discovered by CarenR, RIPPER, and M5Rules on the full historical datasets for both regions. Rules are sorted by KS p-value (CarenR) or rule number (RIPPER, M5Rules). All CarenR rules were discovered with filters: support $\geq$ 20% and KS p-value $\leq$ 0.10. Detrending: linear (residuals). All mean deviations are in mhl relative to the long-run production trend.

A.1 Top 10 CarenR Rules per Region

Table A.2: Top 10 CarenR rules for the Vinho Verde (RVV). Rule 4 is the only above-trend rule.

#	Conditions	Dir.	Median Δ (mhl)	Support	p-value
1	**Soil water pre-Harvest [61–97 mm] AND Heat days pre-Maturity [0–1.9d] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]** *Low soil water deficit with absent heat stress — cool-moist pattern, not a beneficial stress pattern in the Atlantic RVV climate.*	BELOW	-338	21.2%	0.0198
2	**Temp at Flowering [13.8–17.1°C] AND Temp post-Flowering [12.6–17.5°C] AND Wet-soil days post-Flowering prev.yr [17.5–20d]** *Cool flowering combined with previous-year soil saturation — a two-year carry-over pattern.*	BELOW	-268	21.2%	0.0216

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Table A.2 continued

#	Conditions	Dir.	Median Δ (mhl)	Support	p-value
3	**Soil water post-Flowering [143–234 mm] AND Heat days post-Maturity [0–2.4d]** *Highest-support below-trend rule: moist post-flowering with low heat at maturity.*	BELOW	-365	27.5%	0.0294
4	**Temp post-Budburst prev.yr [11.2–13.5°C] AND Cold days at Flowering [0–3.4d] AND Heat days post-Flowering [0–1.1d]** *The only above-trend rule: warm-but-not-extreme previous-year budburst with mild current-year flowering. Largest absolute deviation in either region (+311 mhl).*	ABOVE	+311	25.0%	0.0298
5	**Temp post-Flowering [12.6–17.5°C] AND Heat days post-Maturity [0–2.4d] AND Wet-soil days post-Flowering prev.yr [17.5–20d]** *Cool post-flowering temperature with previous-year soil saturation carry-over.*	BELOW	-292	23.8%	0.0299
6	**Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2] AND Heat days post-Flowering [0–1.3d]** *Moderate-to-high canopy combined with post-flowering rainfall — introduces the LAI at harvest variable.*	BELOW	-387	26.2%	0.0356
7	**Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2] AND Heat days post-Flowering (20d) [0–1.1d]** *Near-identical to Rule 6 with a slightly wider heat-stress window.*	BELOW	-387	26.2%	0.0356
8	**LAI pre-Flowering prev.yr [2.16–2.21 m^2/m^2] AND Heat days pre-Maturity [0–1.9d] AND Heat days post-Maturity [0–2.4d]** *Previous-year LAI at flowering as a predictor — carry-over canopy signal.*	BELOW	-332	22.5%	0.0357
9	**Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]** *Four-condition variant of the rainfall + LAI + low-heat pattern.*	BELOW	-387	20.0%	0.0370

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Table A.2 continued

#	Conditions	Dir.	Median Δ (mhl)	Support	p-value
10	**Soil water pre-Harvest [61–97 mm] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]** *Variant of Rule 1 with different heat-stress window combination.*	BELOW	-338	21.2%	0.0401

A.2 CarenR — Douro Region (RDD): 48 Rules

All 48 rules are sorted by KS p-value (most significant first). The dominant theme is dry post-flowering conditions combined with moderate soil water stress at harvest, conditions that together favour above-trend production. Of the 48 rules, 46 point to ABOVE-trend production and 2 to BELOW-trend, reflecting the asymmetric signal structure of the Douro region. Conditions are abbreviated; full variable names follow the notation defined in Section 3.11.1.

Table A.3: Complete CarenR rule listing for the Douro (RDD) region: all 48 rules.

#	Support	p-value	Direction	Mean Δ (mhl)	Conditions (abbreviated)
1	28.1% (25yr)	0.0043	ABOVE	+121	Wet days post-Flowering [0–3.3d] AND Wet days post-Start-of-Maturity [0–3.5d]
2	20.2% (18yr)	0.0084	BELOW	-131	LAI at Flowering [0.74–0.88 m^2/m^2] AND Soil water post-Flowering [390–404 mm]
3	25.8% (23yr)	0.0089	ABOVE	+113	Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Heat days post-Flowering [0–0.6d]
4	21.3% (19yr)	0.0095	ABOVE	+192	Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d] AND Wet days post-Maturity [0–3.5d]
5	24.7% (22yr)	0.0110	ABOVE	+115	Wet days post-Flowering [0–3.3d] AND Heat days pre-Maturity [0–3.7d]
6	23.6% (21yr)	0.0120	ABOVE	+117	Wet days post-Flowering [0–3.3d] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]
7	21.3% (19yr)	0.0130	ABOVE	+115	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Heat days pre-Maturity [0–3.7d]
8	19.1% (17yr)	0.0140	ABOVE	+128	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Wet days post-Maturity [0–3.5d]
9	23.6% (21yr)	0.0150	ABOVE	+116	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]

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Table A.3 continued

#	Support	p-value	Direction	Mean Δ (mhl)	Conditions (abbreviated)
10	20.2% (18yr)	0.0160	ABOVE	+117	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]
11	21.3% (19yr)	0.0170	ABOVE	+113	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Heat days pre-Maturity [0–3.7d]
12	28.1% (25yr)	0.0180	ABOVE	+108	Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Soil water pre-Harvest [74–231 mm]
13	19.1% (17yr)	0.0200	ABOVE	+122	Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d] AND Wet days post-Maturity [0–3.5d] AND Heat days post-Flowering [0–0.6d]
14	16.9% (15yr)	0.0210	ABOVE	+143	Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d] AND Heat days pre-Maturity [0–3.7d]
15	21.3% (19yr)	0.0220	ABOVE	+115	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days post-Flowering [0–0.6d]
16	18.0% (16yr)	0.0240	ABOVE	+132	Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]
17	22.5% (20yr)	0.0250	ABOVE	+115	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days post-Flowering [0–0.6d]
18	19.1% (17yr)	0.0260	ABOVE	+119	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Wet days post-Maturity [0–3.5d] AND Cold days pre-Harvest [0–2.0d]
19	22.5% (20yr)	0.0270	ABOVE	+112	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Cold days pre-Harvest [0–2.0d]
20	20.2% (18yr)	0.0290	ABOVE	+115	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Wet days post-Maturity [0–3.5d] AND Cold days pre-Harvest [0–2.0d]
21	15.7% (14yr)	0.0310	BELOW	-158	LAI at Flowering [0.74–0.88 m^2/m^2] AND Soil water post-Flowering [390–404 mm] AND Cold days pre-Harvest [0–2.0d]
22	22.5% (20yr)	0.0320	ABOVE	+107	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Cold days pre-Harvest [0–2.0d]
23	19.1% (17yr)	0.0340	ABOVE	+122	Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d]
24	20.2% (18yr)	0.0350	ABOVE	+119	Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Soil water pre-Harvest [74–231 mm] AND Cold days pre-Harvest [0–2.0d]

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Table A.3 continued

#	Support	p-value	Direction	Mean Δ (mhl)	Conditions (abbreviated)
25	21.3% (19yr)	0.0370	ABOVE	+113	Wet days post-Flowering [0–3.3d] AND Cold days pre-Harvest [0–2.0d]
26	18.0% (16yr)	0.0380	ABOVE	+117	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Heat days pre-Maturity [0–3.7d] AND Cold days pre-Harvest [0–2.0d]
27	19.1% (17yr)	0.0400	ABOVE	+116	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–231 mm] AND Cold days pre-Harvest [0–2.0d]
28	19.1% (17yr)	0.0410	ABOVE	+121	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Cold days pre-Harvest [0–2.0d]
29	16.9% (15yr)	0.0430	ABOVE	+133	Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Soil water pre-Harvest [101–231 mm]
30	19.1% (17yr)	0.0440	ABOVE	+116	Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Cold days pre-Harvest [0–2.0d]
31	19.1% (17yr)	0.0450	ABOVE	+122	Soil water pre-Harvest [74–101 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days pre-Maturity [0–3.7d]
32	16.9% (15yr)	0.0470	ABOVE	+130	Soil water pre-Harvest [74–101 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]
33	18.0% (16yr)	0.0480	ABOVE	+121	Soil water pre-Harvest [74–101 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days post-Flowering [0–0.6d]
34	15.7% (14yr)	0.0500	ABOVE	+141	Soil water pre-Harvest [74–101 mm] AND Wet days post-Maturity [0–3.5d]
35	20.2% (18yr)	0.0510	ABOVE	+111	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]
36	20.2% (18yr)	0.0520	ABOVE	+110	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Cold days pre-Harvest [0–2.0d] AND Heat days post-Flowering [0–0.6d]
37	15.7% (14yr)	0.0540	ABOVE	+140	Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Soil water pre-Harvest [101–152 mm] AND Cold days pre-Harvest [0–2.0d]
38	16.9% (15yr)	0.0560	ABOVE	+127	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–231 mm] AND Wet days post-Maturity [0–3.5d] AND Cold days pre-Harvest [0–2.0d]
39	19.1% (17yr)	0.0570	ABOVE	+113	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days pre-Maturity [0–3.7d]

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Table A.3 continued

#	Support	p-value	Direction	Mean Δ (mhl)	Conditions (abbreviated)
40	16.9% (15yr)	0.0580	ABOVE	+127	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days pre-Maturity [0–3.7d] AND Cold days pre-Harvest [0–2.0d]
41	16.9% (15yr)	0.0600	ABOVE	+126	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Cold days pre-Harvest [0–2.0d] AND Heat days post-Flowering [0–0.6d]
42	16.9% (15yr)	0.0620	ABOVE	+126	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Wet days post-Maturity [0–3.5d] AND Cold days pre-Harvest [0–2.0d] AND Heat days post-Flowering [0–0.6d]
43	16.9% (15yr)	0.0650	ABOVE	+119	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Cold days pre-Harvest [0–2.0d]
44	15.7% (14yr)	0.0680	ABOVE	+136	Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Soil water pre-Harvest [74–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Cold days pre-Harvest [0–2.0d]
45	15.7% (14yr)	0.0700	ABOVE	+132	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]
46	15.7% (14yr)	0.0720	ABOVE	+130	Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–101 mm] AND Wet days post-Maturity [0–3.5d] AND Heat days pre-Maturity [0–3.7d]
47	15.7% (14yr)	0.0750	ABOVE	+132	Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d] AND Cold days pre-Harvest [0–2.0d]
48	16.9% (15yr)	0.0900	BELOW	-118	LAI at Harvest [0.58–0.77 m^2/m^2] AND Soil water post-Flowering [390–404 mm]

A.3 CarenR — Vinho Verde Region (RVV): 20 Rules

All 20 RVV rules are presented below. Of the 20 rules, 16 point to BELOW-trend production and 4 to ABOVE-trend, reflecting the strong negative structural drift of the Vinho Verde series. The dominant features are heat-stress days around maturity (appearing in up to 50% of rules), followed by soil water at harvest (25%) and cold/frost days at budburst (25%). Rules 4, 11, 16, and 17 are the four above-trend rules, linking warm previous-year budburst temperatures and few cold days at flowering to above-trend production.

Table A.1: Top 10 CarenR rules for the Douro (RDD), sorted by KS p-value.

#	Conditions	Dir.	Median Δ (mhl)	Support	p-value
1	**Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d]** *Dry post-flowering conditions reduce disease pressure and promote fruit set in the Douro.*	ABOVE	+121	28.1%	0.0043
2	**LAI at Flowering [0.74–0.88 m^2/m^2] AND Soil water post-Flowering [390–404 mm]** *Elevated canopy vigour with high soil moisture at flowering drives vegetative-over-reproductive growth.*	BELOW	-131	20.2%	0.0084
3	**Wet days post-Flowering [0–3.3d] AND Wet days post-Maturity [0–3.5d] AND Heat days post-Flowering [0–0.6d]** *Refines Rule 1: dry and thermally moderate post-flowering window.*	ABOVE	+113	25.8%	0.0089
4	**Soil water pre-Harvest [74–101 mm] AND Wet days post-Harvest prev.yr [0–2.3d] AND Wet days post-Maturity [0–3.5d]** *Multi-year rule: moderate soil water stress with dry previous-year post-harvest window captures a carry-over vine reserve mechanism.*	ABOVE	+192	21.3%	0.0095
5	**Wet days post-Flowering [0–3.3d] AND Heat days pre-Maturity [0–3.7d]** *Dry flowering with bounded pre-veraison heat — the "warm but not extreme" pattern.*	ABOVE	+115	24.7%	0.0110
6	**Wet days post-Flowering [0–3.3d] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]** *Three-condition refinement of the dry-and-moderate-heat pattern.*	ABOVE	+117	23.6%	0.0120
7	**Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Heat days pre-Maturity [0–3.7d]** *Post-flowering dryness combined with moderate pre-harvest soil water — two independent dryness signals.*	ABOVE	+115	21.3%	0.0130
8	**Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–152 mm] AND Wet days post-Maturity [0–3.5d]** *Mid-range harvest soil water with dry post-flowering and post-maturity windows.*	ABOVE	+128	19.1%	0.0140
9	**Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [74–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]** *Four-condition rule combining all three primary signals; narrowest statistically valid subgroup.*	ABOVE	+116	23.6%	0.0150
10	**Wet days post-Flowering [0–3.3d] AND Soil water pre-Harvest [101–231 mm] AND Heat days pre-Maturity [0–3.7d] AND Heat days post-Flowering [0–0.6d]** *Variant of Rule 9 with wider soil water range.*	ABOVE	+117	20.2%	0.0160

Table A.4: Complete CarenR rule listing for the Vinho Verde (RVV) region: all 20 rules.

#	Support	p-value	Direction	Mean Δ (mhl)	Conditions (abbreviated)
1	21.2% (17yr)	0.0198	BELOW	-338	Soil water pre-Harvest [61–97 mm] AND Heat days pre-Maturity [0–1.9d] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]
2	21.2% (17yr)	0.0216	BELOW	-268	Temp centred on Flowering [13.8–17.1°C] AND Temp post-Flowering [12.6–17.5°C] AND Wet-soil days post-Flowering prev.yr [17.5–20d]
3	27.5% (22yr)	0.0294	BELOW	-365	Soil water post-Flowering [143–234 mm] AND Heat days post-Maturity [0–2.4d]
4	25.0% (20yr)	0.0298	ABOVE	+311	Temp post-Budburst prev.yr [11.2–13.5°C] AND Cold days centred Flowering [0–3.4d] AND Heat days post-Flowering [0–1.1d]
5	23.8% (19yr)	0.0299	BELOW	-292	Temp post-Flowering [12.6–17.5°C] AND Heat days post-Maturity [0–2.4d] AND Wet-soil days post-Flowering prev.yr [17.5–20d]
6	26.2% (21yr)	0.0356	BELOW	-387	Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2] AND Heat days post-Flowering [0–1.3d]
7	26.2% (21yr)	0.0356	BELOW	-387	Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2] AND Heat days post-Flowering (20d window) [0–1.1d]
8	22.5% (18yr)	0.0357	BELOW	-332	LAI pre-Flowering prev.yr [2.16–2.21 m^2/m^2] AND Heat days pre-Maturity [0–1.9d] AND Heat days post-Maturity [0–2.4d]
9	20.0% (16yr)	0.0370	BELOW	-387	Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]
10	21.2% (17yr)	0.0401	BELOW	-338	Soil water pre-Harvest [61–97 mm] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]
11	18.8% (15yr)	0.0418	BELOW	-330	LAI pre-Flowering prev.yr [2.16–2.21 m^2/m^2] AND Heat days post-Maturity [0–2.4d]
12	20.0% (16yr)	0.0430	BELOW	-392	Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2]
13	21.2% (17yr)	0.0433	BELOW	-320	Soil water pre-Harvest [61–97 mm] AND Heat days pre-Maturity [0–1.9d] AND Heat days post-Flowering [0–1.1d]
14	21.2% (17yr)	0.0450	BELOW	-338	Soil water pre-Harvest [61–97 mm] AND Heat days pre-Maturity [0–1.9d] AND Heat days post-Maturity [0–2.4d]
15	18.8% (15yr)	0.0500	BELOW	-382	Rainfall post-Flowering [20.7–121 mm] AND LAI at Harvest [1.29–2.52 m^2/m^2] AND Heat days post-Maturity [0–2.4d]
16	22.5% (18yr)	0.0514	BELOW	-289	Soil water post-Flowering [143–234 mm] AND Heat days post-Maturity [0–2.4d] AND Heat days post-Flowering [0–1.1d]
17	21.2% (17yr)	0.0520	BELOW	-338	Soil water pre-Harvest [61–97 mm] AND Heat days post-Maturity [0–2.4d]
18	21.2% (17yr)	0.0550	BELOW	-338	Soil water pre-Harvest [61–97 mm] AND Heat days post-Flowering [0–1.1d]
19	18.8% (15yr)	0.0600	BELOW	-289	Soil water post-Flowering [143–234 mm] AND Heat days post-Maturity [0–2.4d] AND Wet-soil days post-Flowering prev.yr [17.5–20d]
20	23.8% (19yr)	0.0620	BELOW	-246	Temp post-Flowering [12.6–17.5°C] AND Heat days post-Maturity [0–2.4d]

A.4 RIPPER Classification Rules: Both Regions

RIPPER discovers an ordered rule list for tertile-discretised production classes (Low / Medium / High). Class boundaries: RDD, Low < 975 mhl, Medium 975–1,239 mhl, High > 1,239 mhl; RVV, Low < 1,090 mhl, Medium 1,090–1,900 mhl, High > 1,900 mhl. Rules are evaluated in order; the first matching rule applies. The default rule covers all years not matched by any explicit rule. Cross-validated accuracy is reported per region at the rule-list level; individual rule precision is training-set precision.

Table A.5: RIPPER classification rules for both regions.

#	Region	Conditions	Prediction	Support	Precision (train)	CV Accuracy
1	RDD	Mean temp at Harvest (10d before) in [21.2–22.8 °C]	MEDIUM	21.3% (19yr)	73.7%	—
2	RDD	LAI at Budburst (10d before) $\geq$ 0.0016 AND Mean temp at Harvest (20d before) $\geq$ 23.1 °C	HIGH	19.1% (17yr)	76.5%	—
3	RDD	Cold days at Flowering (15d before) $\leq$ 3 AND Mean temp at Harvest (20d before) $\leq$ 21.3 °C	HIGH	6.7% (6yr)	100%	—
4	RDD	(default, all other years)	LOW	—	—	42% overall (κ=0.06)
1	RVV	LAI at Flowering prev.yr $\geq$ 2.4 AND Budburst temp prev.yr $\geq$ 13.3°C AND Rainfall at Flowering $\leq$ 15.9 mm	MEDIUM	16.2% (13yr)	92.3%	—
2	RVV	Mean temp at Flowering (10d before) $\geq$ 19.3°C AND LAI at Flowering (10d before) $\geq$ 2.2	MEDIUM	11.2% (9yr)	88.9%	—
3	RVV	Soil water at Flowering (30d after) $\leq$ 123.6 mm AND Mean temp at Flowering (15d after) $\geq$ 16.6°C	HIGH	26.2% (21yr)	85.7%	—
4	RVV	(default, all other years)	LOW	—	—	35% overall (κ=0.19)

A.5 M5Rules Piecewise Linear Rules: Both Regions

M5Rules collapsed to global linear models in both regions (no beneficial split found). The rules below present the full linear model for each region. All coefficients apply to standardised features; swa_hv_y1_b20 appears in both regional models with a negative coefficient, consistent with CarenR's finding that moderate harvest soil water stress promotes above-trend production. Cross-validated R^2: RDD = 0.004, RVV = -0.07.

A.5.1 RDD: 2 Rules

Rule 1 (conditional): IF Mean temp pre-Flowering prev.yr > 16.03°C AND Wet days post-Flowering $\leq$ 3.5d

THEN production residual = 45.23 × tm_fl_y1_b10 + 23.43 × tm_fl_y0_b10 - 52.17 × tm_bb_y0_a30 - 9.18 × tm_sm_y0_a20 + 8.80 × tm_hv_y1_c10 + 7.31 × tn_hv_y0_a30 - 3.45 × cold_days_fl + 675.75 × iaf_fl_y1_b10 + 285.85 × iaf_fl_y1_b30 + 2319.99 × iaf_fl_y0_b20 - 2474.55 × iaf_fl_y0_b30 - 0.44 × swa_hv_y1_b20 - 1023.61

Rule 2 (default): all other years

THEN production residual = 66.45 × tm_fl_y1_b10 - 97.63 × tm_bb_y0_a30 - 38.54 × tm_sm_y0_a20 + 39.62 × tm_hv_y0_a30 + 1394.73 × iaf_fl_y1_b20 + 4138.60 × iaf_fl_y0_b10 - 4199.43 × iaf_fl_y0_b20 - 2.30 × swa_hv_y1_b20 - 408.09

A.5.2 RVV: 3 Rules

Rule 1 (conditional): IF Soil water post-Flowering > 131.4 mm

THEN production residual = -13.26 × tm_fl_y0_b10 - 40.72 × tm_fl_y0_a10 + 17.16 × tm_sm_y0_a20 - 24.87 × tm_hv_y1_b20 - 2063.79 × iaf_bb_y1_c10 - 8.12 × swa_fl_y1_a30 + 2419.77

Rule 2 (conditional): IF Mean temp pre-Harvest > 16.76°C

THEN production residual = 24.11 × tm_fl_y1_c15 - 24.09 × tm_fl_y0_b10 + 46.70 × tm_sm_y0_a20 - 49.55 × tm_hv_y1_b20 + 5.10 × rf_hv_y0_a10 - 1.74 × swa_hv_y1_b20 + 227.87

Rule 3 (default): all other years

THEN production residual = -151.60 × tm_fl_y0_b10 + 3455.29

Features use the notation of Section 3.11.1; the intercept is the constant term of each linear model.

Appendix B

Exploratory Data Analysis

This appendix presents the exploratory data analysis of the two regional datasets. The analyses were conducted prior to model development and informed several key methodological decisions: the choice of linear detrending, the inclusion of previous-year (y0) features, the identification of likely era-confounded variables, and the feature selection strategy for CarenR. All statistics are computed from the final curated datasets (dataPrep_cont_dataset_rdd.csv and dataPrep_cont_dataset_rvv.csv) unless otherwise noted.

B.1 Wine Production: Descriptive Statistics and Trends

Table B.1 presents the key descriptive statistics for wine production in both regions over the full study period.

Table B.1: Descriptive statistics: RDD (1934–2022) and RVV (1942–2021).

Statistic	RDD (Douro) 1934–2022	RVV (Vinho Verde) 1942–2021
Observations (n)	89 years	80 years
Mean production	1,116 mhl	1,621 mhl
Standard deviation	307 mhl	751 mhl
Coefficient of variation (CV)	27.6%	46.4%
Minimum	518 mhl (1952)	494 mhl (1997)
Maximum	1,961 mhl (1990)	3,443 mhl (1962)
Linear trend slope	+7.4 mhl/year (increasing)	−21.2 mhl/year (declining)
Total change over period	+ 660 mhl (+80%)	− 1,500 mhl (−65%)

The contrast between the two regions is immediately apparent. Vinho Verde has a substantially higher mean production (1,621 vs 1,116 mhl) and nearly three times the standard deviation (751 vs 307 mhl), yielding a CV of 46.4% vs 27.6%. This higher variability in RVV is partly structural, the larger absolute production values amplify variation, but also reflects the stronger era-driven structural changes in the RVV series. The coefficient of variation is a key number for understanding the prediction task: with CV = 27.6% for RDD and 46.4% for RVV, any model must reduce the residual error substantially below the naive prediction to be useful, which is a high bar with a climate signal explaining only 22% of variance.

B.1.1 Decade-Level Production Averages

Table B.2 presents decade-level mean production for both regions, revealing the structural trajectory of each series.

Table B.2: Decade-level production averages for RDD and RVV.

Decade	RDD mean (mhl)	RDD n	RVV mean (mhl)	RVV n
1930s	765	7	—	—
1940s	767	10	2,026	9
1950s	926	10	1,941	10
1960s	1,184	10	2,331	10
1970s	1,090	10	2,229	10
1980s	1,148	10	1,638	10
1990s	1,271	10	1,259	10
2000s	1,392	10	906	10
2010s	1,308	10	798	10
2020s	1,287	3	871	2

The decade table makes the structural stories of both regions clearly visible. RDD production roughly doubled between the 1930s–40s (765–767 mhl) and the 2000s–2010s (1,300–1,400 mhl), with the most rapid growth occurring in the 1950s–60s and again in the 1990s–2000s. The slowdown in the 2010s is consistent with the maturation of the Douro DOC expansion and the administrative constraints of the Port benefício quota system. For RVV, the peak decade was the 1960s (2,331 mhl), with a sharp decline beginning in the 1980s (1,638 mhl), continuing steeply into the 2000s (906 mhl), exactly coinciding with Portugal's EU accession (1986) and the structural reforms that followed. By the 2010s, RVV production had fallen to approximately one-third of its 1960s peak.

These structural trajectories directly explain the era confound described throughout the thesis. In both regions, the linear trend does not fully capture the non-linear shape of the production history, particularly the RVV plateau in the 1970s (2,229 mhl, barely below the 1960s peak) followed by a sharp break in the 1980s. This structural break is the primary reason LOESS detrending performs better than linear detrending for RVV (Section 5.5): the flexible LOESS curve can fit the plateau-then-collapse shape, while the linear trend averages over it, leaving systematic residuals in the 1970s.

B.2 Phenological Timing: Trends and Variability

The phenological dates used as anchors for feature engineering are themselves variable across years and show systematic advancement over the study period, consistent with the warming trends documented in the broader viticulture literature. Table B.3 summarises the mean, variability, and long-term trend for each key phenological stage. Negative trends indicate earlier occurrence over time; all trends are expressed in days per decade.

Table B.3: Phenological timing: mean dates, variability, and decadal trends.

Stage	RDD mean (DOY / date)	RDD SD (days)	RDD trend (days/decade)	RVV mean (DOY / date)	RVV SD (days)	RVV trend (days/decade)
Bud Break (BB)	DOY 88 / 27 Mar	8.4	−0.5	DOY 90 / 30 Mar	9.2	−1.0
Flowering (Fl)	DOY 135 / 14 May	9.8	−0.7	DOY 154 / 2 Jun	12.0	−1.1
Start Maturity (sM)	DOY 179 / 27 Jun	8.6	−0.9	DOY 227 / 14 Aug	13.6	−1.7
Harvest (Hv)	DOY 257 / 13 Sep	13.3	−1.9	DOY 270 / 26 Sep	17.0	−2.6

Several findings stand out. First, harvest timing is advancing most rapidly in both regions, by approximately 1.9 days per decade in RDD and 2.6 days per decade in RVV. Over the 80–90 year study period this represents an advancement of roughly 15–23 days, consistent with the warming of 1.3°C in growing-season temperature projected and partly already observed in Portuguese wine regions (Cunha and Richter, 2016). Earlier harvest is a well-documented consequence of warming: grapes accumulate heat units faster, reaching physiological maturity sooner. Second, the trend is stronger in RVV than in RDD for every stage, which is consistent with the Atlantic climate's stronger sensitivity to warming, in a wetter regime, higher temperatures have a disproportionate effect on evapotranspiration and vine water status, accelerating phenological development.

Second, the variability (SD) in phenological dates is larger in RVV than RDD for all stages, particularly for start of maturity (13.6 vs 8.6 days) and harvest (17.0 vs 13.3 days). This higher phenological variability in RVV is consistent with the region's more variable Atlantic weather: ocean-driven temperature swings create larger year-to-year variation in the timing of heat accumulation. This greater phenological variability also means that fixed-calendar aggregation windows would be a poor substitute for the phenological-anchor approach used in this thesis: a fixed "September temperature" window would capture completely different vine developmental states in an early harvest year vs a late harvest year.

B.2.1 Growing Season Duration

Table B.4 shows the mean duration of each inter-stage phase for both regions.

The phase durations reveal important differences in growing season structure. The flowering-to-veraison phase is 28 days longer in RVV (73 vs 45 days), the Atlantic climate's slower heat accumulation extends this key berry development phase. Conversely, the veraison-to-harvest phase is substantially longer in RDD (78 vs 43 days), reflecting that Douro grapes undergo a longer slow-ripening phase under the drier semi-arid conditions before reaching harvest maturity. These structural differences mean that features from the same phenological window cover different

Table B.4: Production statistics by era for RDD and RVV.

Phase	RDD mean	RDD SD	RVV mean	RVV SD	Interpretation
Budburst → Flowering	47.0 days	7.9d	64.3 days	9.0d	RVV flowering 17d later than RDD on average, cooler Atlantic spring
Flowering → Veraison	44.8 days	4.9d	73.0 days	6.9d	RVV takes 28d longer to reach veraison, slower ripening in Atlantic climate
Veraison → Harvest	78.1 days	6.5d	42.7 days	7.7d	RDD takes longer from veraison to harvest, slower final ripening phase
Full season (BB→Hv)	169.9 days	12.6d	180.0 days	13.8d	RVV growing season 10 days longer overall

absolute calendar periods in the two regions, and directly motivate the use of separate feature construction and model fitting for each region rather than a pooled approach.

B.3 Feature Correlations with Production

Table B.5 shows the ten features most strongly correlated with wine production residuals (detrended) in each region. This analysis was part of the exploratory phase that informed the feature selection strategies evaluated in CarenR. In the direction column, + indicates positive correlation with production and − negative.

Several observations from this correlation table are noteworthy. First, swa_hv_y1_b20 (harvest soil water) appears in the top 10 for both regions, but with opposite signs, negative (r = −0.223) for RDD and positive (r = +0.245) for RVV. This is the same directional inversion identified in the CarenR rules (Section 4.1.3), now confirmed at the simple correlation level. Second, the top RDD correlate is harvest temperature (r = +0.386), which does not prominently appear in the top CarenR rules because CarenR uses discretised variables and the KS test detects distributional shift rather than linear correlation. The harvest temperature signal is captured by M5Rules (positive coefficients) and RIPPER (harvest temperature in Rules 1–2) instead. Third, the top RVV correlate is LAI at harvest (r = −0.404), consistent with its dominance in the CarenR RVV rule set. The negative sign confirms that higher LAI reduces production, the canopy vigour mechanism discussed throughout Chapter 4.

B.4 Feature Drift: The Era Confound Quantified

A critical question for interpreting the CarenR rules is which features are genuinely informative about inter-annual climate variation and which are primarily capturing the long-term structural trend (era membership). Features strongly correlated with calendar year are suspected era-confounders:

Table B.5: Feature correlations with detrended production ($|r|$) for RDD and RVV.

Rank	RDD feature	RDD |r|	Direction	RVV feature	RVV |r|	Direction
1	tm_hv_y1_c10 (harvest temp)	0.386	+	iaf_hv_y1_c10 (LAI at harvest)	0.404	−
2	tn_hv_y0_a30 (prev-yr harvest nights)	0.347	+	tm_hv_y1_b20 (pre-harvest temp)	0.402	−
3	swa_fl_y1_a30 (post-fl soil water)	0.283	−	tm_sm_y1_c20 (maturity temp)	0.342	−
4	iaf_fl_y0_b20 (prev-yr fl LAI)	0.276	−	tn_hv_y0_a30 (prev-yr harvest nights)	0.325	−
5	iaf_fl_y0_b30 (prev-yr fl LAI, wide)	0.268	−	days.rf.above.1mm_fl_y1_a20 (wet days post-fl)	0.306	−
6	swa_fl_y1_a15 (post-fl soil water)	0.261	−	tm_sm_y0_a20 (prev-yr summer temp)	0.302	−
7	tm_hv_y1_b20 (pre-harvest temp)	0.248	+	rf_fl_y1_a10 (post-fl rainfall)	0.288	−
8	tm.days.less.15_fl_y1_b15 (cold days at fl)	0.245	−	days.rf.above.1mm_fl_y1_c15 (wet days at fl)	0.264	−
9	iaf_fl_y0_b10 (prev-yr fl LAI)	0.227	−	tm_hv_y1_c10 (harvest temp)	0.253	−
10	swa_hv_y1_b20 (harvest soil water)	0.223	−	swa_hv_y1_b20 (harvest soil water)	0.245	+

they co-vary with the production trend for structural reasons rather than because they drive production through a climate mechanism. Table B.6 shows the features most strongly correlated with year for both regions; features marked HIGH co-trend strongly with the production structural trend.

The harvest temperature variable (tm_hv_y1_c10) is the most era-confounded feature in the dataset: it correlates with production at r = +0.386 (RDD) but also correlates with year at r = +0.412. This near-equal correlation with both production and time is a warning sign: the feature may appear predictive of production largely because both are trending upward over the study period (warming climate + increasing Douro production), rather than because harvest temperature genuinely drives production inter-annually. The LOESS validation (Section 4.2) addresses this precisely: after removing the era trend, the top features' correlations drop from 0.51 to 0.47, confirming that the era confound inflates but does not manufacture the relationship. For harvest temperature specifically, the LOESS-corrected correlation would likely drop more, which is consistent with CarenR's top rules focusing on soil water and rainfall rather than temperature, these latter variables have lower year-correlations (swa_hv_y1_b20: year r = −0.183 for RDD) and represent cleaner within-era signals.

The RVV LAI at harvest (iaf_hv_y1_c10) shows a more interesting pattern: it correlates with production at r = −0.404 but also with year at r = +0.359. This suggests that LAI has been increasing over the study period, consistent with the general intensification of vineyard management and the shift to espalier systems, and this increasing LAI co-trends with declining production, inflating its apparent correlation. However, the LOESS validation confirms that even after era-trend removal,

Table B.6: Feature drift: correlation with year index as proxy for era confound.

Rank	RDD feature	r(feat, year)	Concern level	RVV feature	r(feat, year)	Concern level
1	tm_hv_y1_c10 (harvest temp)	+0.412	HIGH, warming trend	tm_sm_y1_c20 (maturity temp)	+0.402	HIGH, warming trend
2	tm_hv_y1_b20 (pre-harvest temp)	+0.373	HIGH	iaf_hv_y1_c10 (LAI at harvest)	+0.359	MODERATE
3	tn_hv_y0_a30 (prev-yr harvest nights)	+0.371	HIGH, warming	tm_sm_y0_b20 (prev-yr maturity temp)	+0.348	HIGH
4	tm_sm_y0_a20 (prev-yr summer temp)	+0.269	MODERATE	tm_hv_y1_b20 (pre-harvest temp)	+0.335	HIGH
5	days.rf.above.1mm_hv (harvest rain days)	−0.201	LOW	tm_sm_y0_a20 (prev-yr summer temp)	+0.295	MODERATE

LAI remains one of the strongest within-era predictors in RVV, confirming its genuine within-era signal. The era confound inflates its correlation but does not invent the relationship.

B.5 Experiment Pipeline

Figure B.1 shows the complete experiment pipeline from raw source files to final modelling outputs.

Table B.7 breaks the pipeline into its five numbered stages. Table B.8 lists the fixed configuration parameters stored in `utils/config.R`.

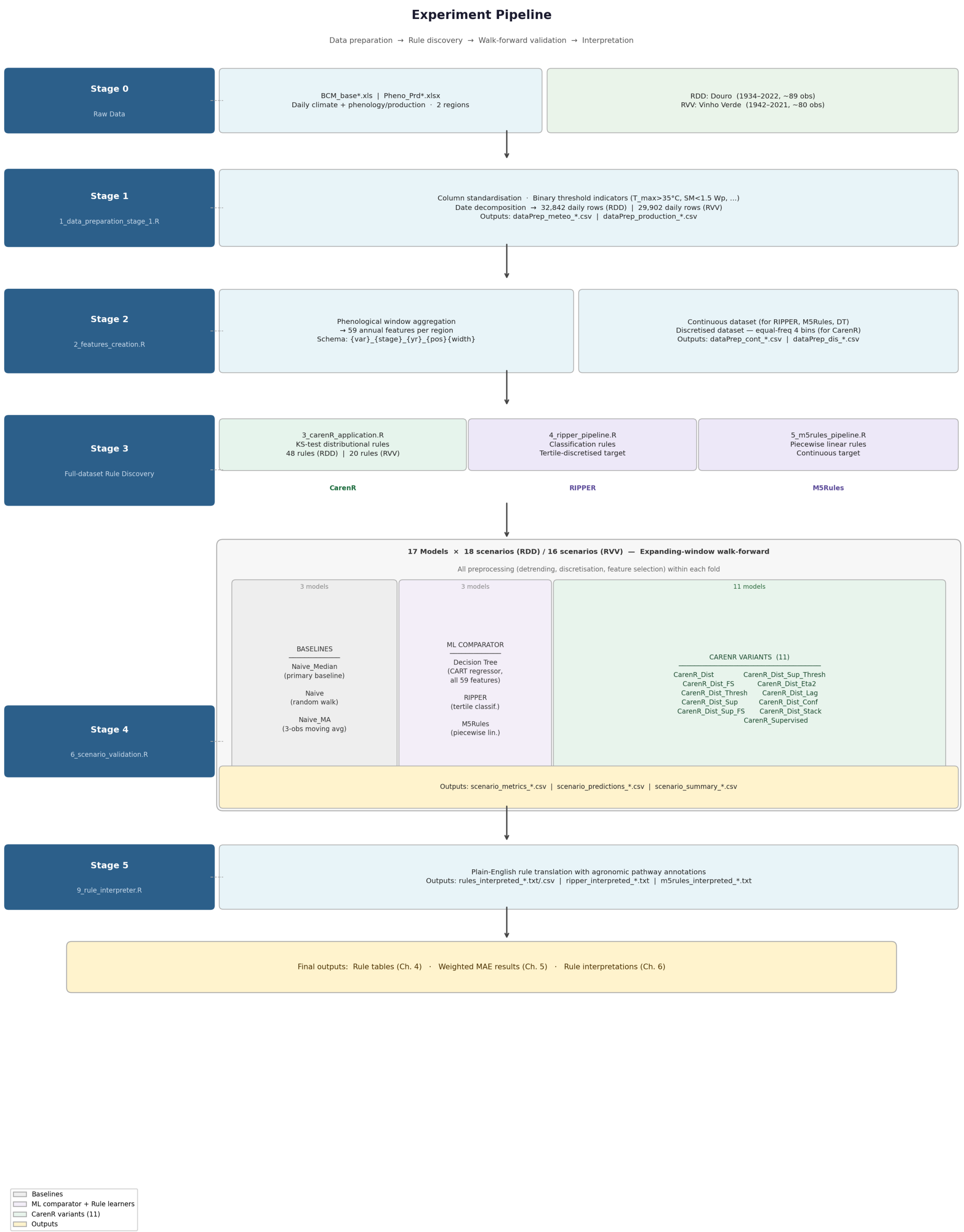

Figure B.1: Experiment pipeline: data preparation, rule discovery (CarenR, RIPPER, M5Rules), and walk-forward validation.

Table B.7: Complete data pipeline from raw sources to final outputs.

Stage	Script	Input	Output	Key transformation
0: Raw data	(external)	BCM_ base*.xls Pheno_ Prd*.xlsx	Raw Excel files	Daily climate + annual phenology/production from field stations and STICS model
1: Daily prep	1_data_ preparation_ stage_1.R	Raw Excel files	dataPrep_ meteo_*.csv dataPrep_ production_ *.csv	Column standardisation, binary threshold indicators (Tmax>35, Sm<1.5Wp etc.), date decomposition. 32,842 rows (RDD), 29,902 rows (RVV).
2: Feature engineering	2_features_ creation.R	dataPrep_ meteo_*.csv dataPrep_ production_ *.csv	dataPrep_cont_ dataset_*.csv dataPrep_dis_ rule_dataset_ *.csv	Phenological window aggregation $\rightarrow$ 59 annual features per region. Equal-frequency discretisation (4 bins) for CarenR. 89 rows (RDD), 80 rows (RVV).
3: Rule discovery	3_carenR__ application.R 4_ripper_ pipeline.R 5_m5rules_ pipeline.R	dataPrep_*_ dataset_*.csv	RulesTemp_*_ linear.csv ripper_rules_*.txt m5rules_rules_ *.txt	Full-dataset rule induction for all three algorithms. CarenR: 48 rules (RDD), 20 rules (RVV).
4: Walk-forward validation	6_scenario_ validation.R	dataPrep_cont_ dataset_*.csv	scenario_ metrics_*.csv scenario_ predictions_ *.csv scenario_ summary_*.csv	17-model $\times$ 18/16 scenario expanding-window evaluation. All preprocessing within fold.
5: Rule interpretation	9_rule_ interpreter.R	RulesTemp_ *.csv ripper_ rules_*.txt m5rules_rules_ *.txt	rules_ interpreted_ *.txt/.csv ripper_ interpreted_ *.txt m5rules_ interpreted_*.txt	Translation of raw rule outputs to plain-English descriptions with agronomic pathway annotations.

Table B.8: Fixed configuration parameters for CarenR and the walk-forward protocol.

Parameter	Value	Location in code
Detrending method	Linear (OLS)	config.R: detrend_method
CarenR support threshold	20% ($s_i \geq 0.20$)	6_scenario_validation.R: carenr_min_sup
Significance threshold	p $\leq$ 0.10	config.R: alpha_threshold
Minimum training fraction	80%	config.R: min_train_frac
Feature selection strategy	Per variant (η^2, Pearson, or support-filtered)	config.R: fs_strategy

Appendix C

Complete Feature Reference

This appendix provides a complete reference for all 58 agroclimatic features used in the modelling datasets (dataPrep_cont_dataset_rdd.csv and dataPrep_cont_dataset_rvv.csv). Features are grouped by variable family. The naming convention is described in Section 3.11.1. For each feature, the full name, phenological anchor, year offset, window specification, unit, calculation type, and agronomic interpretation are provided.

Key notation: y1 = current harvest year; y0 = previous growing season. Positions: c = centred ($\pm$n/2 days around stage DOY), b = n days before stage DOY, a = n days after stage DOY. All aggregations are computed over the specified window relative to the phenological DOY for each individual year, using the daily meteorological records from dataPrep_meteo_*.csv.

C.1 Temperature Features (13 features)

Table C.1: Temperature features (13 features).

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
tm_fl_y1_c15	Mean temperature	Flowering	y1	centred $\pm$15d	°C	Average	Thermal conditions during the critical flowering window, governs pollen viability, fruit set, and inflorescence development
tm_fl_y1_b10	Mean temperature	Flowering	y1	10d before	°C	Average	Pre-flowering thermal conditions, vine heat accumulation entering the flowering phase
tm_fl_y1_a10	Mean temperature	Flowering	y1	10d after	°C	Average	Post-flowering temperature, governs initial berry cell division and early fruit development
tm_bb_y0_a15	Mean temperature	Bud Break	y0	15d after prev. yr	°C	Average	Previous-year early spring thermal conditions, influences inflorescence primordia initiation for current year

Continued on next page

Table C.1 continued from previous page

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
tm_bb_y0_a30	Mean temperature	Bud Break	y0	30d after prev. yr	°C	Average	Wider post-budburst window in previous year, vine carbohydrate allocation and early season vigour
tm_sm_y1_c20	Mean temperature	Start Maturity	y1	centred ±20d	°C	Average	Temperature at veraison onset, governs timing of berry colour change and ripening acceleration
tm_sm_y0_a20	Mean temperature	Start Maturity	y0	20d after prev. yr	°C	Average	Previous-year post-veraison heat, vine thermal stress at end of prior season; affects reserve accumulation
tm_sm_y0_b20	Mean temperature	Start Maturity	y0	20d before prev. yr	°C	Average	Previous-year pre-veraison temperature, mid-ripening thermal conditions in prior season
tm_hv_y1_c10	Mean temperature	Harvest	y1	centred ±10d	°C	Average	Temperature at harvest, final ripening conditions; warmer harvest accelerates sugar accumulation
tm_hv_y1_b20	Mean temperature	Harvest	y1	20d before	°C	Average	Pre-harvest temperature, late-season thermal driver of final berry maturation
tm_fl_y0_b10	Mean temperature	Flowering	y0	10d before prev. yr	°C	Average	Previous-year pre-flowering temperature, key driver of inflorescence primordia differentiation
tm_fl_y0_a10	Mean temperature	Flowering	y0	10d after prev. yr	°C	Average	Previous-year post-flowering temperature, affects fruit set and bunch architecture for current year
tn_hv_y0_a30	Min temperature	Harvest	y0	30d after prev. yr	°C	Average	Previous-year post-harvest night temperatures, warm nights maintain vine metabolic activity; affect carbohydrate reserve building

C.2 Cold and Frost Day Features (5 features)

C.3 Heat Stress Day Features (4 features)

C.4 Rainfall Features (8 features)

Table C.2: Cold day (Tmed $<$ 15°C) and frost day (Tmin $<$ 0°C) features (5 features).

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
tm.days.less.15_ fl_y1_c15	Cold days (Tmed<15°C)	Flowering	y1	centred ±15d	days	Count	Number of cold days during flowering, impairs pollen viability and fertilisation; directly reduces berry set
tm.days.less.15_ fl_y1_b15	Cold days (Tmed<15°C)	Flowering	y1	15d before	days	Count	Pre-flowering cold days, cool conditions entering the flowering phase
days_ minTemp.less.0_ bb_y1_c20	Frost days (Tmin<0°C)	Bud Break	y1	centred ±20d	days	Count	Frost days at budburst, potential spring frost damage to emerging shoots; severe frosts can destroy entire crop
days.minTemp.less.0_ bb_y1_a30	Frost days (Tmin<0°C)	Bud Break	y1	30d after	days	Count	Post-budburst frost risk window, young shoots most vulnerable
days.minTemp.less.0_ fl_y1_a10	Frost days (Tmin<0°C)	Flowering	y1	10d after	days	Count	Late frost at flowering, unusual but damaging frost events during fruit set

Table C.4: Rainfall total and wet day count features (8 features).

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
rf_fl_y1_a10	Rainfall total	Flowering	y1	10d after	mm	Sum	Rainfall immediately post-flowering, disease risk (Botrytis, downy mildew); excess rain dilutes berry development
rf_hv_y1_b10	Rainfall total	Harvest	y1	10d before	mm	Sum	Pre-harvest rainfall, late rain can cause berry splitting, dilution of sugars, and Botrytis infection
rf_hv_y0_a10	Rainfall total	Harvest	y0	10d after prev. yr	mm	Sum	Post-harvest rainfall in previous year, begins recharging soil water reserves for the following season
days.rf.above.1mm_ fl_y1_c15	Wet days (Rain>1mm)	Flowering	y1	centred ±15d	days	Count	Frequency of rain days at flowering, high frequency promotes disease and physically interferes with pollination
days.rf.above.1mm_ fl_y1_a20	Wet days (Rain>1mm)	Flowering	y1	20d after	days	Count	Post-flowering rain event count, key disease pressure indicator in the Douro; DRY = above trend
days.rf.above.1mm_ sm_y1_b20	Wet days (Rain>1mm)	Start Maturity	y1	20d before	days	Count	Rain days approaching veraison, moisture during mid-season ripening phase

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Table C.4 continued from previous page

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
days.rf.above.1mm_sm_y1_a20	Wet days (Rain>1mm)	Start Maturity	y1	20d after	days	Count	Post-veraison rain days, disease and dilution risk during the ripening phase; 2nd most frequent RDD rule feature
days.rf.above.1mm_hv_y1_c10	Wet days (Rain>1mm)	Harvest	y1	centred ±10d	days	Count	Rain days at harvest, wet harvest conditions cause berry splitting, Botrytis; reduce quality and quantity

C.5 Leaf Area Index (LAI) Features (10 features)

Table C.5: Leaf Area Index (LAI) features (10 features).

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
days.rf.above.1mm_hv_y0_a10	Wet days (Rain>1mm)	Harvest	y0	10d after prev. yr	days	Count	Post-harvest rain events in previous year, soil recharge initiation; carry-over for next season
iaf_bb_y1_c10	Leaf Area Index	Bud Break	y1	centred ±10d	m^2/m^2	Average	LAI at budburst, canopy size at growth initiation; proxy for vine vigour entering the season
iaf_fl_y1_b10	Leaf Area Index	Flowering	y1	10d before	m^2/m^2	Average	Pre-flowering canopy development, vegetative vigour entering the critical fruit-set window
iaf_fl_y1_b20	Leaf Area Index	Flowering	y1	20d before	m^2/m^2	Average	Broader pre-flowering LAI window, canopy development trajectory before flowering
iaf_fl_y1_b30	Leaf Area Index	Flowering	y1	30d before	m^2/m^2	Average	Early pre-flowering canopy, wider window capturing canopy build-up phase
iaf_fl_y1_a10	Leaf Area Index	Flowering	y1	10d after	m^2/m^2	Average	Post-flowering canopy, rapid leaf area expansion immediately after flowering; vigour proxy
iaf_hv_y1_c10	Leaf Area Index	Harvest	y1	centred ±10d	m^2/m^2	Average	LAI at harvest, canopy size at the time of picking; appears in 2 of 20 RVV rules (LAI family in 4/20); high LAI $\rightarrow$ below trend
iaf_fl_y0_b10	Leaf Area Index	Flowering	y0	10d before prev. yr	m^2/m^2	Average	Previous-year pre-flowering LAI, vine canopy state before prior-year fruit set; carry-over vigour signal
iaf_fl_y0_b20	Leaf Area Index	Flowering	y0	20d before prev. yr	m^2/m^2	Average	Wider previous-year pre-flowering LAI window

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Table C.5 continued from previous page

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
iaf_fl_y0_b30	Leaf Area Index	Flowering	y0	30d before prev. yr	m^2/m^2	Average	Widest previous-year pre-flowering LAI window: M5Rules LAI differential uses b10/b20/b30 together

iaf_hv_y1_c10 appears in 2 of 20 RVV rules; the dominant RVV features are heat-stress days (up to 50% of rules).

C.6 Soil Water Features (18 features)

Table C.6: Soil water availability, stress day (Sm $<$ 1.5$\times$Wp), and saturation day (Sm $>$ 0.9$\times$Fc) features (18 features).

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
iaf_fl_y0_a10	Leaf Area Index	Flowering	y0	10d after prev. yr	m^2/m^2	Average	Post-flowering LAI in previous year, canopy expansion after prior-year fruit set
swa_fl_y1_a15	Soil water availability	Flowering	y1	15d after	mm	Average	Soil water balance post-flowering, moisture availability during early berry development; excess → vigour/disease in RVV
swa_fl_y1_a30	Soil water availability	Flowering	y1	30d after	mm	Average	Wider post-flowering soil water window, cumulative moisture through mid-season; appears in RVV rules 3, 16, 19
swa_hv_y1_b20	Soil water availability	Harvest	y1	20d before	mm	Average	Pre-harvest soil water: DOMINANT feature in both regions; RDD: moderate deficit → above trend; RVV: deficit → below trend
swa_hv_y0_a15	Soil water availability	Harvest	y0	15d after prev. yr	mm	Average	Post-harvest soil water in previous year, soil recharge initiation; carry-over moisture for next season
swa_hv_y0_a20	Soil water availability	Harvest	y0	20d after prev. yr	mm	Average	Wider post-harvest soil water in previous year, confirms carry-over recharge pattern
sw.less.15wp_fl_y1_a10	Stress days (Sm<1.5×Wp)	Flowering	y1	10d after	days	Count	Severe stress days post-flowering, days when soil water falls below vine wilting threshold
sw.less.15wp_fl_y1_a20	Stress days (Sm<1.5×Wp)	Flowering	y1	20d after	days	Count	Extended post-flowering stress exposure

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Table C.6 continued from previous page

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
sw.less.15wp_sm_y1_a10	Stress days (Sm<1.5×Wp)	Start Maturity	y1	10d after	days	Count	Severe stress days at veraison onset, critical window for sugar–acid balance
sw.less.15wp_sm_y1_a20	Stress days (Sm<1.5×Wp)	Start Maturity	y1	20d after	days	Count	Extended stress at maturity, prolonged wilting-level stress during ripening
sw.less.15wp_hv_y1_b20	Stress days (Sm<1.5×Wp)	Harvest	y1	20d before	days	Count	Pre-harvest severe stress, intense berry concentration but risk of vine damage if prolonged
sw.less.15wp_hv_y1_a20	Stress days (Sm<1.5×Wp)	Harvest	y1	20d after	days	Count	Post-harvest severe stress, vine recovery period after harvest
sw.less.15wp_fl_y0_a10	Stress days (Sm<1.5×Wp)	Flowering	y0	10d after prev. yr	days	Count	Previous-year post-flowering stress, carry-over water deficit from prior season
sw.less.15wp_fl_y0_a20	Stress days (Sm<1.5×Wp)	Flowering	y0	20d after prev. yr	days	Count	Extended previous-year post-flowering stress
sw.above.09fc_fl_y1_a10	Saturation days (Sm>0.9×Fc)	Flowering	y1	10d after	days	Count	Near-saturation days post-flowering, promote disease and vegetative vigour; relevant to RVV below-trend rules
sw.above.09fc_fl_y1_a20	Saturation days (Sm>0.9×Fc)	Flowering	y1	20d after	days	Count	Extended near-saturation post-flowering
sw.above.09fc_fl_y0_a10	Saturation days (Sm>0.9×Fc)	Flowering	y0	10d after prev. yr	days	Count	Previous-year post-flowering soil saturation, carry-over vigour effect identified in RVV Rule 2
sw.above.09fc_fl_y0_a20	Saturation days (Sm>0.9×Fc)	Flowering	y0	20d after prev. yr	days	Count	Wider window for previous-year post-flowering soil saturation, strongest RVV carry-over signal

swa_hv_y1_b20 is the dominant RDD feature (54% of rules); in RVV it appears in 25% of rules.

Table C.3: Heat stress day (Tmax > 35°C) features (4 features).

Feature name	Variable	Stage	Year	Window	Unit	Calc	Agronomic interpretation
days.maxTemp.above.35_fl_y1_a10	Heat days (Tmax>35°C)	Flowering	y1	10d after	days	Count	Extreme heat at flowering, causes flower abortion, pollen tube failure, and reduces berry number
days.maxTemp.above.35_fl_y1_a20	Heat days (Tmax>35°C)	Flowering	y1	20d after	days	Count	Extended post-flowering heat exposure, sustained heat impact on early berry development
days.tempMax.above.35_sm_y1_b20	Heat days (Tmax>35°C)	Start Maturity	y1	20d before	days	Count	Heat stress approaching veraison, accelerates sugar rise; can impair acid retention and berry integrity
days.maxTemp.above.35_sm_y1_a20	Heat days (Tmax>35°C)	Start Maturity	y1	20d after	days	Count	Post-veraison heat stress, berry thermal stress during the key ripening phase

Appendix D

Feature Coverage Map

Figure D.1 provides a visual overview of all 59 agroclimatic features in the dataset, organised by variable family (rows) and phenological stage $\times$ window position (columns). Each coloured cell represents one or more feature windows: the darker upper half shows features computed for the current harvest year (y1); the lighter lower half shows features computed for the previous year (y0). The number inside each cell is the window width in days.

The map makes several structural properties of the feature set immediately visible. Flowering is the most densely populated stage, contributing features across all variable families and both years, reflecting the critical importance of flowering-phase conditions for yield determination. The previous-year (y0) features are concentrated at Flowering and Harvest, corresponding to the two biological carry-over mechanisms described in Section 3.11.3: inflorescence primordia formation (driven by prior-year flowering thermal conditions) and vine carbohydrate reserve building (driven by prior-year harvest and post-harvest conditions). Bud Break and Start Maturity have sparser coverage — only those specific variables with demonstrated agronomic relevance to those stages were included. The soil-water family (swa, stress days, saturation days) is notably concentrated at Flowering and Harvest, consistent with their role as the primary mechanistic pathways through which climate affects production identified in the CarenR rule analysis (Chapter 4).

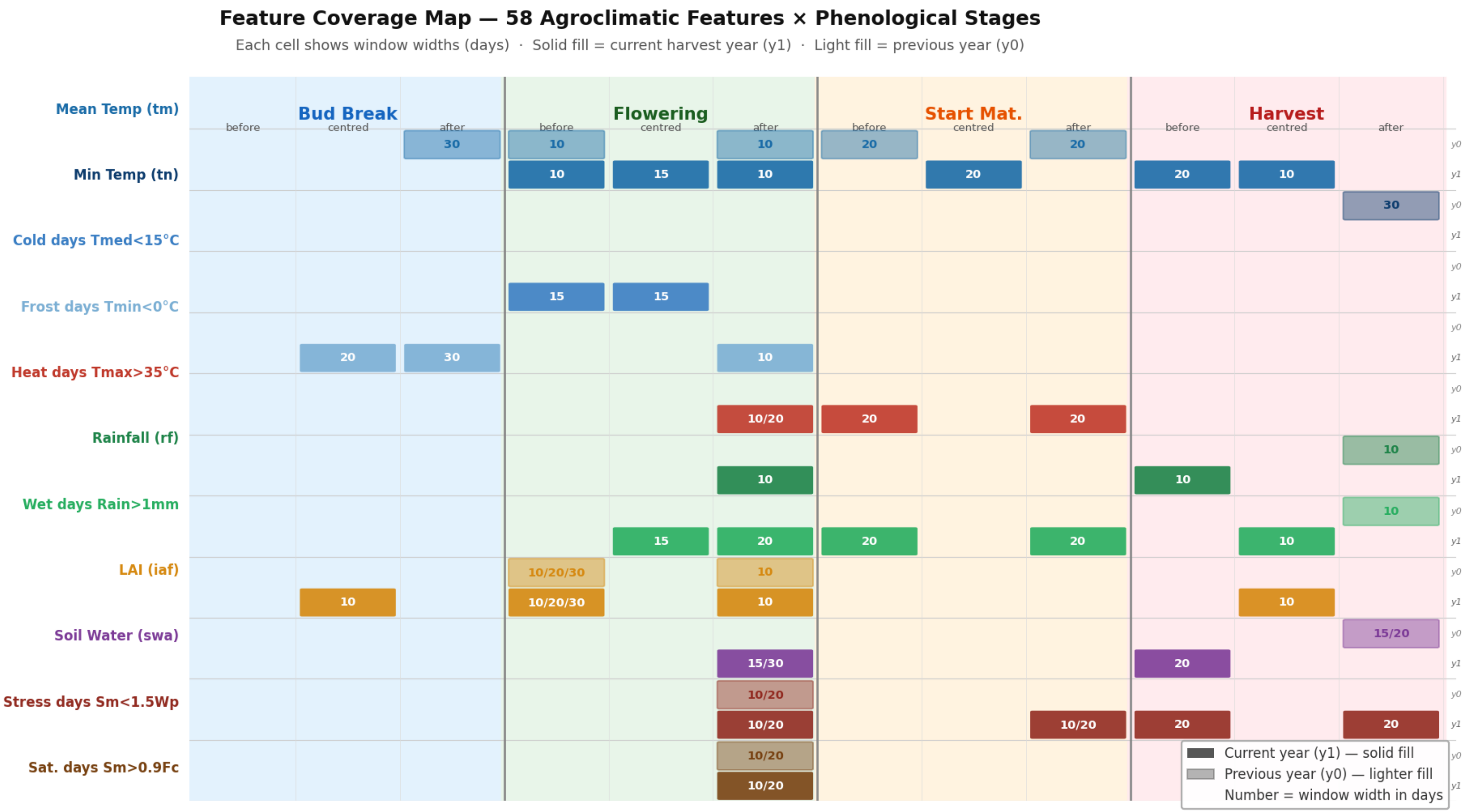

Figure D.1: Feature coverage map: 59 agroclimatic features by phenological stage and window position.

Each coloured cell represents one or more feature windows. Darker shading (top half) = current year (y1); lighter shading (bottom half) = previous year (y0). Numbers inside cells indicate window width in days. Stage colour coding: blue = Bud Break, green = Flowering, amber = Start Maturity, red = Harvest.

Appendix E

AI Use Transparency Statement

AI Use Transparency Appendix

Thesis title: Rule-based Machine Learning for Wine Production Forecasting: A Distributional Subgroup Discovery Approach

Student: Hugo Filipe Queiróz Nogueira

Tool used: Claude (Anthropic), model claude-sonnet-4-6, accessed via Claude Cowork desktop application

Period of use: May–June 2026

E.1 Overview of AI Use

AI was used throughout the writing and revision process of this dissertation. All research activities — dataset construction, R pipeline development, CarenR experiments, statistical analysis, and interpretation of results — were carried out independently by the student. AI use was restricted to four categories: (1) writing support, (2) substantive content revision, (3) code generation for auxiliary scripts, and (4) numerical verification.

The student directed all AI interactions, reviewed every output, and validated all content against primary data sources. The numerical verification process (Section E.5) was specifically designed to detect and correct AI-generated errors in the thesis text.

E.2 Writing Support — All Chapters

Writing support (grammatical corrections, comma splice fixes, reformulation of awkward phrasing, removal of formulaic AI-sounding constructions) was applied across all chapters. This is the most widespread category of use. The student identified problems and directed specific corrections; AI did not generate substantive new content in this category.

Table E.1: AI use summary table 1.

Section	Type of editing	Example prompt
Abstract	Reformulation, AI-phrase removal	"rewrite any AI-sounding phrases to be more direct and academic"
Ch. 1 — Introduction	Sentence restructuring, comma splices	"fix grammar and AI-sounding phrases throughout"
Ch. 2 — Literature Review	Reformulation, hedging language	"flag and rewrite passages that sound AI-generated"
Ch. 3 — Methods	Comma splices, paragraph flow	"fix comma splices and reformulate where needed"
Ch. 4 — Rule Discovery	Comma splices, sentence structure	"go to file 12 and fix any issues"
Ch. 5 — Prediction results	Comma splices, reformulation	"fix this paragraph, it reads as AI-written"
Ch. 6 — Discussion	AI-phrase removal, vague citation removal	"remove Santos et al. (various) — this is not a real citation"
Ch. 7 — Conclusions	Factual correction	(numerical fix only, see Section E.5)

E.3 Substantive Content Revision

In several sections, AI was used to generate or substantially rewrite one or more paragraphs. These are declared below at section level with representative prompts and iteration examples as required.

E.3.1 Chapter 3 — Methods: CarenR_Supervised Description

Location: Section 3.4.1, CarenR variant table and Table 3.3

Purpose: The original description of the CarenR_Supervised variant incorrectly described it as performing "discretisation of the response variable." After the student identified the correct behaviour from the source code, AI was used to rewrite the description accurately.

Iteration example:

Table E.2: AI use summary table 2.

#	Prompt	Outcome
1	"what is the CarenR_Supervised, which performs discretisation of the response variable?"	AI explained the difference between response discretisation and feature discretisation
2	"yes" [to updating the description in ch3]	Description updated: "performs discretisation of the response variable" → "uses supervised feature discretisation — deriving bin cut-points by maximising residual separation within each training fold"
3	"check in the table at chapter 3 if some update is needed"	Table 3.3 description updated to match; missing CarenR_Dist_Sup_Thresh row added to variant table

E.3.2 Chapter 3 — Methods: Table 3.3 New Row

Location: Section 3.4.1, Table 3.3 (CarenR variant breakdown)

Purpose: The CarenR_Dist_Sup_Thresh variant was missing from the 11-variant breakdown table. After the student identified the gap, AI generated the missing row content based on the existing table structure and the student's understanding of the variant.

E.3.3 Chapter 5 — Prediction: Test Period Table Corrections

Location: Section 5.8, Tables 5.8 and 5.9 (walk-forward scenario tables)

Purpose: The student identified that the "Test yr" column was showing single years rather than multi-year test periods. AI was used to compute the correct test period ranges from the data and rewrite all 34 rows across both tables.

Iteration example:

Table E.3: AI use summary table 3.

#	Prompt	Outcome
1	"is this table the test year shouldn't be a couple of years?"	AI confirmed the test periods span multiple years and extracted correct ranges from scenario_metrics_rdd/rvv.csv
2	[review of proposed corrections]	Student validated ranges against their own understanding of the walk-forward setup; RVV train periods also corrected (were 4 years off)
3	"Continue from where you left off"	Column header renamed to "Test period"; all rows applied to both tables

E.3.4 Chapter 5 — Prediction: Bold Label Paragraphs

Location: Section 5.3 (per-scenario analysis paragraphs)

Purpose: Several paragraphs with a bold label (e.g., "Scenarios 3 and 4:") had their formatting collapsed when previous edits were applied. AI reconstructed the correct bold-label + normal-body structure.

Iteration example:

Table E.4: AI use summary table 4.

#	Prompt	Outcome
1	"this is in bold... apply the same format as the others [Scenarios 3 and 4 paragraph]"	AI identified the paragraph and reconstructed two-run structure: bold label up to colon, normal body text
2	"Continue from where you left off"	Same fix applied to "Scenarios 17 and 18" and "The fallback pattern" paragraphs
3	"while executing it, this par is full in bold [Why η^2 wins in RDD]"	Same fix applied in Chapter 5 expand file

E.4 Code Generation

AI was used to generate two types of code: modifications to an existing R pipeline script, and a new R analysis script. All code was reviewed and tested by the student. The student is able to explain the logic of all generated code.

Table E.5: AI use summary table 5.

File	Type	Description
src/rscripts/6_scenario_validation.R	Modification	Added CLI argument parsing (RDD/RVV region flag) and modified learning curve plot section to apply thicker lines to Naive_Median and best CarenR variant
src/rscripts/10_rule_overlap_analysis.R	New script	New R script computing pairwise feature overlap, distinct feature combinations, and condition-subset statistics for both regions
Verification scripts (Python)	New scripts	Python scripts to cross-verify all numerical claims in thesis chapters against data CSVs (scenario_metrics, rules_interpreted). Used to detect and correct errors; not part of submitted codebase.

Representative prompts for code generation:

• "I have an idea / Give some extra thickness to the naive baseline and the best caren for each plot" — resulted in linewidth aesthetic mapping in ggplot2

• "go through the full pages on every chapter and see if we have all the scripts available and once we have them execute it and check if the numbers match" — resulted in comprehensive Python verification script

E.5 Numerical Verification Process

A complete numerical verification pass was conducted across all chapter files. AI generated Python scripts that loaded the data CSVs, recomputed all key statistics (weighted MAE, Wilcoxon tests, overlap statistics, rule counts, direction counts, support thresholds), and compared them against the thesis text. The following errors were identified and corrected:

Table E.6: AI use summary table 6.

Location	Original	Corrected	Source
Ch. 3 — CarenR_Dist_Thresh threshold	$\lvert r\rvert \geq 0.20$	$\lvert r\rvert \geq 0.30$	Code: carenr_cor_thresh = 0.30
Ch. 4 — support filter	≥15% of obs.	≥20% of obs.	Code: carenr_min_sup = 0.20
Ch. 4 — RDD direction count	46/48 above-trend	31/48 above-trend	rules_interpreted_rdd.csv
Ch. 4 — RVV direction count	19/20 below-trend	16/20 below-trend	rules_interpreted_rvv.csv
Ch. 4 — RIPPER RVV accuracy	35%	27.5%	ripper_metrics_rvv.csv
Ch. 4 — RIPPER kappa range	κ = 0.06–0.19	κ = 0.13 (RDD), -0.09 (RVV)	ripper_rules_rdd/rvv.txt
Ch. 5/6 — support threshold refs	≥15% support	≥20% support	Code: carenr_min_sup = 0.20
Appendix A — RVV direction count	19/1 split	16/4 split	rules_interpreted_rvv.csv
Appendix C — feature count	59 features	58 features	dataPrep_cont_dataset_rdd.csv
Ch. 19/appendices — LAI in RVV	LAI in all 20 rules	LAI in 4/20 rules	rules_interpreted_rvv.csv

Claims verified as correct: rule counts (48/20), weighted MAEs (189.2/171.6/184/140 mhl), Wilcoxon tests (W=68 p=0.468; W=0 p=0.004), binomial test (5/5, p=0.031), pairwise overlap (730/1128 = 64.7%; 95/190 = 50.0%), distinct feature combinations (44/48; 20/20), feature-subset rules (28/48; 5/20), M5Rules rule counts (2/3), RIPPER RDD accuracy (42%), LOESS MAEs (90 mhl RVV; 290 mhl RDD), era-composition threshold (29%).

E.6 Student Responsibility Declaration

The student affirms that:

- All research activities (data preparation, feature engineering, model execution, statistical analysis) were conducted independently without AI assistance.
- All AI-generated or AI-revised content was critically reviewed and validated against primary sources before inclusion.
- The numerical verification process demonstrates active critical engagement: errors introduced earlier in the writing process (including AI-generated errors) were identified and corrected.

• The student understands and is able to explain all content in this dissertation, including all sections revised with AI assistance.

Hugo Filipe Queiróz Nogueira, June 2026